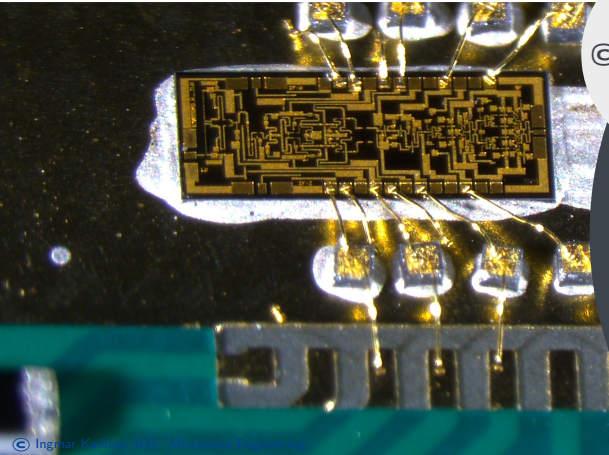
University of Stuttgart
Germany

Institute of
Robust Power
Semiconductor Systems



© Ingmar Kallfass

Microwave Engineering

Table of Contents I

1. Course Outline
2. The Millimeterwave Frequency Spectrum: Applications and Technologies
3. Functional Buildung Blocks of Monolithic Microwave Integrated Circuit (MMIC)s
4. Low Noise Amplifier
5. Broadband Amplifiers
6. Power Amplifiers
7. Microwave Network Analysis (Appendix)
8. Planar Transmission Lines

Abbreviations I

Notation	Description
ADC	analog-to-digital converter
AM-PM	amplitude modulation to phase modulation
ASIC	application-specific integrated circuit
AWGN	additive white gaussian noise
BCB	benzocyclobutene
BiCMOS	bipolar complementary metal-oxide-semiconductor
BJT	bipolar junction transistor
CAD	computer-aided design
CDR	clock-and-data recovery
CG	common-gate
CMOS	Complementary Metal-Oxide-Semiconductor

Abbreviations II

CPU	central processing unit
CPW	coplanar waveguide
CS	common-source
DAC	digital-to-analog converter
DHBT	double-heterojunction bipolar transistor
DR	dynamic range
ECTS	European Credit Transfer System
EM	electro-magnetic
FET	field-effect-transistor
FMCW	Frequency-Modulated-Continuous-Wave
FPGA	Field Programmable Gate Arrays
FR	flame retardant

Abbreviations III

GaAs	Gallium Arsenide
GaN	gallium nitride
GCPW	grounded coplanar waveguide
HBT	heterojunction bipolar transistor
HEMT	high-electron-mobility transistor
IC	integrated circuit
IF	Intermediate Frequency
IIP3	Input related third-order intercept point
InP	indium phosphide
IP	intellectual property
IP3	Third-order intercept point
IQ	inphase-quadratur
IR mixers	image-rejection mixers
IV	current-voltage

Abbreviations IV

LO	local-oscillator
MDS	minimum detectable signal
MHEMT	metamorphic-high-electron-mobility transistor
MIC	MIC
MIM	Metal-Insulator-Metal
MIMO	multiple input multiple output
MMIC	Monolithic Microwave Integrated Circuit
MMICs	Monolithic Microwave Integrated Circuits
mmW	millimeter-wave
MOSFET	metal–oxide–semiconductor field-effect transistor
MSL	Microstrip Transmission Lines
MW	Microwave
MWE	Microwave Engineering

Abbreviations V

NRZ	non return to zero
OFDM	orthogonal frequency division multiplex
OIP3	Output related third-order intercept point
OOK	on-off keying
PA	power amplifier
PAE	power-added-efficiency
PDK	Process Design Kit
pHEMT	pseudomorphic high-electron-mobility transistor
PLL	phase-locked loop
PM-AM	phase modulation to amplitude modulation
RF	Radio Frequency
RX	Receiver

Abbreviations VI

SFDR	Spurious-free dynamic range
Si	silicon
SiGe	silicon germanium
SiN	silicon nitride
SiO_2	silicon oxide
SISO	single input single output
SNR	signal to noise ratio
SoC	System-on-a-Chip
SOI	Silicon on isolator
ssb	single-sideband
SSPA	solid state power amplifier
TIA	transimpedance amplifier
TRL	transmission line
TSV	Through-Substrate-Via
TWA	Travelling-Wave Amplifier

Abbreviations VII

TX	Transmitter
VNA	vector network analyzer
VSWR	voltage standing wave ratio
WLAN	wireless local area network
WPAN	wireless personal area network

Symbols I

Symbol	Description	Unit
A_0	low-frequency voltage gain	1
β	phase constant	°
BW	absolute bandwidth	Hz
δR	range resolution	m
FBW	fractional bandwidth	%
f_c	center frequency	Hz
f_l	lower corner frequency	Hz
f_u	upper corner frequency	Hz
f_max	Maximum frequency of oscillation	GHz
f_T	transition-frequency	GHz

Symbols II

ICP_{1dB}	input-related 1-dB compression point	W
MAG	maximum available gain	dB
MSG	maximum stable gain	dB
OCP_{1dB}	output-related 1-dB compression point	W
P_{sat}	saturated output power	W
Q	quality factor	1
B	bandwidth	Hz
C	channel capacity	b/s
N_0	noise power spectral density	W
P_{r}	received power	W

Symbols III

c_0	speed of light	m/s
τ	group delay	s
V_d	drain voltage	V
V_ds	drain-source voltage	V
V_gs	gate-source voltage	V
V_th	threshold voltage	V
Z_0	characteristic impedance	Ω

University of Stuttgart
Germany

Institute of
Robust Power
Semiconductor Systems

Course Outline

1

1 Course Outline

2 The Millimeterwave Frequency Spectrum: Applicatio and Technologies

3 Functional Buildung Blocks of MMICs

4 Low Noise Amplifier

5 Broadband Amplifiers

6 Power Amplifiers

7 Microwave Network Analysis (Appendix)

1.1 Course Outline
1.2 Related Courses
1.3 Organisational Issues
1.4 Literature Recommendations

Course Outline

1.1

What This Course Is About

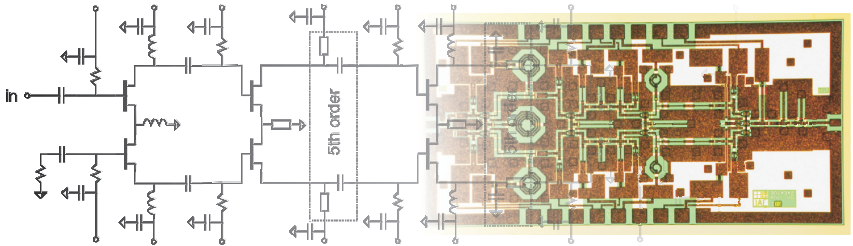


Figure 1: From a schematic to a layout in a MMIC.

Millimeterwave Monolithic Integrated Circuits (MMIC)

- The lecture Microwave Engineering (MWE) conveys the theory and design of microwave and Monolithic Microwave Integrated Circuits (MMICs), covering the areas of microwave network analysis, component modelling, microwave circuit design, analysis and layout. The focus is on low-noise, broadband and power amplifier design.

Course Contents

1. The Millimeterwave Spectrum: MMIC Applications and Technologies
   - An Introduction to MMICs
   - The Millimeterwave Frequency Spectrum
   - Millimeter-Wave Semiconductor Technologies
   - MMIC Applications
   - Market and Cost Considerations

Course Contents

2. Building Blocks of MMICs
   - Transistors and their Cutoff Frequencies
   - MMIC Process

Course Contents

3. Linear Circuits I: Low-Noise Amplifiers
   - Key Applications and Figures of Merit
   - Noise in Cascaded Stages
   - Noise Matching
   - Simultaneous Noise- and Power Matching
   - Bias Networks

Course Contents

4. Linear Circuits II: Broadband Amplifiers
   - Key Applications and Figures of Merit
   - Wideband Matching
   - Inductive Peaking
   - Transimpedance Amplifier
   - Travelling-Wave Amplifiers
   - The Cascode

Course Contents

5. Nonlinear Circuits I: Microwave Power Amplifiers
   - Key Applications and Figures of Merit
   - Amplifier Classes
   - Load-Pull Analysis
   - Parallel Stages
   - Alternative power amplifier (PA) Topologies

Course Contents

Appendix 1: Microwave Network Analysis

- N-Port Parameters
- Scattering Parameter
- The Smith-Chart
- Rollet Stability Factor and Power Gain Definitions
- Power Matching and Impedance Transformations

Course Contents

Appendix 2: Planar Transmission Line Theory

- Microstrip Transmission Lines
- Coplanar Waveguides
- Transmission Line Theory
- TRL Applications in Circuit Design
- Impedance Transformation
- Couplers Based on Lambda/4 Lines

1.1 Course Outline
1.2 Related Courses
1.3 Organisational Issues
1.4 Literature Recommendations

Related Courses

1.2

Lab Course Microwave Circuit Design

- 4 European Credit Transfer System (ECTS), winter term
- The purpose of this lab course is to experience a full design cycle for microwave monolithic integrated circuits from design to measurement with the aid of a state-of the-art CAD environment and measurement equipment.
- The participants will learn how to **design, simulate and layout integrated circuits** for a broadband microwave transmit and receive analog frontend.
- A **state-of-the-art semiconductor foundry process** and its associated Process Design Kit (PDK) are employed.
- Finally, the frontend is employed to build and **characterize a high data rate wireless point-to-point communication link**.
- After completion of this course, successful students will obtain the **Keysight Radio Frequency (RF)/Microwave (MW) Industry-Ready Student Program Certificate**.

Microwave Analog Frontend Design 1 I

- 6 ECTS, summer term
- Lecture
  - Applications of Nonlinear Microwave Circuits
    - Technologies
    - Applications
    - Cost Considerations
  - RF Frontend Non-Idealities
    - Receiver sensitivity
    - Transmitter linearity
    - local-oscillator (LO) stability
  - Frequency Multiplies
    - Frequency multiplier theory, figures of merit and state of the art
    - Even- and odd-order field-effect-transistor (FET) multipliers
    - Broadband multiplier techniques
    - Harmonic control in multiplier chains
  - Mixers Part 1
    - Mixer theory and figures of merit
    - Diode mixers
    - Passive and Active FET Mixers
    - Single-balanced, double-balanced, inphase-quadratur (IQ), image-rejection mixers (IR mixers) mixers

Microwave Analog Frontend Design 1 II

- Mixers Part 2
  - The Gilbert cell
  - heterojunction bipolar transistor (HBT)-based Gilbert cell circuit analysis
  - Applications of Gilbert cells
  - Design measures for speed enhancement
- Exercises
  - Tutorials
  - computer-aided design (CAD) hands-on

Microwave Analog Frontend Design 2

- 6 ECTS, summer term
- Lecture
  - Nonlinear Circuit Analysis
    - Nonlinear transistor modeling and time domain circuit analysis
    - Nonlinear frequency domain circuit analysis
    - Dynamic current-voltage (IV) waveforms as circuit design tool
  - Design for Manufacturing
    - Statistical Design: Monte Carlo, Yield, Sensitivity
    - Yield Centering
    - Stability Control
  - Packaging, Mounting, Integration Techniques
  - Link Budget Calculations
    - Communication Power Budget
    - Radar Power Budget
    - Atmospheric Attenuation
- Exercises
  - Tutorials
  - Seminar

Seminar Selected Topics on Power and Microwave Electronics

- summer term, 6 ECTS
- Students acquire knowledge and experience on writing and presenting a scientific publication through the **study and preparation of an individually chosen topic** from among a given list of topics in the area of power and microwave electronics.
- In the **tutored self-study** part students delve into an individually selected topic and prepare a scientific publication in the form of a conference paper and give an oral presentation of the paper.
- During an **intermediate review** at about half-term students submit and present their paper draft and get feedback and a paper review from the tutor.
- After final submission the student gets a **review on the final paper** and **presents the paper**.
- The self-study part is supplemented by **lectures on scientific writing and presentation**.

1.1 Course Outline
1.2 Related Courses
1.3 Organisational Issues
1.4 Literature Recommendations

Organisational Issues

1.3

Contact

- Ingmar Kallfass, Prof. Dr.-Ing.
  - Institute of Robust Power Semiconductor Systems (ILH)
  - Pfaffenwaldring 47, 70569 Stuttgart
  - phone: 0711-685-68747, fax: 0711-685-58747
  - ingmar.kallfass@ilh.uni-stuttgart.de
  - http://www.ilh.uni-stuttgart.de/en
  - http://www.uni-stuttgart.de/en
- Benjamin Schoch, M.Sc.
  - Pfaffenwaldring 31, 70569 Stuttgart
  - phone: 0711-685-67430
  - benjamin.schoch@ilh.uni-stuttgart.de
- Thomas Ufschlag, M.Sc.
  - Pfaffenwaldring 31, 70569 Stuttgart
  - Telefon: 0711-685-68611
  - thomas.ufschlag@ilh.uni-stuttgart.de

MWE Course Organisation

- Lectures
  - winter term, Thursday 14:00 - 15:30, V7.04
- Exercises
  - winter term, Thursday 15:45 - 17:15, V7.04
  - hands-on MMIC design and layout
- Examination
  - 6 ECTS credits
  - written/oral for $> 16/\leq 16$ candidates, offered in English and in German
  - oral: 10 min presentation of an individually attributed scientific publication + 20 min questions
  - written: ca. 50% questions on a scientific publication + ca. 50% general questions
  - papers incl. presentation template distributed about 4 weeks prior to exam.
- ILIAS
  - course schedule
  - all material (slides, script, recordings) are provided in English and in German (multilingual online offer)
  - annotated slides and recordings will be uploaded

Pre-Requisites

- Knowledge
  - a knowledge of high-frequency techniques is helpful
  - a knowledge of analog circuit design and semiconductor devices is helpful

1.1 Course Outline
1.2 Related Courses
1.3 Organisational Issues
1.4 Literature Recommendations

Literature Recommendations

1.4

Supplementary Literature

- D. Pozar, Microwave Engineering. Wiley, 2004
- G. Vendelin, A. Pavio, and U. Rohde, Microwave Circuit Design Using Linear and Nonlinear Techniques. Wiley, 2005
- Stephen A. Maas, Nonlinear Microwave and RF Circuits, ser. 2nd ed. Artech House, London, 2003
- O. Zinke, H. Brunswig, Hochfrequenztechnik I und II
- M. Hoffmann, Hochfrequenztechnik: Ein Systemtheoretischer Zugang

University of Stuttgart
Germany

Institute of
Robust Power
Semiconductor Systems

The Millimeter-wave Frequency Spectrum: Applications and Technologies

2

1 Course Outline

2 The Millimeterwave Frequency Spectrum: Applicatio and Technologies

3 Functional Buildung Blocks of MMICs

4 Low Noise Amplifier

5 Broadband Amplifiers

6 Power Amplifiers

7 Microwave Network Analysis (Appendix)

Lecture Goals

in this part of the lecture, we will...

- highlight the major applications of MMICs,
- compare the competing high-speed semiconductor technologies,
- consider markets and cost

2.1 The Millimeterwave Frequency Spectrum
2.2 Semiconductor Technologies
2.3 Applications of Monolithic Millimeterwave Integrated Circuits
2.4 An Introduction to MMICs
2.5 Markets and Costs

The Millimeter-wave Frequency Spectrum

2.1

Electromagnetic Spectrum: The “THz Gap”

frequency ranges (Fig. 2)

- microwave: 3 - 30 GHz, 10 - 1 cm free-space wavelength
- millimeter-wave: 30 - 300 GHz, 10 - 1 mm free-space wavelength
- sub-millimeter-wave: >300 GHz
- “Terahertz”: loosely used in the literature, typ. 0.1 - 10 THz

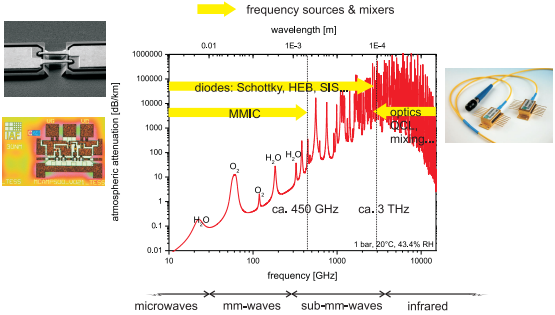


Figure 2: Atmospheric attenuation chart.

Electromagnetic Spectrum: The “THz Gap”

”THz gap” (Fig. 3)

- for atmospheric opacity
- for limited solid-state output power and sensitivity

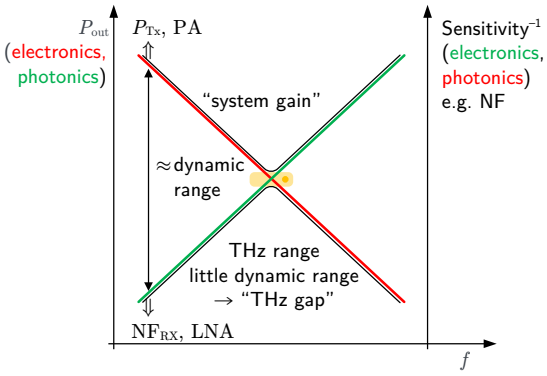


Figure 3: The THz gap.

Why mm-Waves?

spectroscopy

- detect and monitor molecule absorption frequencies in mm-wave region
- applications in
  - meteorology,
  - atmospheric sensing,
  - medical...

Why mm-Waves?

mm-wave ranging and imaging

- motivation
  - non-ionizing radiation penetrates clothing, adverse weather, packaging material
  - high bandwidth for high radar range resolution
    - cp. range resolution δR of a FMCW radar:

$$\delta R = \frac{c_0}{2 \cdot B} \tag{1}$$

  - low wavelength for high radiometric resolution
  - reflectance, transmittance and absorption of different media
- applications in
  - safety & security
  - concealed weapon detection
  - avionics (landing aid, runway control)...
- radiometric imaging (w/ or w/o illumination)

Why mm-Waves?

mm-wave communication

- motivation
  - high absolute bandwidth for high data rate
  - cp. Shannon theorem for the capacity C of an AWGN channel:

$$C = B \cdot \log_2 (1 + \mathsf{SNR}) = B \cdot \log_2 \left(1 + \frac{P_\mathrm{r}}{N_0 B}\right) \tag{2}$$

    B ...bandwidth, P_r...receive power, N_0...noise power spectral density
  - directivity and low interference
- applications in
  - intra-machine, board-2-board communication
  - WPAN, WLAN, media kiosque, home theatre, smart office
  - data centers
  - fixed wireless links in telecom 5G/6G fronthaul and backhaul, access networks

2.1 The Millimeterwave Frequency Spectrum
2.2 Semiconductor Technologies
2.3 Applications of Monolithic Millimeterwave Integrated Circuits
2.4 An Introduction to MMICs
2.5 Markets and Costs

Semiconductor Technologies

2.2

Which Semiconductor Technology?

	Field-effect transistor	Bipolar transistor	Diode
III-V-based	GaAs pHEMT InP HEMT GaAs MHEMT GaN HEMT	GaAs HBT InP (D-)HBT	GaAs Schottky
Si-based	nmos ($< 100\,\mathrm{nm}$ L_g) RF-CMOS	SiGe HBT	

Table 3: The major millimeter-wave technology candidates

Choice of Semiconductor Technology

	Si CMOS	SiGe HBT	InP HBT	InP HEMT	GaN HEMT
Cutoff frequencies	$> 200\,\mathrm{GHz}$	$> 200\,\mathrm{GHz}$	$> 400\,\mathrm{GHz}$	$> 600\,\mathrm{GHz}$	$> 200\,\mathrm{GHz}$
Output power	low	medium	medium	medium	high
Breakdown voltage	$\sim 1\,\mathrm{V}$	$\sim 2\,\mathrm{V}$	$\sim 4\,\mathrm{V}$	$\sim 2\,\mathrm{V}$	$> 20\,\mathrm{V}$
Intrinsic gain	low	high	high	low	low
Transconductance	$\sim 2\,\mathrm{S/mm}$	$\sim 60\,\mathrm{S/mm}$	$\sim 60\,\mathrm{S/mm}$	$\sim 2\,\mathrm{S/mm}$	$\sim 1\,\mathrm{S/mm}$
Homogeneity	high	high	high	low	low
Threshold voltage Δ	$\sim 2\,\mathrm{mV}$	$\sim 2\,\mathrm{mV}$	$\sim 2\,\mathrm{mV}$	$\sim 25\,\mathrm{mV}$	$\sim 25\,\mathrm{mV}$
1/f noise	high	low	low	high	high
RF noise	high	high	high	low	low
Yield	high	high	medium	low	low
Transistor count	∞	∞	~ 1000	~ 100	~ 100
Mixed-signal capable	yes	yes	yes	no	no
Market	mass	mass	niche	niche	niche

Table 4: Performance comparison of high-speed semiconductor families

MMIC Frequency Coverage

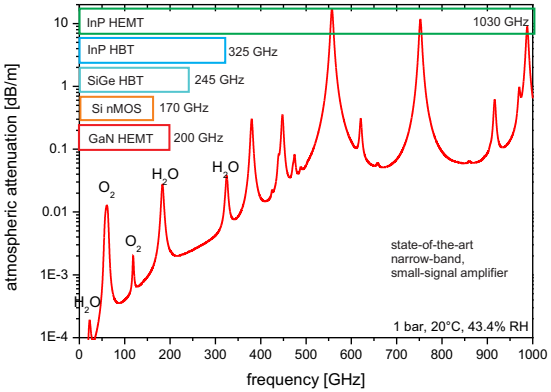


Figure 4: Record MMIC performance (narrow-band, small-signal gain amplifiers).

World Record: The First 1 THz Amplifier

DARPA-funded InP MMIC reaches record 1.03 THz, earns Guinness World Record

DARPA

October 28, 2014

No Comments

KEYWORDS electronics / Guinness / program / record / terahertz / world

EMAIL / PRINT / REPRINTS / MORE / TEXT SIZE+

Related Articles

Leibniz research institutes hold key to faster, powerful THz chips

Lake Shore enhances software capabilities of prototype turnkey THz characterization system

Related Events

MMIC Design Overview

COMSOL Training Courses

Guinness World Records has recognized **DARPA**'s Terahertz Electronics program for creating the fastest solid-state amplifier integrated circuit ever measured. The 10-stage common-source MMIC achieves 9 dB gain at 1.03 THz. Noise figure and output power have not yet been measured.

The MMIC was designed and fabricated by Northrop Grumman Corp. using 25 nm InP HEMTs as the active devices. Researchers used E-beam lithography to define the gates on the MMIC, which was 18 µm thick and no bigger than a grain of kosher salt, according to Bill Deal, manager of the Terahertz Electronics program at Northrop Grumman.

The previous high frequency record for a solid-state amplifier was 850 GHz, set in 2012 by Northrop Grumman, on the same DARPA program.

The objective of the Terahertz Electronics program is to develop the device and integration technologies to realize compact, high-performance transceivers that operate above a THz. The program is funding the development of InP HBT and HEMT transistor technologies as well as low-loss interconnects to integrate multiple MMICs into compact transmitter and receiver modules.

"Terahertz circuits promise to open up new areas of research and unforeseen applications in the sub-millimeter-wave spectrum, in addition to bringing unprecedented performance to circuits operating at more conventional frequencies," said Dev Palmer, DARPA program manager. "This breakthrough could lead to revolutionary technologies such as high-resolution security imaging systems, improved collision-avoidance radar, communications networks with many times the capacity of current systems and spectrometers that could detect potentially dangerous chemicals and explosives with much greater sensitivity."

Figure 5: http://www.microwavejournal.com

High-Speed Semiconductor Technologies

	country	technology	variant	node	f_T/f_{max}
IBM	US	silicon germanium (SiGe) bipolar complementary metal-oxide-semiconductor (BiCMOS)	8HP	0.13 µm	200/250
IHP	D	SiGe BiCMOS	SG13G2	0.13 µm	300/500
Infineon	D	SiGe BiCMOS	B7HF200	0.18 µm	170/250
NXP	NL	SiGe BiCMOS	QUBiC4Xi	0.25 µm	216/177
ST Microel.	F/I	SiGe BiCMOS	STBiCMOS9	0.13 µm	160/180
TowerJazz	US/Isr	SiGe BiCMOS	SBC18H3	0.13 µm	240/280
FhG IAF	D	Gallium Arsenide (GaAs) metamorphic-high-electron-mobility transistor (MHEMT)	n/a	35 nm	515/900
OHMMIC	F	GaAs MHEMT	D007IH	70 nm	300/460
NGC	US	indium phosphide (InP) high-electron-mobility transistor (HEMT)	n/a	30 nm	600/¿1000
Qorvo (ex-Triquint)	US	GaAs pseudomorphic high-electron-mobility transistor (pHEMT)	TPQ13-N	130 nm	95/-
UMS	F/D	GaAs pHEMT	PH10	100 nm	130/-
Win	TW	GaAs pHEMT	PP10	100 nm	130/-
Teledyne	US	InP HBT	n/a	0.25 µm	392/859

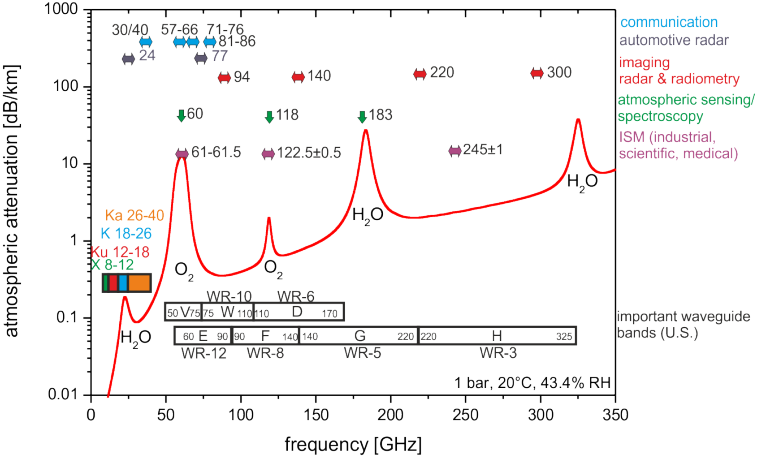
Figure 6: Complementary Metal-Oxide-Semiconductor (CMOS) excluded; major semiconductor manufacturers (no claim to completeness).

Applications of Monolithic Millimeterwave Integrated Circuits

2.3

2.1 The Millimeterwave Frequency Spectrum
2.2 Semiconductor Technologies
2.3 Applications of Monolithic Millimeterwave Integrated Circuits
2.4 An Introduction to MMICs
2.5 Markets and Costs

Frequency of Operation (7)



Generic Analog Frontend Architecture (8)

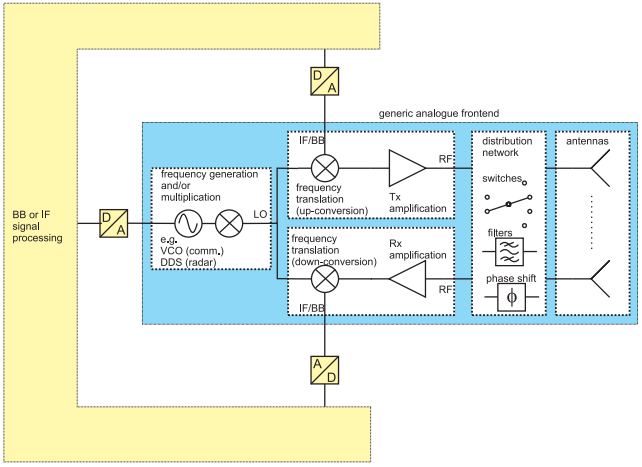


Figure 8: Functional blocks of an analog RF frontend.

Analog Frontend Functional Blocks I

Frequency Generation

1. Radar
   - transmission signal (Transmitter (TX)) and local oscillator signal (Receiver (RX))
   - modulated radar
     - FMCW "chirp"
     - current research: orthogonal frequency division multiplex (orthogonal frequency division multiplex (OFDM)) radar
   - pulsed radar
2. Communication
   - local oscillator (LO) for up- and down-conversion in mixers
   - fixed frequency or multiple sub-carriers
3. Architectures
   - fundamental oscillator with phase-locked loop (phase-locked loop (PLL)) stabilization (phase noise reduction of free-running oscillator)
   - subharmonic oscillator with frequency multiplication

Analog Frontend Functional Blocks II

Frequency Translation

1. Down-Conversion
   - in communication systems: mixer to convert from RF to IF
   - in sensor systems (active and passive)
   - homodyne (mix with transmit signal, e.g. radar) or heterodyne (mix with LO)
   - zero-IF or super-heterodyne receivers
   - direct detection: total power measurement by detector diode with (= zero-IF) or without down-conversion
2. Up-Conversion
   - in communication systems: mixer to convert from IF to RF
   - in radar: up-convert FMCW chirp signal generated at lower frequencies

Analog Frontend Functional Blocks III

Antenna Distribution Network

1. Radiometers
   - Dicke switch
   - active cold loads
   - polarization selectivity
2. Radar
   - mono- or bi-static
   - phased-arrays
   - electronic beam steering
3. Communication
   - single input single output (single input single output (SISO)), multiple input multiple output (multiple input multiple output (MIMO))
   - antenna switch (RX, TX)

Satellite Communication

- traditional up to Ku-band (12-18 GHz)
- current state of the art (SoA): Ka-band (26-40 GHz)
- research and development (R&D): Q-band (33-50 GHz), V-band (40-75 GHz), E-band (60-90 GHz)

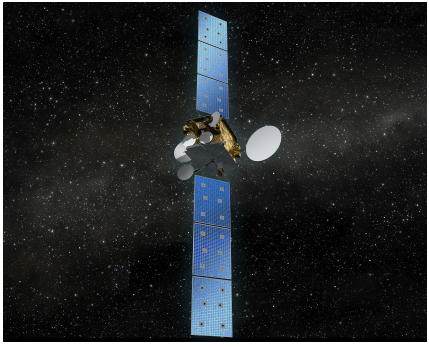


Figure 9: Application example: satellite communication.

60 GHz wireless communication Fig.(10)

- worldwide frequency allocation: 57-66 GHz
- IEEE802.11ad standard
- application in
  - home theatre (HD)
  - Gbit WLAN
  - “media kiosk” / “data shower”
  - fast synchronisation
  - p2p links

Figure 10: Application example: Notebook with IEEE802.11ad 60 GHz wireless connectivity.

E-band fixed wireless links Fig.(11)

- Frequency standard: 71-76 GHz and 81-86 GHz
- application in
  - telecom backhaul
  - last mile access
  - fiber extension

Figure 11: Application example: E-Band fixed wireless links. photo: www.ericsson.com.

24 and 77 GHz Automotive Radar Fig.(12)

- traditionally III-V, today dominated by SiGe
- recent developments
  - self-test and self-calibration
  - phased-arrays
  - R&D interest in higher frequencies

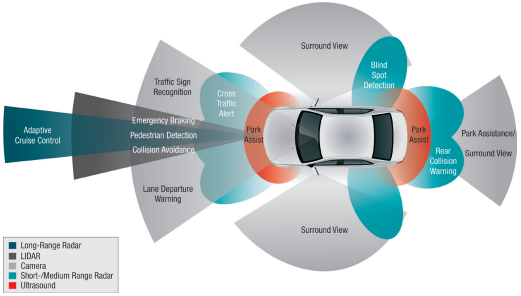


Figure 12: Application example: automotive radar.

Automotive Radar Chip Sets

- GaAs chip sets, (generation 1, replaced by SiGe)
  - III-V, typ. GaAs
  - analog frontend only
  - multi-chip integration and assembly
- SiGe chip sets
  - SiGe BiCMOS technologies
  - multi-channel,FMCW ramp generator, ADC
- CMOS chip sets
  - typ. 45 nm CMOS
  - System on chip (SOC) incl. DSP, µC

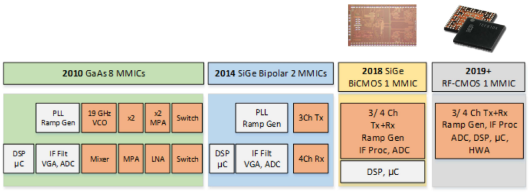


Figure 13: Application example: automotive radar development.

Imaging in Saftey & Security

- stand-off detetction and localisation of hidden weapons
- portal scanners at airports and cirtical infrastructure
- e.g. "ProVision" by L3-Communictaions
  - 1D transmit/receive array with mechanical rotation
- portal scanners by Rhode&Schwarz
  - 2D radar array at 80 GHz in 130 nm SiGe BiCMOS technology Fig.(14)

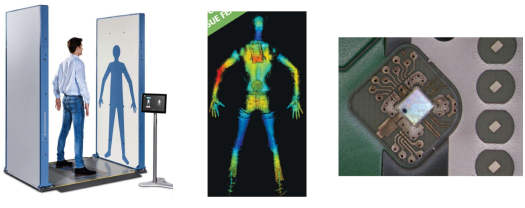


Figure 14: Application example: Rhode&Schwarz portal scanner.

High Resolution Radar

- helicopter landing aid against "brown out" or "white out" Fig.(15)
- surveillance, e.g. in unmanned aerial vehicles(UAV)
- material inspection, e.g. detection of cracks and delaminations

Figure 15: Application example: radar for a helicopter landing aid.

Imaging Radiometry

- direct detetcion radiometers Fig.(16)
- applictaions in saftey&security, enviromental,...
- require high temperature contrasts (indoors iwth illumination
- typically mechanical pixel scan
- more information from polarimetric systems

Figure 16: Application example: 94 GHz Imaging Fraunhofer FHR, Wachtberg.

Radar Imaging in Safety& Security

- motivation: higher radar bandwidth results in higher geometrical resolution
- example: 300 GHz FMCW ISAR by Fraunhofer FHR Fig.(17)

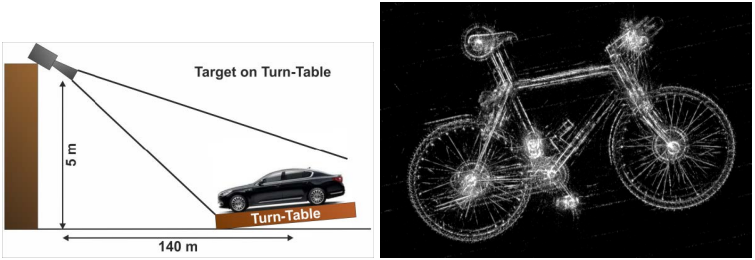


Figure 17: Application example: 300 GHz FMCW ISAR imaging radar.

2.1 The Millimeterwave Frequency Spectrum
2.2 Semiconductor Technologies
2.3 Applications of Monolithic Millimeterwave Integrated Circuits
2.4 An Introduction to MMICs
2.5 Markets and Costs

An Introduction to MMICs

2.4

The “classic” MMIC

first MMIC: Ray Pengelly and James Turner, "Monolithic Broadband GaAs FET Amplifiers", 1975 (Microwaves101) characteristic features

- III-V-based (GaAs, InP, gallium nitride (GaN))
- low integration density / large chip size
- extensive use of reactive elements (matching, coupling, biasing...)
- interstage matching
- passives as distributed elements (transmission lines)
- few levels (typically two) of metal interconnect layers

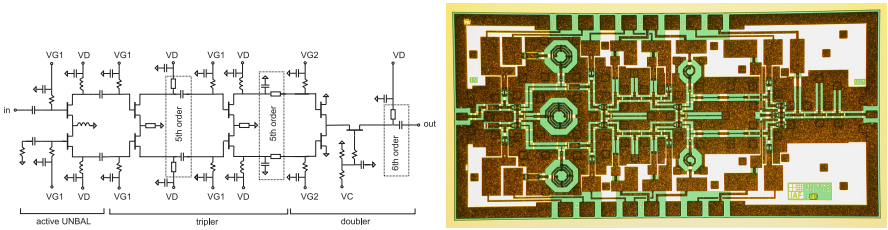
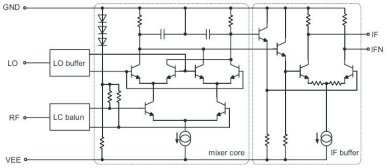


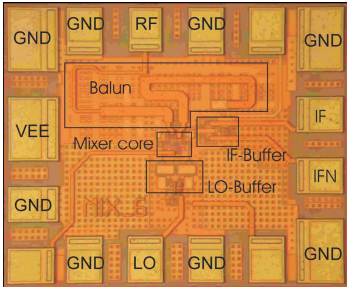
Figure 18: W-band frequency multiplier-by-six, Fraunhofer IAF MHEMT, chip size: $1.5 \times 4\,\mathrm{mm}^2$.

The “new” MMIC

characteristic features

- Si-based (SiGe HBT, silicon (Si) CMOS)
- high integration density / small chip size
- limited use of reactive elements and transmission lines
- typically differential topologies
- in highly compact circuits (e.g. Gilbert cells), use of active impedance transformers (source/emitter follower)





0.55 x 0.45 mm² Perndl et.al. IMS 2004 Infineon SiGe HBT

Figure 19: 77 GHz down-converter for automotive radar, Infineon SiGe HBT, 2004. Chip size: $0.55 \times 0.45\,\mathrm{mm}^2$.

Mixed-Signal Integrated Circuits

Mixed-Signal Integrated Circuits

- a circuit which handles both analog and digital signals on a single chip
- a circuit (digital or analog architecture) which handles very high speed digital signals

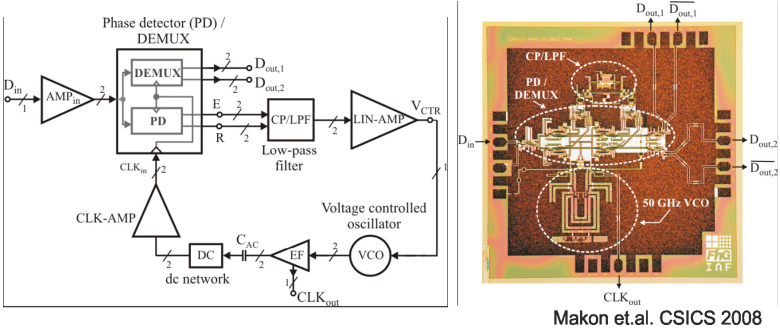


Figure 20: 100 Gbit/s on-off keying (OOK) CDR, Fraunhofer IAF InP double-heterojunction bipolar transistor (DHBT), 2008. [**Makon:01**]

Mixed-Signal Integrated Circuits

examples

- ADC and DAC
- optical communication systems
  - CDR
  - frequency dividers
  - (de-)multiplexers

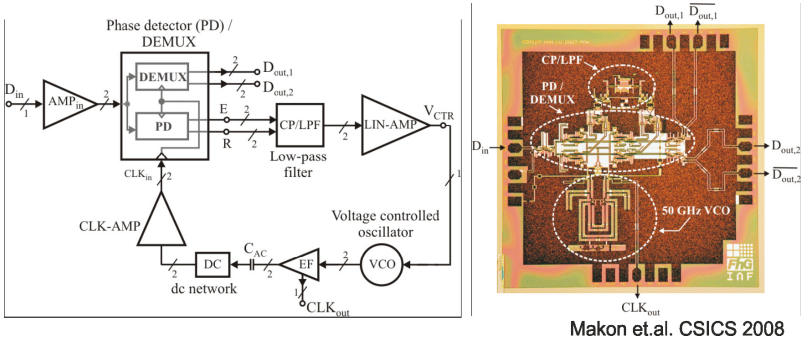


Figure 20: 100 Gbit/s OOK CDR, Fraunhofer IAF InP DHBT, 2008. [**Makon:01**]

2.1 The Millimeterwave Frequency Spectrum
2.2 Semiconductor Technologies
2.3 Applications of Monolithic Millimeterwave Integrated Circuits
2.4 An Introduction to MMICs
2.5 Markets and Costs

Markets and Costs

Kommt nicht dran!!!

2.5

Companies and Institutions Developing Microwave Products

big companies

- Airbus (Munich)
- Balluff (Stuttgart)
- Bosch (Stuttgart)
- Continental (Friedrichshafen, Munich)
- Hensoldt (Ulm)
- Huawei (Munich)
- Infineon (Munich)
- Innosent (Würzburg)
- Kathrein (Rosenheim)
- Keysight (Böblingen)
- Nokia Bell-Labs (Stuttgart)
- Rohde&Schwarz (Munich)
- Siemens (Munich, Nürnberg)
- Sony Europe (Stuttgart)
- TES Electronic Solutions (Stuttgart)
- Tesat Spacecom (Backnang)
- Valeo (Bietigheim)
- and many more...

Companies and Institutions Developing Microwave Products

SME / small and medium enterprises

- start-ups
- spin-offs
- often from universities
- just some examples
  - EESY-IC (Uni Nuremberg)
  - ACST (TU Darmstadt)
  - AdMOS (Stuttgart)
  - TeraSense (UK)
  - and many more...

Companies and Institutions Developing Microwave Products

research institutions

- Fraunhofer Gesellschaft (IAF, HHI, IIS, ...)
- Max-Planck Gesellschaft (MPI-FKF, ...)
- Leibnitz Gesellschaft (FBH, IHP, ...)
- German Aerospace Center
- Helmholtz Gesellschaft (KIT Campus North, ...)
- FZ Jülich

Market Considerations

millimeterwave market structure

- highly fragmented
- many niche applications: earth, life and material science...
- many professional/industrial applications: production...
- many military applications, some dual-use: aeronautics, safety & security...
- few consumer, mass market applications

the potential “killer” application: candidates for mass markets

- automotive radar
- 60 GHz wireless (“home theatre” etc.)

Cost considerations: Production cost

substrate	device	cost [\$/mm^2]
Si	CMOS	$\sim$0.01
SiGe	HBT	0.1 - 0.5
GaAs	HEMT	1 - 2
InP	HEMT	10

Table 5: technology cost per unit area

- What can be done in Si, will be done in Si!
- further considerations beyond mere semiconductor cost
  - cost for assembly and testing is comparable for all technologies
  - at millimeter-wave (mmW), cost for mounting, packaging and assembly becomes dominant
  - mmW applications for mass markets require low-cost packaging
  - Si allows for on-chip integration of digital baseband functionalities

Cost considerations: Mask set cost

device	cost [€/mask set]
Si-CMOS	>1 Mio.
SiGe-HBT	>100000
III-V HEMT	10000

Table 6: Mask set costs

- Si-based technologies require mass market (communication, automotive) to be price-competitive
- further considerations
  - Design cycle: roughly 2-4 months for design, production, testing is comparable for all technologies
  - Si-based technologies are processed in batches (20 wafers, 8"-12") per run
  - GaAs-based technologies are processed by lots (4-6 wafers, 3"-6") per run, e-beam writing of individual wafers

Cost considerations

- R&D: 10^2
- Prototyping: 10^3
- Products
  - consumer: $> 10^6$ per year
  - professional: $> 10^5$ per year

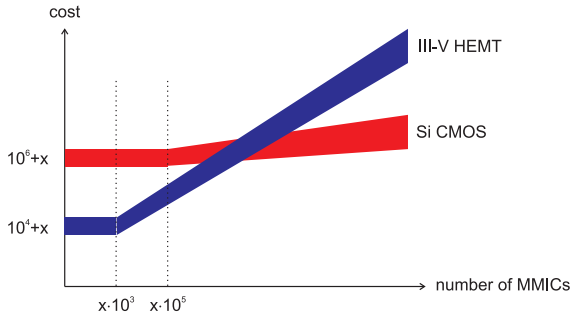


Figure 21: Cost versus number of MMICs sold

Development Cost of Aggressively Scaled CMOS

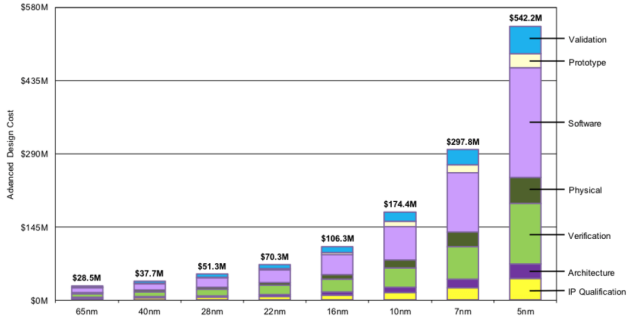


Figure 22: Development Cost of Aggressively Scaled CMOS

- new technology nodes
  - the introduction of new prodcuts/integrated circuit (IC)s in new technolog nodes (smaller gate legnths) requires extremely high software, hardware and personnel cost
- intellectual property (IP) qualification
  - IP = “semiconductor intellectual property core”, “IP core” or “IP block”
  - IP blocks are functional groups of ICs (memory, central processing unit (CPU), ...), which are part of application-specific integrated circuit (ASIC)s and Field Programmable Gate Arrays (FPGA)s
- architecture
  - increasing complexity (integration density, transistor count) of System-on-a-Chip (SoC) chip architectures
- verification
  - increasing effort for quality control of new IP blocks and architectures, before mass production can start
- physical = fabrication
  - exponentially increasing cost for mask sets at aggressively scaled technology nodes
    - sub-micron structures require hihgly precise lithography technologies and techniques
- Software
  - development of software for the operation of the ICs
- Prototype
  - fabrication

- Validation
  - functional test and characterization
- exponential increase of cost
  - the cost of all individual development steps scale approximately exponential with technology scaling
- consequence: consolidation
  - the introduction of new technology nodes can only be sustained by fewer and fewer companies
  - currently, only three semiconductor foundries worldwide offer 7-nm technology nodes: Intel, Samsung and TSMC (wikichip.org).
- consequence: re-use
  - chip in new technology nodes must be used in more products in order for return of investment.
  - niche markets are not economically viable for application-specific products.

Economy of Scale

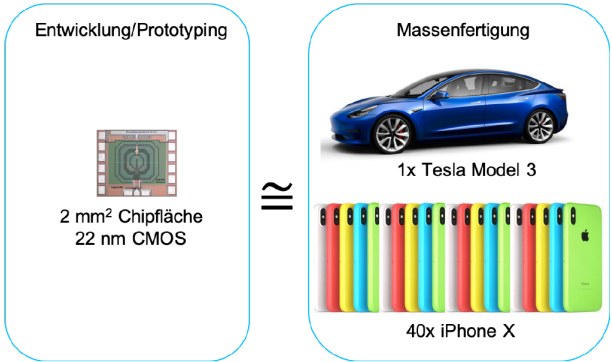


Figure 23: Economy of scale in Si-based technologies.

University of Stuttgart
Germany

Institute of
Robust Power
Semiconductor Systems

Functional Buildung Blocks of MMICs

3

1 Course Outline

2 The Millimeterwave Frequency Spectrum: Applicatio
and Technologies

3 Functional Buildung Blocks of MMICs

4 Low Noise Amplifier

5 Broadband Amplifiers

6 Power Amplifiers

7 Microwave Network Analysis (Appendix)

3.1 Transistors
3.2 Lumped Element Passives
3.3 Integrated Circuit Process

Transistors

3.1

Active Components I

- metal–oxide–semiconductor field-effect transistor (MOSFET), bipolar junction transistor (BJT), HBT, HEMT...

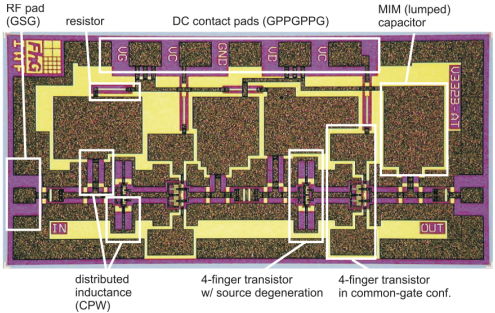


Figure 24: MMIC building elements.

e.g. 4 fingers

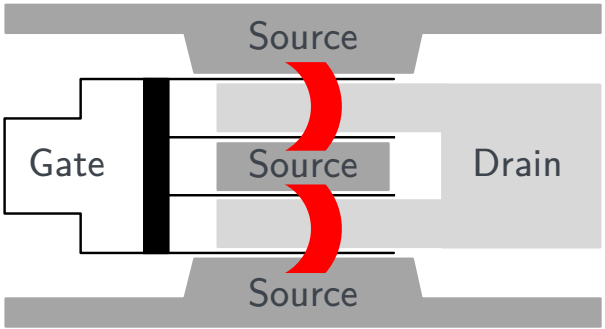


Figure 25: Transistor gate fingers.

Transit frequency f_T I

- frequency, where short-circuit current gain (h_{21}) becomes unity

$$f_T = f\,\big|_{|h_{21}|=1} \tag{3}$$

$$h_{21} = \left.\frac{i_2}{i_1}\right|_{v_2=0V} \tag{4}$$

- extrapolation of measurement data with -20 dB/decade slope

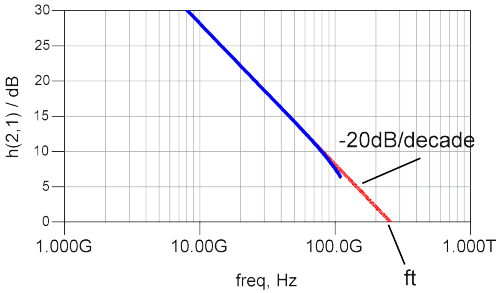


Figure 26: Derivation of transit frequency f_T.

Approximation Formula for Transit Frequency I

transit frequency f_T derived from $g_\mathrm{m} - C_\mathrm{gs}$

$$f_\mathrm{T} \approx \frac{g_\mathrm{m}}{2\pi C_\mathrm{gs}} \tag{5}$$

independency of f_T on gate resistance

- devices with high gate resistance (FET) or high base resistance (BJT) may still have a high f_T.
- cp. aggressively scaled CMOS

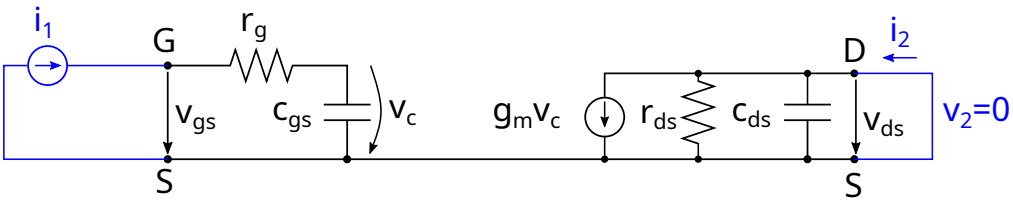


Figure 27: First-order transistor model approximation for f_T derivation.

Maximum frequency of oscillation $f_{\max}$ I

- derived from unilateral power gain U

$$f_{\max,\mathrm{U}} = f\,\big|_{|U|=1} \tag{6}$$

- derived from MSG/MAG

$$f_{\max,\mathrm{MSG/MAG}} = f\,\big|_{|\mathrm{MSG/MAG}|=1} \tag{7}$$

- extrapolation of measurement data with -20 dB/decade slope

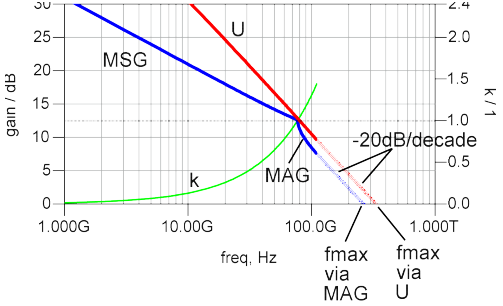


Figure 28: Derivation of maximum frequency of oscillation $f_{\max}$ $f_{\max}$.

FET Small-Signal Model: Approximation Formula for $f_{\max}$ I

- including effect of series ("access") resistances R_g, R_s, R_d (but excluding access inductance L_g etc.) (DiLorenzo 1997)

$$f_{\max} \approx \frac{1}{2} \frac{f_\mathrm{T}}{\sqrt{\frac{R_\mathrm{g}+R_\mathrm{gs}+R_\mathrm{s}}{r_\mathrm{ds}}} + \pi f_\mathrm{T} C_\mathrm{gd} \left(2R_\mathrm{g} + R_\mathrm{gs} + R_\mathrm{s}\right)} \tag{8}$$

FET Small-Signal Model: Approximation Formula for $f_{\max}$ II

- $f_{\max}$ is impacted by the parasitic effects of a transistor device, e.g., gate resistance and feedback elements like C_{gd} and R_{s}.

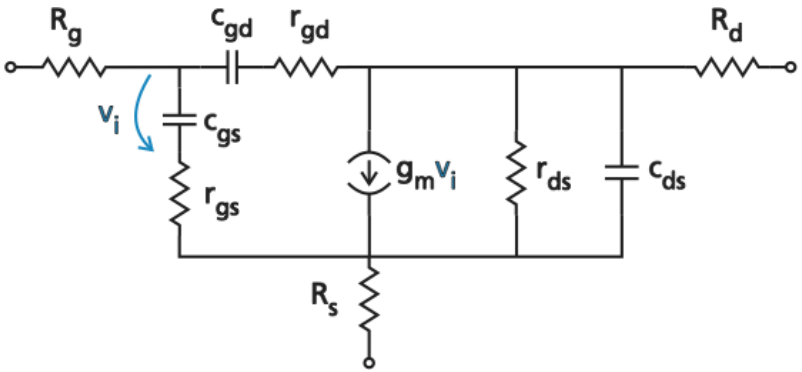


Figure 29: Small-signal model for derivation of $f_{\max}$.

Speed: Which Semiconductor Technology? I

f_T versus $f_{\max}$

- f_{T} is mostly relevant in digital and mixed-signal circuits (current steering in capacitive loads) and mmW power amps (low impedance loads)
- $f_{\max}$ is mostly relevant in analog circuits (conjugate complex impedance matching)
- devices can be tailored for high $f_{\max}$ *OR* high f_{T}
- a large variety of processes with less speed available in all technologies

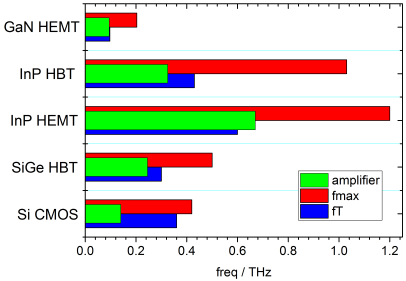


Figure 30: Speed comparison of high-speed semiconductor families.

3.1 Transistors
3.2 Lumped Element Passives
3.3 Integrated Circuit Process

Lumped Element Passives

3.2

lumped vs distributed I

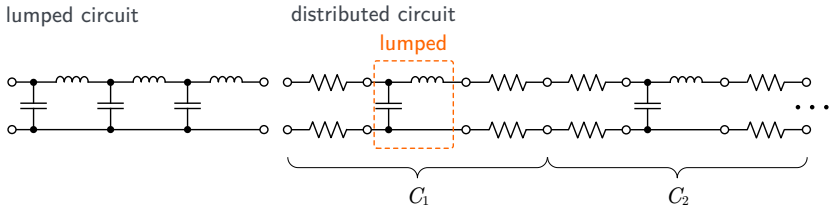


Figure 31: Concentrated vs. lumped building elements.

- "lumped" / "concentrated": effect of phase delay is negligible
- Reminder:

$$\lambda' = \frac{\lambda}{\sqrt{\epsilon_{\text{eff}}}} = \frac{c_o}{\sqrt{\epsilon_{\text{eff}}} f} \tag{9}$$

- Rule of thumb: An element can be considered as lumped, if its geometrical extention d is less than $\lambda'/16$

lumped vs distributed II

- Distributed effects (reflections, radiation, standing waves, impedance tranformation etc.) must be considered when the geometrical extention exceeds $\lambda'/16$

$$d_{\text{geometrical}} > \frac{\lambda'}{16} = \frac{1}{16} \cdot \frac{1}{\sqrt{\epsilon_{\text{eff}}}} \cdot \frac{c_o}{f_{\text{max.spectrum}}} \tag{10}$$

lumped passives I

- capacitors
- (spiral) inductors
- resistors

Quality Factor Q I

defined via transfer function of 2^{nd} order resonance circuit

- series RLC resonance circuit transfer function $H(s)$:

$$H(s) = c\frac{s}{s^2 + 2\alpha s + \omega_0^2} \tag{11}$$

  - α ... damping constant
  - ω_0... resonance frequency
  - $s = \sigma + j\omega$... complex angular frequency

- quality factor:

$$Q = \frac{\omega_0}{2\alpha} \tag{12}$$

Quality Factor Q II

interpretations of Q

$$Q = 2\pi \frac{\text{stored energy}}{\text{dissipated energy per cycle}} \tag{13}$$

- 3-dB resonance bandwidth (fig. 32):

$$\triangle f = \frac{f_0}{Q} \tag{14}$$

Quality Factor Q III

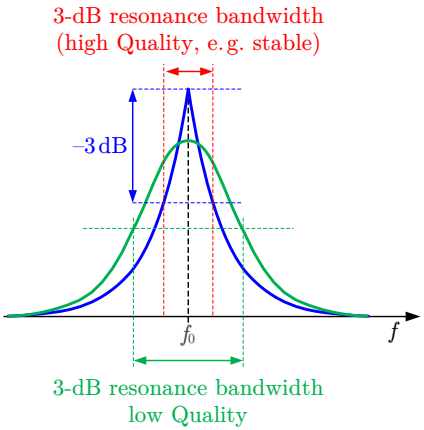


Figure 32: Relation of 3-dB bandwidth and quality factor.

Quality Factor Q IV

integrated passives' Q

- dependent on substrate conductivity
- on GaAs: Q's up to 20 possible

Capacitors - Use in MMICs I

Areas of application for capacitors (fig. 33)

- DC-blocking at input, output, interstage
- RF-shunting at bias pads
- Shunt- or series-C matching in the RF path

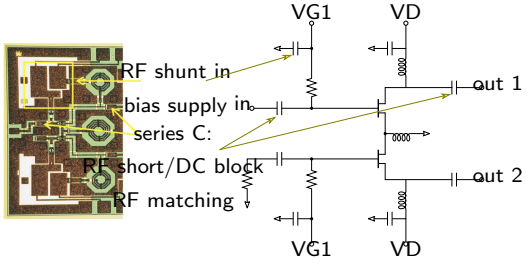


Figure 33: Use of lumped capacitors in microwave circuits. Example: active UNBAL stage of a frequency-multiplier MMIC in W-band.

Metal-Insulator-Metal (MIM) Capacitors I

consider Metal-Insulator-Metal (MIM) stack as plate capacitor

$$C = \epsilon_r \epsilon_0 \frac{A}{t} \tag{15}$$

- ϵ_r... insulator/dielectric permittivity
- t... dielectric thickness
- capacitance density c:

$$c = \frac{\epsilon_r \epsilon_0}{t} \tag{16}$$

- $[c] = \mathrm{F}/\mu m^2$

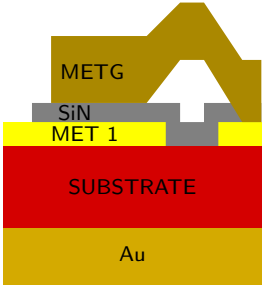


Figure 34: MIM capacitor layer stack.

Metal-Insulator-Metal (MIM) Capacitors II

example: Si_3N_4

- $\epsilon_r = 6.4,\ t = 250\,\mathrm{nm}$
- capacitance density $c = \frac{\epsilon_r \epsilon_0}{t} = 0.225\,\mathrm{fF}/\mu\mathrm{m}^2$

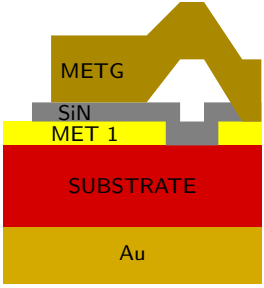


Figure 35: MIM capacitor layer stack.

Semiconductor Junction Capacitors I

reverse-biased semiconductor diodes fig. 37

- pn-junction or Schottky-junction (FET-gate)
- pro's: easy to realize, use existing actives
- con's: bias-dependent / nonlinear, less accurate

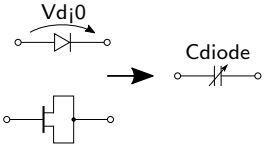


Figure 36: Semiconductor junction capacitors.

Semiconductor Junction Capacitors II

others

- planar inter-digitated structures or coupling lines
- pro's: accurate (lithographic resolution)
- con's: only very small values realisable

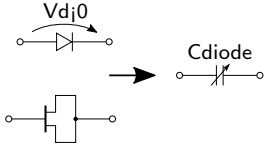


Figure 37: Semiconductor junction capacitors.

MIM Capacitors - CAD Models I

lumped element model with parasitics fig. 38

$$C_{\mathrm{MIM}} = \epsilon_r \epsilon_0 \frac{A}{t} \tag{17}$$

- C_{MIM}...intentional capacitance
- R_{ser}...series resistance

$$R_{\mathrm{ser}} \approx \frac{R_{\mathrm{plate1}} + R_{\mathrm{plate2}}}{3} \tag{18}$$

- L_{ser}...inductance (leads to resonance)
- C_{par}...substrate capacitance

- dielectric loss tangent

$$\tan\delta = \frac{1}{R_{\mathrm{par}}\omega C} \tag{19}$$

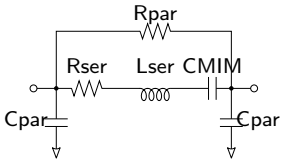


Figure 38: RF equivalent circuit of a capacitor (in series connection = twoport).

MIM Capacitors - CAD Models II

Quality factor of one-port (capacitor to ground)

- consider parallel RC circuit:

$$Q_C = \frac{\mathrm{imag}(Y)}{\mathrm{real}(Y)} \tag{20}$$

- resonance due to L

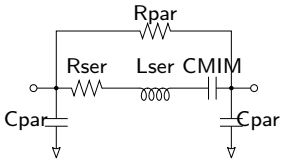


Figure 39: RF equivalent circuit of a capacitor (in series connection = twoport).

MIM Capacitors - CAD Models III

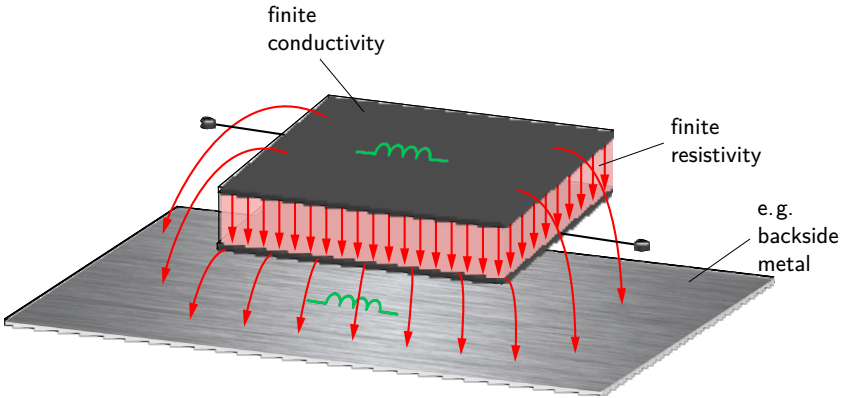


Figure 40: 3D view of the capacitor.

MIM Capacitors - Typical S-Parameters I

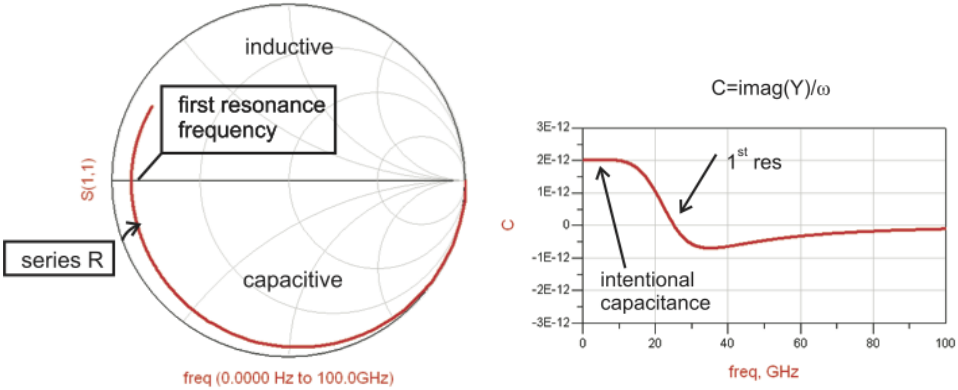


Figure 41: Typical S-parameters and effective capacitance of a capacitor (to ground = one-port).

MIM Capacitors - Application for Decoupling I

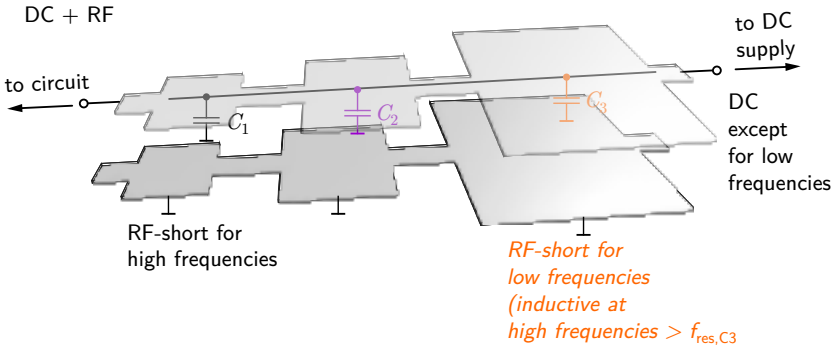


Figure 42: Distributed capacitor layout for broadband decoupling.

MIM Capacitors - Application for Decoupling II

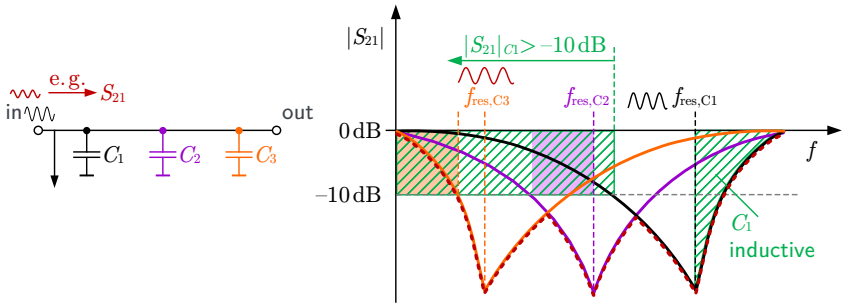


Figure 43: Transmission of the distributed capacitor layout for broadband decoupling.

Spiral Inductors - Use in MMICs I

Inductor: circuit functions (fig. 44)

- biasing (RF block)
- inductive peaking
- current-sources in differential stages
- matching (use transmission line (TRL) at mm-waves)

other inductances

- bond-wire ($\sim$1 nH/mm wire length)
- narrow TRLs

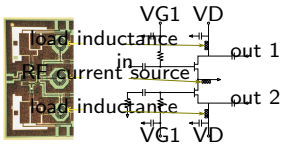


Figure 44: Use of spiral inductors in microwave circuits.

Spiral Inductors I

geometrical parameters (fig. 45)

- number of turns
- width and spacing of turns
- shape: rounded, square etc.

approximation formula (Mohan et.al.)

$$L \approx K_1 \mu_0 \frac{n^2 d_{\text{avg}}}{1 + K_2 \rho} \qquad (21)$$

- $d_{\text{avg}} = \frac{1}{2}(d_o + d_i)$...average diameter
- $\rho = \frac{d_o - d_i}{d_o + d_i}$...fill ratio

shape	K_1	K_2
square	2.34	2.75
hexagonal	2.33	3.82
octagonal	2.25	3.55

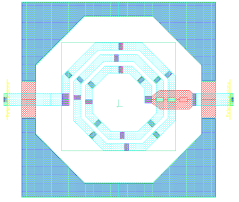


Figure 45: Example layout of a planar spiral inductor.

Spiral Inductors - CAD model I

inductor equivalent circuit (fig. 46)

- L...intentional ind. (conductor self-ind. + mutual ind.)
- R_{ser}...metal sheet resistance (incl. skin effect)
- R_{par}...substrate loss
- C_{par}...turn coupling, introduces resonance!
- $C_{\mathrm{sub.}}$...substrate capacitance

quality factor of one-port

- consider series R-L circuit

$$Q_L = \frac{\mathrm{imag}(Z)}{\mathrm{real}(Z)} \approx \frac{\omega L}{R_{\mathrm{ser}}} \tag{22}$$

effective inductance

$$L' = \frac{L}{1 - \omega^2 L C_{\mathrm{par}}} \tag{23}$$

- due to the parasitic coupling capacitance(s) resonance frequency(ies) occur in the impedance of a real inductor implementation

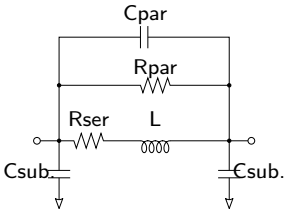


Figure 46: RF equivalent circuit of an inductor (in series connection = twoport).

Spiral Inductors - Typical S-Parameters I

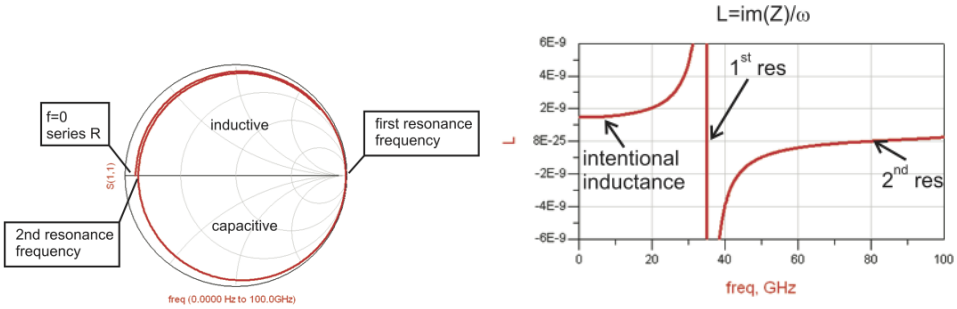


Figure 47: Typical S-parameters and effective inductance of a inductor (to ground = one-port).

spiral inductor resonance frequency (fig. 47)

- useful frequency range only up to 1st resonance
- the larger the inductor, the lower the resonance frequency

Spiral Inductors - Typical S-Parameters II

- use at microwave frequencies requires careful modeling, typically via electro-magnetic (EM) simulation
- at millimeter-wave frequencies, spiral inductors are mostly replaced by (inductive) transmission lines

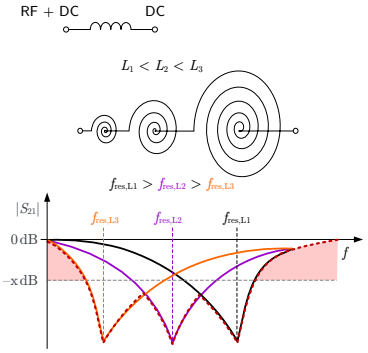


Figure 48: Transmission of the distriuted inductor for broadband RF blocking.

Resistors - Use in MMICs I

- Biasing
- Stability
- “Forced” matching
- resistive load

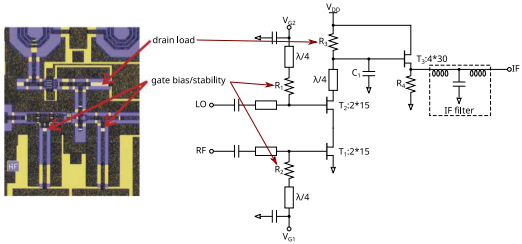


Figure 49: Use of resistors in microwave circuits.

Resistors - Use in MMICs II

- metal layers (e.g. NiCr)
- semiconductor layers w/ doping
- hetero-structure epitaxial layers

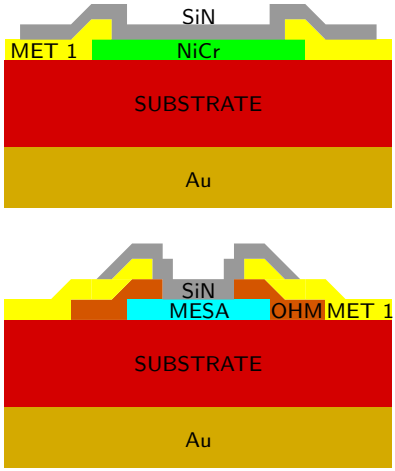


Figure 50: Layer stack of a thin film resistor. Top: NiCr. Bottom: epi-mesa.).

Resistors - CAD Models I

sheet resistance

$$R = \rho \frac{L}{tW} = \rho_{\mathrm{sheet}} \frac{L}{W} \tag{24}$$

- $[\rho_{\mathrm{sheet}}] = \Omega_{\square}$ ("Ohm-square")

parasitic

- includes phase shift via transmission line elements
- contact resistance (metal to resistance layer)
- less important: substrate capacitance, inductance

further important parameters

- maximum static current density: typ. $I_{\max} = 1\mathrm{A/mm}_{\mathrm{width}}$
- x-wafer and x-batch resistance spread (large for semiconductor layers)

Figure 51: RF equivalent circuit of a thin-film resistor (in series connection = twoport).

A General Note on CAD Models I

- Equivalent circuit based models
  - possibly scalable with respect to geometry
  - frequency extrapolation possible
  - model parameters
    - extracted from measurement (fitting), e.g. $L = \frac{\mathrm{imag}(Z)}{j\omega}$, $R = \mathrm{real}(Z)$
    - based on physical parameters
  - incl. nonlinearities, i.e. parameters = function of voltage/current
- EM-simulation or measurement based models
  - small-signal only (S-parameter)
  - accurate
  - not scalable (table-based or geometry library)
  - time consuming simulation

3.1 Transistors
3.2 Lumped Element Passives
3.3 Integrated Circuit Process

Integrated Circuit Process

3.3

Typical Layer Stack of III-V based MMIC Process (fig. 52) I

Metal interconnects in III-V technology

- few layers of interconnect (typ. 2-3), typ. Au
- often with airbridge technology (low parasitic capacitive coupling)
- dielectric layer (typ. SiN) for MIM capacitors and for transistor and surface passivation

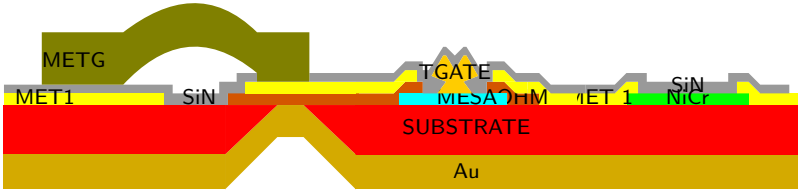


Figure 52: MMIC process layer stack example: Fraunhofer IAF GaAs mHEMT technology with two metal interconnect layers (Au).

Typical Layer Stack of Si-Based MMIC Process (fig. 193) I

Metal interconnects in Silicon technology

- many layers of interconnects (e.g. Cu)
- planarisation after each deposition step
- dielectric openings (vias)

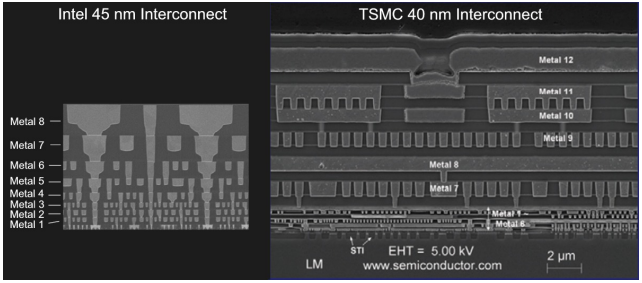


Figure 53: MMIC process layer stack examples: Si CMOS with several metal interconnect layers. Source: www.chip-architect.com

Evolution of CMOS Metal Interconnect Technology (fig. 54) I

"Aggressively scaled" or "deep sub-micron" CMOS

- 12+ metal interconnect layers
- "density rules": min/max filling ratio per metal layer
- high layouting effort and need for EM simulations for RF circuit design

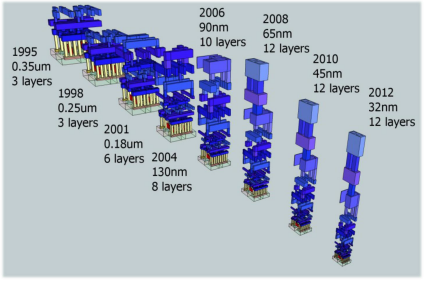


Figure 54: Evolution over time of a CMOS technology node citeEsposito$_2$016

Process Design Rules I

Design rules (fig. 55)

- intra-layer and inter-layer minimum and maximum dimensions
- width, spacing, overlap, enclosure
- maximum aspect ratio length/width (e.g. air-bridges)

CAD automatic checks

- DRCDRCdesign rule check: design rule check
- ERCERCelectrical rule check: electrical rule check (electrical shorts, floating nodes)
- LVSLVSlayout versus schematic: layout versus schematic (all component parameters and connections)

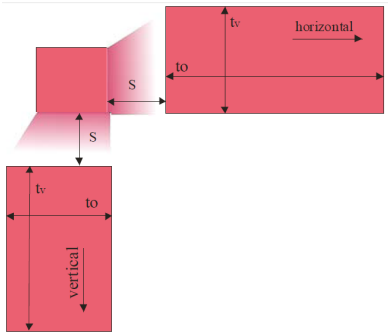


Figure 55: MMIC process design rules.

Low Noise Amplifier

4

1 Course Outline

2 The Millimeterwave Frequency Spectrum: Applicatio
and Technologies

3 Functional Buildung Blocks of MMICs

4 Low Noise Amplifier

5 Broadband Amplifiers

6 Power Amplifiers

7 Microwave Network Analysis (Appendix)

4.1 LNA Figures of Merit
4.2 LNA Design Methodology

LNA Figures of Merit

4.1

Noise Definitions I

Degredation of the signal-to-noise ratio

- noise factor:

$$F = \frac{(S/N)_{\text{input}}}{(S/N)_{\text{output}}} > 1 \tag{25}$$

- noise figure:

$$NF = 10 \log F \tag{26}$$

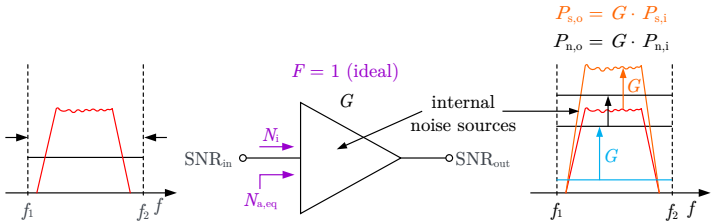


Figure 56: Noise figure definition.

Noise Definitions II

- noiseless amplifier with gain G and additional noise power N_a referred to input::

$$F = \frac{S_i/N_i}{GS_i/\left(G\left(N_i + N_a\right)\right)} = 1 + \frac{N_a}{N_i} \tag{27}$$

Noise Definitions III

Equivalent noise temperature

- thermal noise power from a resistor at temperatur T_e

$$N = kT_eB \tag{28}$$

$$F = 1 + \frac{kT_eB}{kT_0B} = 1 + \frac{T_e}{T_0} \tag{29}$$

  - k...Boltzmann constant
  - B...bandwidth
  - T_0...reference temperature (convention: $T_0 = 290\,\mathrm{K}$)

Noise sources I

Thermal noise

- white noise
- noise power spectral density

$$v_n^2 = 4kTR \tag{30}$$

$$[v_n^2] = \mathrm{V}^2/\mathrm{Hz} \tag{31}$$

- noise power under impedance match, e.g. in $50\ \Omega$ environment

$$P = kTB \tag{32}$$

$$[P] = \mathrm{W} \tag{33}$$

Noise sources II

Thermal noise modeling on circuit design level

- noise voltage source, root-mean-square (RMS) or peak for sine wave analysis:

$$v_{\text{n,RMS}} = \sqrt{4kTBR} \tag{34}$$

$$v_{\text{n,peak}} = \sqrt{8kTBR} \tag{35}$$

$$[v_n] = \text{V} \tag{36}$$

- noise current source RMS:

$$i_n = \sqrt{\frac{4kTB}{R}} \tag{37}$$

Noise sources III

Shot noise

- random fluctuations in current flow
- noise power:

$$P = 2qIBR \tag{38}$$

- $P \propto I$, shot noise increases proportionally to current

Flicker noise (1/f noise)

- due to carrier generation/recombination, charge + interface traps, etc.

$$N(f) \propto 1/f^{\alpha}, \alpha = 0..2 \tag{39}$$

- N... noise power spectral density

Noise sources IV

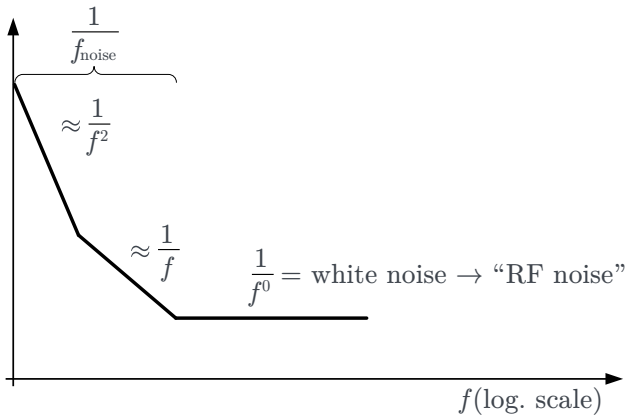


Figure 57: Noise spectrum.

FET Noise Figure I

RF noise figure versus frequency (Fig. 58)

- increase with frequency
  - increased impact of gate resistance thermal noise on gate-induced channel noise
- increase with gate width
  - shot noise
  - increased gate resistance

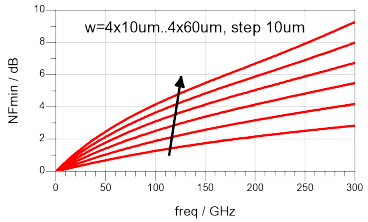


Figure 58: FET RF noise figure versus frequency. Simulation using IAF 100 nm mHEMT (neglecting 1/f noise).

FET Noise Figure II

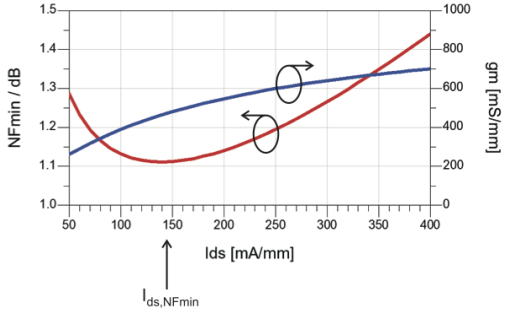


Figure 59: FET noise figure versus current density. Simulation using IAF 100 nm mHEMT @ 94 GHz.

noise figure versus drain current density (Fig. 59)

- current density for minimum noise figure is different from current density for maximum gain: $I_{\mathrm{ds,NFmin}} \neq I_{\mathrm{ds,gmmax}}$

Noise Figure: Which Semiconductor Technology? (Fig. 60)

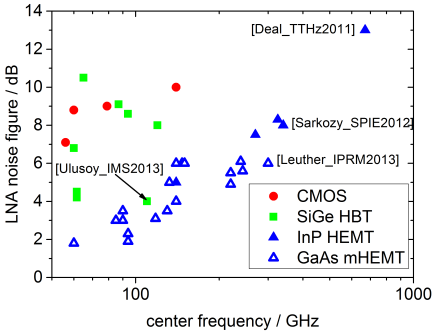


Figure 60: State of the art: LNA noise figure.

Noise Figure of Cascaded Stages (Fig. 61) I

Friis formula

- output spectral noise density: $N_{\text{out}} = G_1 G_2 \left(N_{\text{in}} + N_{\text{eq,tot}}\right) = G_2 \left(G_1 \left(N_{\text{in}} + N_{\text{eq1}}\right) + N_{\text{eq,2}}\right)$
- $N_{\text{eq,tot}} = N_{\text{eq1}} + \frac{N_{\text{eq,2}}}{G_1}$
- using: $N_{\text{eq,tot}} = \left(F_{\text{tot}} - 1\right) kT_0$

$$F_{\text{tot}} = F_1 + \frac{F_2 - 1}{G_1} \tag{40}$$

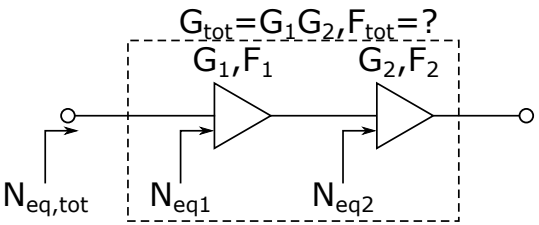
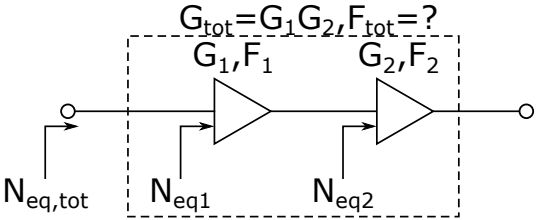


Figure 61: Noise figure of cascaded stages.

Noise Figure of Cascaded Stages (Fig. 61) II

n cascaded stages

$$F_{\text{tot}} = F_1 + \sum_{k=2}^{n} \frac{F_k - 1}{\prod_{l=1}^{k-1} G_l} \tag{41}$$

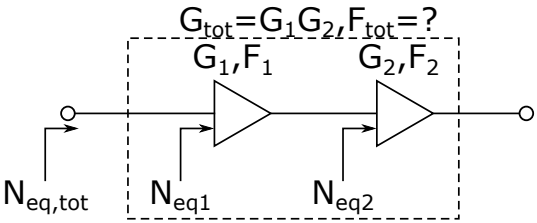


Noise figure of cascaded stages.

Noise Figure of Cascaded Stages (Fig. 61) III

Friis in terms of noise temperature

$$T_{\text{e,sys}} = T_1 + \frac{T_2}{G_1} + \frac{T_3}{G_1 G_2} \ldots \tag{42}$$

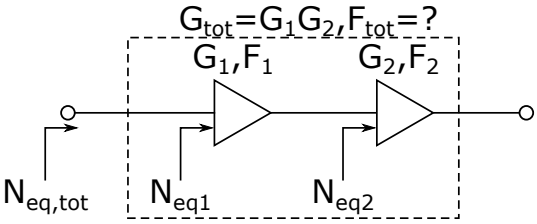


Noise figure of cascaded stages.

Noise Figure of Cascaded Stages (Fig. 61) IV

noise figure of an attenuator A in thermal equilibrium ($T = T_0$)

- output spectral noise density: $T_{\text{out}} = \frac{1}{A}(T_0 + T_{\text{eq}}) = T_0 \ldots$ no noise above T_0
- $T_{\text{eq}} = T_0 (A - 1)$
- $F = 1 + \frac{T_{eq}}{T_0} = A$

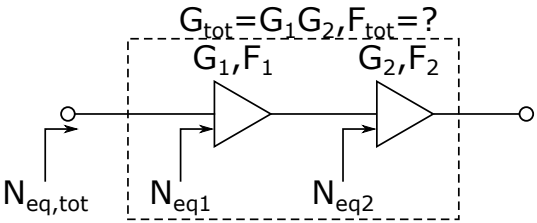


Noise figure of cascaded stages.

Noise Figure of Cascaded Stages (Fig. 61) V

examples

- An LNA with 20 dB gain suppresses the noise contribution of subsequent stage by a factor of 100.
- Loss in the feed network from the antenna to the LNA sets a lower limit to the achievable noise figure.

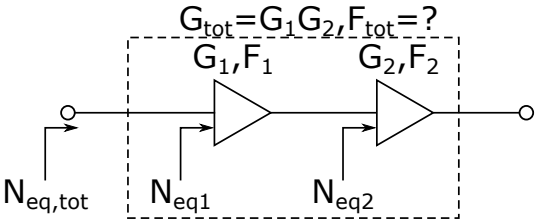


Noise figure of cascaded stages.

Noise Figure of Cascaded Stages (Fig. 61) VI

the "raison d'être" of LNAs

- Provide low noise figure and enough gain, such that the LNA noise figure determines the overall receiver noise figure.
- Provide pre-amplification to weak signals (remote sensing, wireless communication etc.)



Noise figure of cascaded stages.

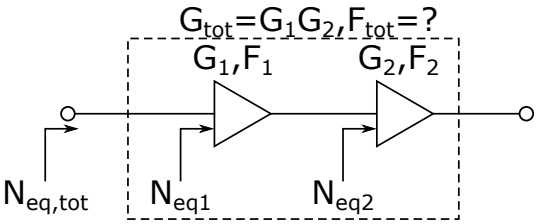
Noise Figure of Cascaded Stages (Fig. 61) VII

gain partitioning

The LNA gain should be high to suppress noise contribution of subsequent stages

but

The LNA gain should not drive the subsequent stage into power compression (linearity)



Noise figure of cascaded stages.

Noise Measure I

Noise measure

- quality factor combing noise figure and gain

$$M = \frac{F-1}{1-\frac{1}{G}} \tag{43}$$

Noise Measure II

consider two cascaded stages

- assume impedance matching such that individual noise and gain is independent of cascade order

$$F_{1-2} = F_1 + \frac{F_2 - 1}{G_1} \tag{44}$$

$$F_{2-1} = F_2 + \frac{F_1 - 1}{G_2} \tag{45}$$

  - if $F_{1-2} < F_{2-1}$, stage 1 should be placed first

$$F_1 + \frac{F_2 - 1}{G_1} < F_2 + \frac{F_1 - 1}{G_2} \Leftrightarrow \frac{F_1 - 1}{1 - \frac{1}{G_1}} < \frac{F_2 - 1}{1 - \frac{1}{G_2}} \Leftrightarrow M_1 < M_2 \tag{46}$$

Noise Measure III

properties of M

- a chain of stage with identical M will have a noise measure of M and overall noise figure of $F = 1 + M$
- cascade should use components with ascending M for minimum overall noise figure

Key LNA Performance Criteria I

noise figure: degradation of signal-to-noise ratio

$$F = \frac{(S/N)_{\text{input}}}{(S/N)_{\text{output}}} > 1 \tag{47}$$

Key LNA Performance Criteria II

gain

- indicated in dB:

$$G_{\mathrm{dB}} = 10 \log \frac{P_{\mathrm{out}}}{P_{\mathrm{in}}} = 20 \log \frac{V_{\mathrm{out}}}{V_{\mathrm{in}}} \tag{48}$$

- example: 20 dB gain is a power amplification of 100 and a voltage amplification of 10.

Key LNA Performance Criteria III

port matching

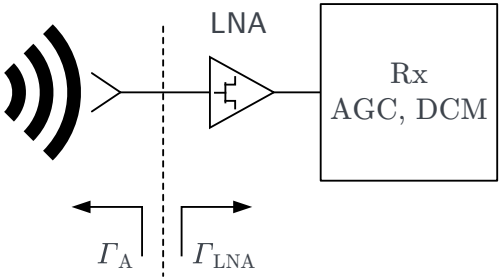
- reflection coefficients:

$$\Gamma_{\text{in}} = S_{11} + \frac{S_{12}S_{21}\Gamma_{\text{load}}}{1 - S_{22}\Gamma_{\text{load}}}, \Gamma_{\text{out}} = S_{22} + \frac{S_{12}S_{21}\Gamma_{\text{source}}}{1 - S_{11}\Gamma_{\text{source}}} \tag{49}$$

- voltage standing-wave-ratio:

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \tag{50}$$

Key LNA Performance Criteria IV



$$\Gamma_{\mathrm{A}} = \Gamma^{*}_{\mathrm{LNA}} = 0$$

$$Z_{\mathrm{A}} = Z^{*}_{\mathrm{LNA}} = 50\,\Omega$$

$\rightarrow$ avoid reflections

Figure 62: Why avoid reflections?

Key LNA Performance Criteria V

bandwidth

- consider constant gain $\times$ bandwidth product

Key LNA Performance Criteria VI

intermodulation, linearity

- typically part of spec, but not a design goal, hence: small-signal analysis
- note: Rx linearity requirements arise e.g. from finite Tx-Rx isolation, strong interferers, reflections

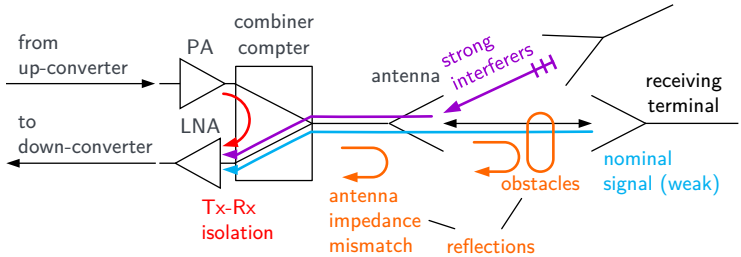


Figure 63: Key LNA performance criteria.

LNA Figure of Merit (Fig. 64)

Figure of Merit

- combine several performance characteristics into a single figure of merit for benchmarking.

$$FoM_{\mathrm{ss}}[\frac{\mathrm{dB}\cdot\mathrm{GHz}}{\mathrm{mW}}] = \frac{f_{\mathrm{c}}}{f_{\mathrm{max}}}\cdot\frac{1}{(NF[1]-1)}\cdot\frac{G[dB]BW[\mathrm{GHz}]}{P_{\mathrm{dc}}[\mathrm{mW}]} \tag{51}$$

- use logarithmic gain, since power consumption is related to the number of cascaded stages, which multiply the gain

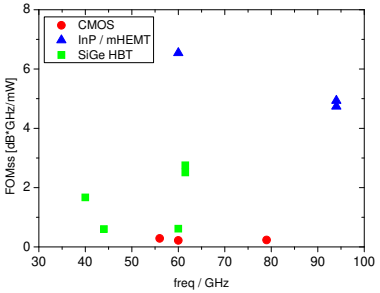


Figure 64: LNA figure of merit.

4.1 LNA Figures of Merit
4.2 LNA Design Methodology

LNA Design Methodology

4.2

The Context of Noise Matching

- Evaluate the impact of sourec impedance on the noise figure of a twoport (amplifier)
- We will find that there exists an optimum source impedance $\Gamma_{\mathrm{s,opt}}$ which leads to a minimum achievable noise figure $NF_{\min}$
- In general, $\Gamma_{\mathrm{s,opt}} \neq \Gamma_1^*$, therefore noise matching $\neq$ power matching

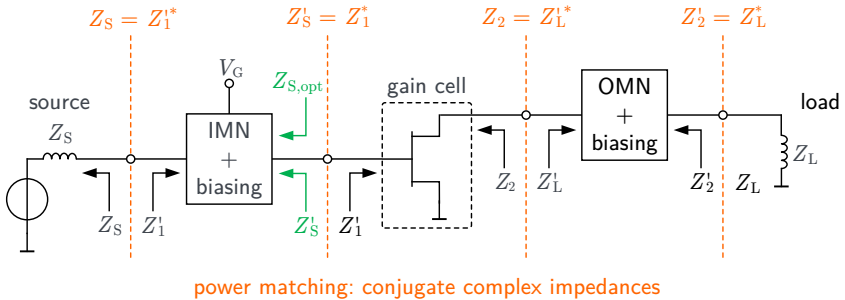


Figure 65: Context of noise matching.

Impact of Source Impedance on Noise Figure (Fig. 66) I

noiseless twoport with equivalent input-referred noise

- all internal noise sources of the twoport are combined into input-referred noise sources
- parallel noise current i and series noise voltage v

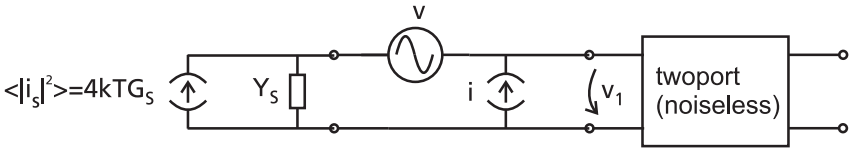


Figure 66: Impact of source impedance on the noise figure of a twoport.

Impact of Source Impedance on Noise Figure (Fig. 66) II

noise correlation admittance

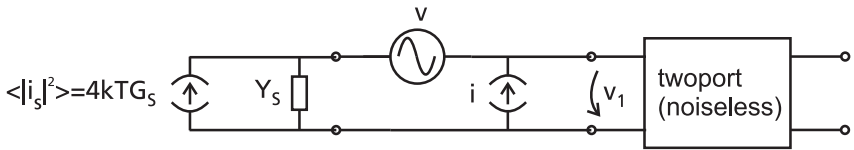
- noise from the two sources i and v may be partially correlated:

$$i = i_u + i_c \tag{52}$$

$$i_c = Y_c \cdot u \tag{53}$$

$$Y_c = G_c + jB_c \tag{54}$$

- Y_c ...correlation admittance

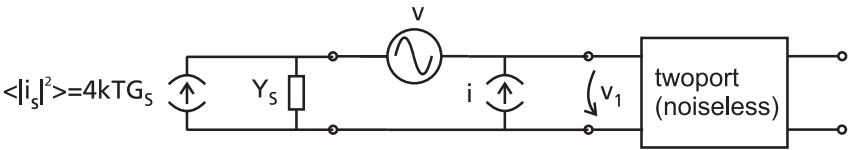


Impact of source impedance on the noise figure of a twoport.

Impact of Source Impedance on Noise Figure (Fig. 66) III

noise power spectral density

$$\langle i \cdot v^* \rangle = \langle (i_u + i_c) \cdot v^* \rangle = \langle i_c \cdot v^* \rangle = Y_c \cdot \left\langle |v|^2 \right\rangle \tag{55}$$



Impact of source impedance on the noise figure of a twoport.

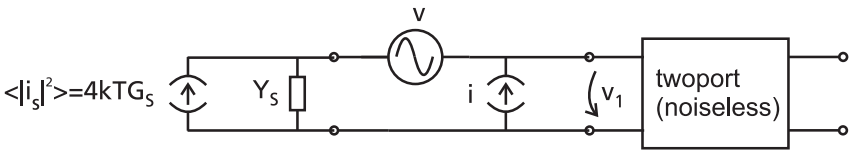
Impact of Source Impedance on Noise Figure (Fig. 66) IV

source admittance

$$Y_S = G_S + jB_S \tag{56}$$

- input noise power spectral density (RMS):

$$\left\langle |i_s|^2 \right\rangle = 4kTG_S \tag{57}$$



Impact of source impedance on the noise figure of a twoport.

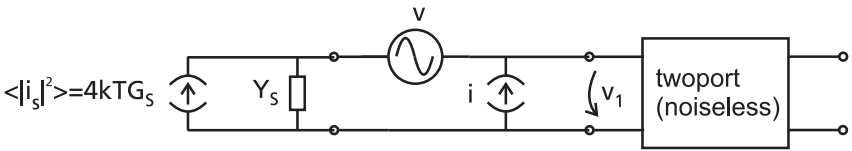
Impact of Source Impedance on Noise Figure (Fig. 66) V

twoport noise figure

$$F = \frac{N_S + N_{eq}}{N_S} \tag{58}$$

- to calculate N_{eq}, consider noiseless twoport as load and calculate available power ($\left\langle |v_1|^2 \right\rangle Y_{in}$) at input

$$F = \frac{\left\langle |v_1|^2 \right\rangle \Big|_{\text{all sources active}}}{\left\langle |v_1|^2 \right\rangle \Big|_{\text{sources noise only}}} \tag{59}$$



Impact of source impedance on the noise figure of a twoport.

Impact of Source Impedance on Noise Figure (Fig. 66) VI

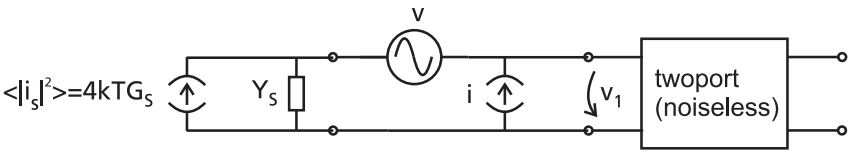
result after nodal analysis

$$F = F_{\min} + \frac{R_\mathrm{n}}{G_\mathrm{s}} \left((G_\mathrm{s} - G_{\mathrm{s,opt}})^2 + (B_\mathrm{s} - B_{\mathrm{s,opt}})^2 \right) = F_{\min} + \frac{R_\mathrm{n}}{G_\mathrm{s}} \left| Y_\mathrm{s} - Y_{\mathrm{s,opt}} \right|^2 \tag{60}$$

$$F_{\min} = 1 + 2R_\mathrm{n} \left(G_\mathrm{c} + G_{\mathrm{s,opt}} \right) \tag{61}$$

$$B_{\mathrm{s,opt}} = -B_\mathrm{c} \tag{62}$$

$$G_{\mathrm{s,opt}} = \sqrt{\frac{g_\mathrm{n}}{R_\mathrm{n}} - B_\mathrm{c}^2} \tag{63}$$



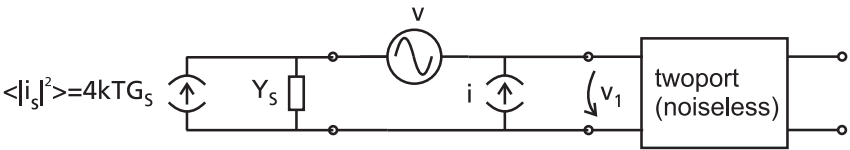
Impact of source impedance on the noise figure of a twoport.

Impact of Source Impedance on Noise Figure (Fig. 66) VII

noise parameters

$$F = F_{\min} + \text{penalty for deviating from } Y_{\mathrm{s,opt}} \tag{64}$$

- "noise parameters": $Y_{\mathrm{s,opt}}$, $F_{\min}$ and R_{n}
- the larger R_{n}, the higher the penalty



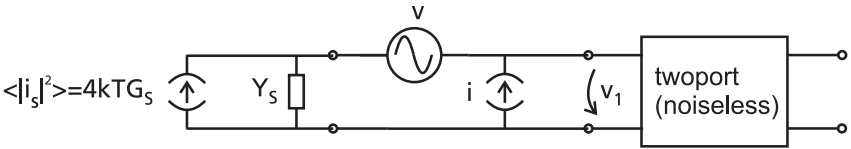
Impact of source impedance on the noise figure of a twoport.

Impact of Source Impedance on Noise Figure (Fig. 66) VIII

express impedances/admittances in terms of reflection coefficients

$$Z = Z_0 \frac{1+\Gamma}{1-\Gamma} \tag{65}$$

$$Y = Y_0 \frac{1-\Gamma}{1+\Gamma} \tag{66}$$



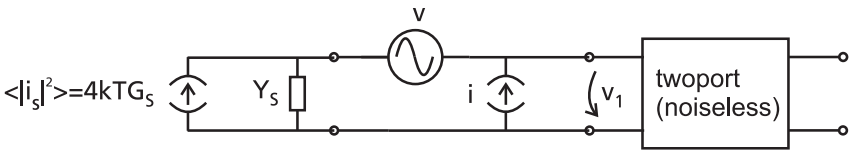
Impact of source impedance on the noise figure of a twoport.

Impact of Source Impedance on Noise Figure (Fig. 66) IX

noise figure

$$F = F_{min} + \frac{R_n}{G_s} |Y_s - Y_{s,opt}|^2 = F_{min} + \frac{4r_n |\Gamma_s - \Gamma_{s,opt}|^2}{\left(1 - |\Gamma_s|^2\right) |1 + \Gamma_{s,opt}|^2} \tag{67}$$

$$r_n = R_n Y_0 \tag{68}$$



Impact of source impedance on the noise figure of a twoport.

Source Pull Measurement for Noise Parameters

- vary source impedance (using impedance tuners) and measure output noise figure
- interpolating the noise circles results in the noise paraboloid (Fig. 67) from which the noise parameters Γ_{opt}, $F_{\min}$ and R_n can be inferred

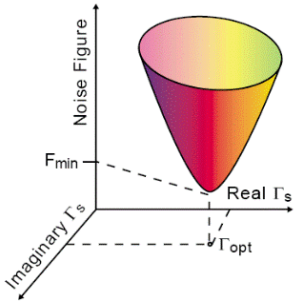


Figure 67: Noise paraboloid. Source: www.microwaves101.com.

LNA Design Goal: Simultaneous Noise and Power Matching (Fig. 162) I

input match

- input matched for low-noise:

$$\Gamma_s^{'} = \Gamma_{\mathrm{s,opt}} \tag{69}$$

- input matched for power:

$$\Gamma_s^{'} = \Gamma_1^* \tag{70}$$

- requires:

$$\Gamma_1 = \Gamma_{\mathrm{s,opt}}^* \tag{71}$$

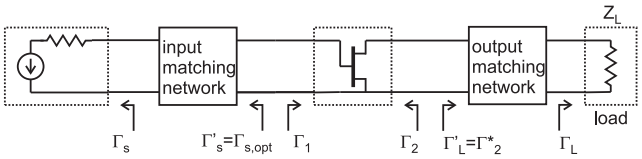


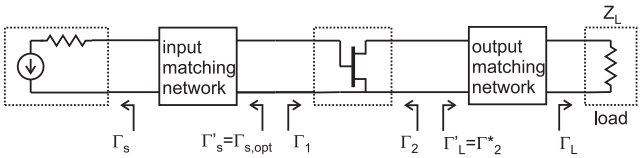
Figure 68: LNA impedance matching.

LNA Design Goal: Simultaneous Noise and Power Matching (Fig. 162) II

output match

- output matched for power:

$$\Gamma_l' = \Gamma_2^* \tag{72}$$



LNA impedance matching.

Simultaneous Noise and Power Matching I

Noise & power match possible, if

$$\Gamma_1 = \Gamma_{\mathrm{s,opt}}^* \tag{73}$$

- the same matching network achieves simultaneous match (Fig. 164)

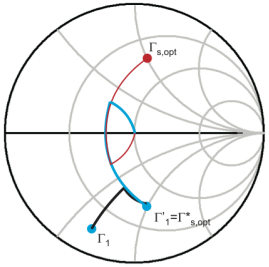
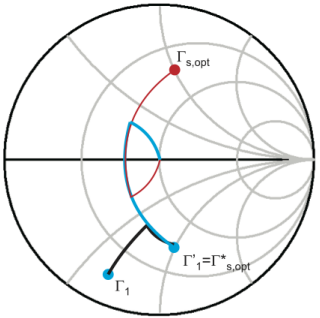


Figure 69: Transformation paths for simultaneous noise and power matching.

Simultaneous Noise and Power Matching II

approach

- modify the transistor input impedance without modifying its optimum source reflection coefficient

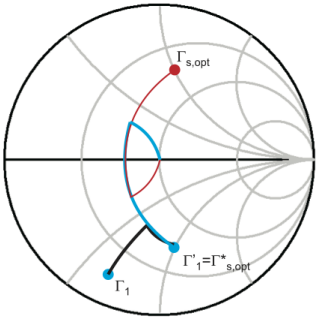


Transformation paths for simultaneous noise and power matching.

Simultaneous Noise and Power Matching III

methods

- feedback, e.g. inductive source degeneration
- deliberate output mismatch (requires $S_{12} \neq 0$)



Transformation paths for simultaneous noise and power matching.

Simultaneous Noise and Power Matching IV

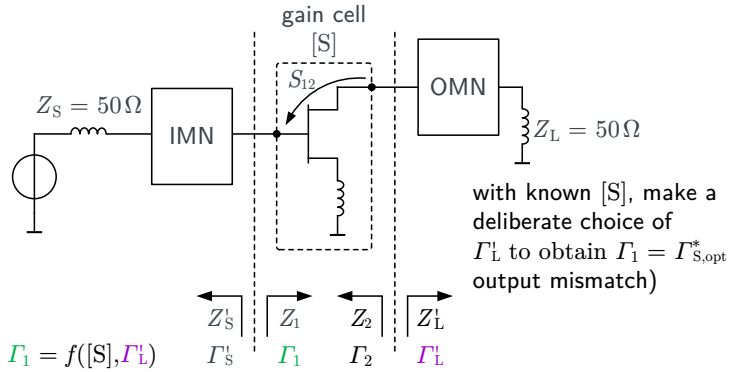


Figure 70: Deliberate output mismatch.

Simultaneous Noise and Power Matching V

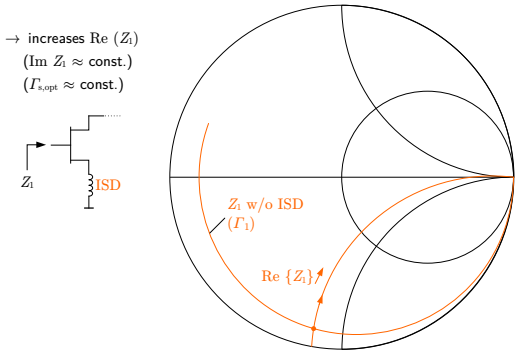


Figure 71: Inductive source degeneration.

Inductive Source Degeneration I

Input impedance with inductive source degeneration (Fig. 72)

$$v_1 \approx v_i + j\omega L_s \cdot i_2 = \frac{i_1}{j\omega C_{gs}} + j\omega L_s g_m \frac{i_1}{j\omega C_{gs}} \tag{74}$$

$$Z_1 = \frac{v_1}{i_1} = L_s \frac{g_m}{C_{gs}} - j\frac{1}{\omega C_{gs}} \approx 2\pi f_t L_s - j\frac{1}{\omega C_{gs}} \tag{75}$$

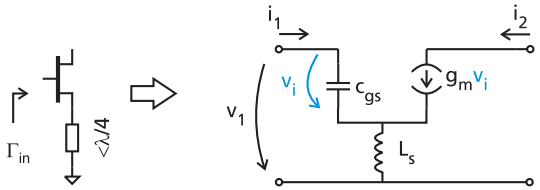
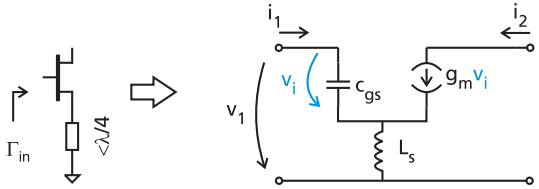


Figure 72: Simplified circuit analysis for inductive source degeneration.

Inductive Source Degeneration II

Effect of inductive source degeneration

- L_s increases the real part of the input impedance
- in mmW-circuits, L_s is realized as a shorted stub



Simplified circuit analysis for inductive source degeneration.

LNA Design Sequence

single-stage LNA design sequence

- choice of transistor bias (drain current density for low noise $I_{d,ln}$)
- choice of transistor dimensions
  - sufficient gain at center frequency
  - low noise figure (number of gate fingers)
  - convenient $\Gamma_{\mathrm{s,opt}}$
- in- and output matching networks (iterative, due to $S_{12} \neq 0$)
  - input matched for low-noise: $\Gamma_s' = \Gamma_{\mathrm{s,opt}}$
  - output matched for power: $\Gamma_l' = \Gamma_2^*$
- check stability
  - in-band
  - out-of-band (e.g. LF terminations)
  - unconditional (arbitrary source and load impedance)

Choice of the Transistor Dimensions - Gain I

Gain versus transistor width (Fig. 73)

- up to $k = 1$, the gain increaes slightly with transistor width
- the $k = 1$ point occurs at lower frequencies for larger transistors

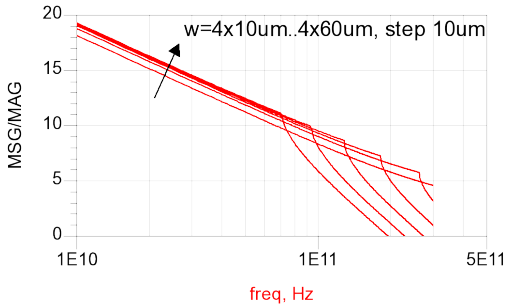
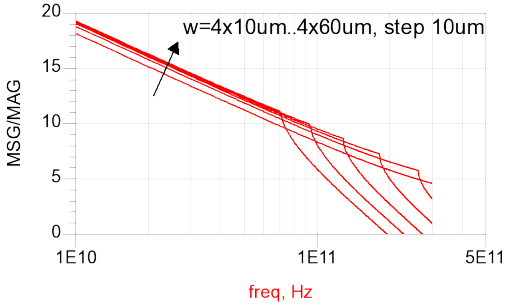


Figure 73: MSG/MAG versus frequency and transistor scaling. Simulation using IAF 100 nm mHEMTs

Choice of the Transistor Dimensions - Gain II

In practice

- chose a transistor size as large as possible (convenient Γ_{opt}, gain, output power), but with $k < 1$ at the operating frequency (for maximum gain) and with some safety margin (to account for fabricartion spread)



MSG/MAG versus frequency and transistor scaling. Simulation using IAF 100 nm mHEMTs

Choice of the Transistor Dimensions - Noise Figure

Minimum noise figure versus transistor width (Fig. 74)

- assumes perfect noise matching ($\Gamma_S' = \Gamma_{\mathrm{S,opt}}$)
- smaller transistor size = lower noise figure, but beware of noise measure and convenient $\Gamma_{\mathrm{S,opt}}$

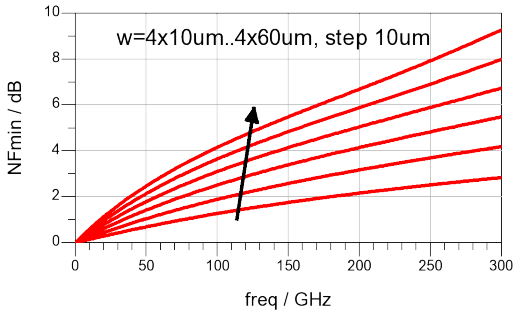


Figure 74: Minimum noise figure versus frequency and transistor width. Simulation using IAF 100 nm mHEMTs, w/o 1/f noise.

Choice of the Transistor Dimensions - Γ_{opt} I

Optimum source impedance versus transistor width (Fig. 75)

- note that the real part of the optimum source impedance is much higher than the real part of the input impedance, hence: inductive source degeneration.

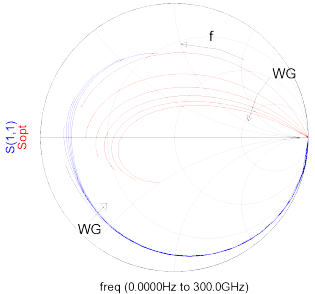
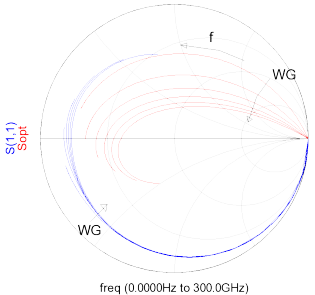


Figure 75: Optimum source impedance $\Gamma_{S,opt}$ and input impedance versus frequency and transistor size. Simulation using IAF 100 nm mHEMTs.

Choice of the Transistor Dimensions - Γ_{opt} II

In practice

- adjust the real part of the input impedance by inductive source degeneration or other feedback methods.
- chose a transistor size for $\mathrm{Im}\left(\Gamma_{\mathrm{S,opt}}\right) \approx -\mathrm{Im}\left(\Gamma_1\right)$



Optimum source impedance $\Gamma_{\mathrm{S,opt}}$ and input impedance versus frequency and transistor size. Simulation using IAF 100 nm mHEMTs.

Multi-Stage LNAs I

interstage matching network (Fig. 76)

- no need to go to intermediate reference level of $Z_0 = 50\Omega$
- matching to a shifted center frequency can improve the bandwidth

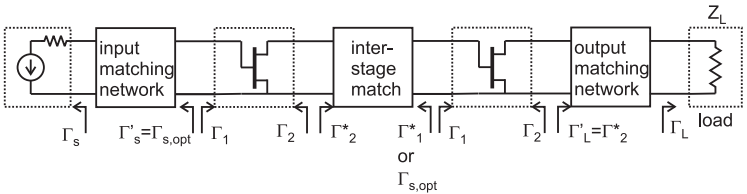
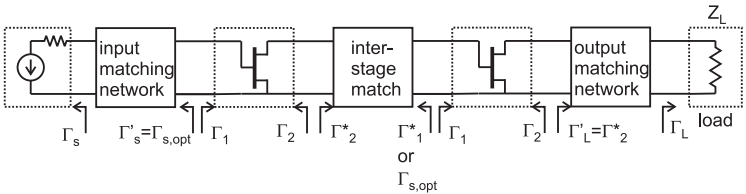


Figure 76: Impedance matching in multi-stage LNAs.

Multi-Stage LNAs II

stability

- carry out GammaProbe test at all active↔passive interfaces, i.e. at all transistor terminals above ground
- vary the source and load impedance over the entire Smith chart



Impedance matching in multi-stage LNAs.

Bias Networks: $\lambda/4$ Stubs (Fig. 77) I

- $\lambda/4$ stubs tranform the RF-short generated by shunt capacitors at the DC supply ports into an RF-open at the RF-path.
- the RF short decouples any off-chip impedance (bond-wire inductance, cables, probes, SMU impedance,...)

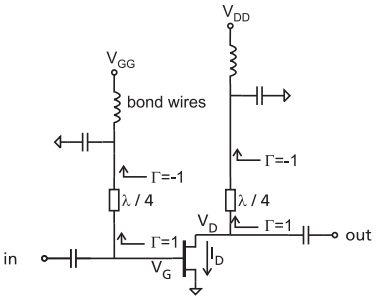


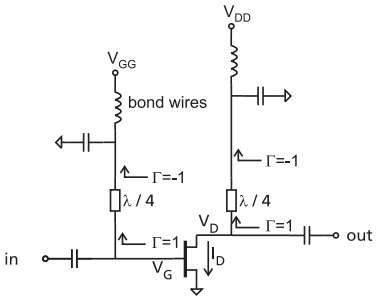
Figure 77: Biasing by $\lambda/4$ transmission lines.

Bias Networks: $\lambda/4$ Stubs (Fig. 77) II

internal voltages

$$V_G = V_{GG} \tag{76}$$

$$V_D = V_{DD} \tag{77}$$

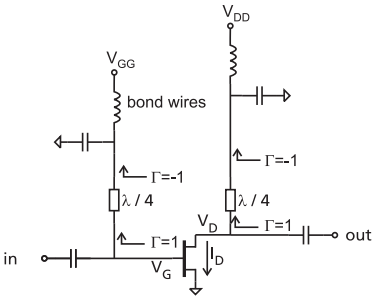


Biasing by $\lambda/4$ transmission lines.

Bias Networks: $\lambda/4$ Stubs (Fig. 77) III

best option at mm-wave frequencies

- no power loss in resistors
- often, $< \lambda/4$ stubs also form part of the matching network (equivalent to parallel inductor)



Biasing by $\lambda/4$ transmission lines.

Bias Network Alternatives - Resistive (Fig. 78) I

internal voltages

$$V_G \approx V_{GG} \tag{78}$$

since $I_g \approx 0$

$$V_D = V_{DD} - I_D R_D \tag{79}$$

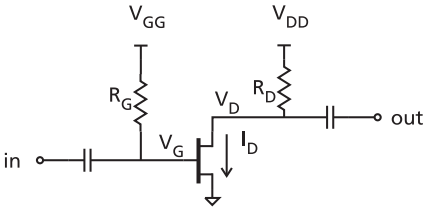


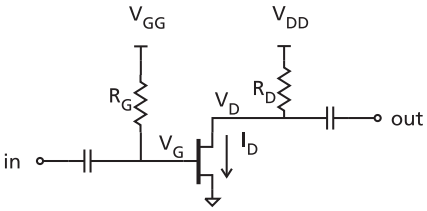
Figure 78: Biasing through resistors.

Bias Network Alternatives - Resistive (Fig. 78) II

typically no option for drain supply

- DC power consumption in R_D
- transistor breakdown may occur during drain supply voltage ramp-up

$$I_{\mathrm{D}} = 0|_{t=0} \rightarrow V_{\mathrm{D}} = V_{\mathrm{DD}}|_{t=0} \tag{80}$$

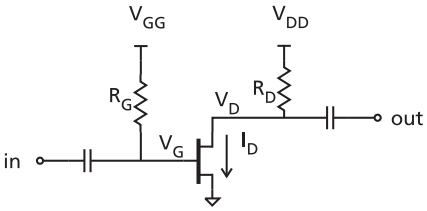


Biasing through resistors.

Bias Network Alternatives - Resistive (Fig. 78) III

"forced" 50Ω-matching

- $R_G = 50\Omega$
- thermal noise from R_G degrades noise figure
- can prevent breakdown due to gate current e.g. in power amplifiers



Biasing through resistors.

Bias Network Alternatives - Self-biasing (Fig. 79) I

internal voltages

$$V_s = R_s I_d \tag{81}$$

$$V_g = \frac{R_1}{R_1 + R_2} V_d \tag{82}$$

- bypass capacitor

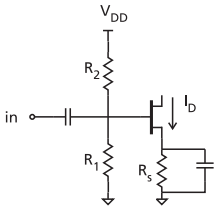
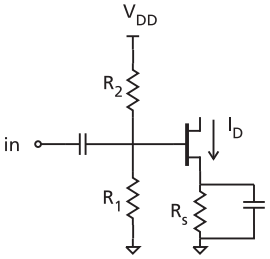


Figure 79: Gate bias through self-biasing.

Bias Network Alternatives - Self-biasing (Fig. 79) II

discussion

- allows for single supply voltage
- allows for all-positive supply voltage, if negative V_{gs} is required
  - class B bias circuits
  - class A: if $V_{gs,gmmax} < 0$ in depletion mode transistors



Gate bias through self-biasing.

Bias Network Alternatives - Active Load (Fig. 80) I

active load technique

- use the high output impedance as load (ideally a current source)
- requires conducting channel with $g_m = 0 \rightarrow v_{gs} = 0$
- D-mode HEMT: connected gate-source
- MOSFET: connected gate-drain with RF shorting capacitance between gate-source or a current mirror circuit

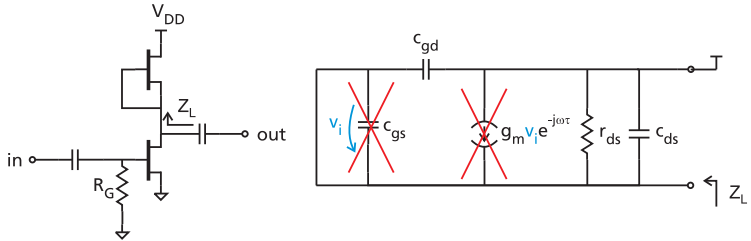
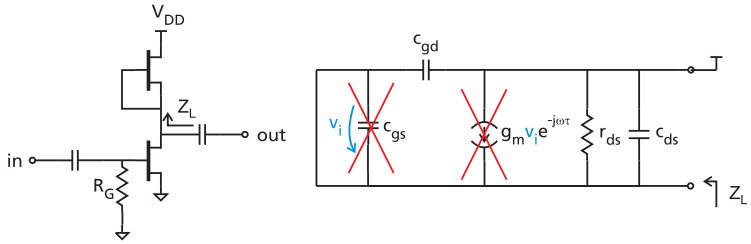


Figure 80: Drain biasing through active load.

Bias Network Alternatives - Active Load (Fig. 80) II

DC

- DC voltage drop across load = drain-source voltage V_{ds}, typically $\approx 1\,\mathrm{V}$
- DC power consumed in load: $P_{load} = I_d V_{ds}$



Drain biasing through active load.

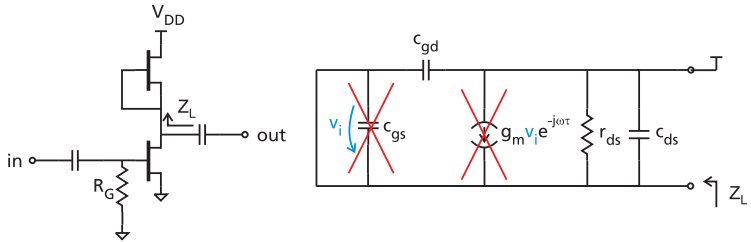
Bias Network Alternatives - Active Load (Fig. 80) III

RF

- RF load impedance:

$$Z_l = \frac{1}{\frac{1}{r_{ds}} + j\omega\left(C_{gd} + C_{ds}\right)} = \frac{r_{ds}}{1 + j\omega r_{ds}\left(C_{gd} + C_{ds}\right)} \tag{83}$$

- advantage: $Z_l \gg \frac{V_{ds}}{I_d}$ below frequency roll-off
- additional capacitive load



Drain biasing through active load.

University of Stuttgart
Germany

Institute of
Robust Power
Semiconductor Systems

Broadband Amplifiers

5

1 Course Outline

2 The Millimeterwave Frequency Spectrum: Applicatio
and Technologies

3 Functional Buildung Blocks of MMICs

4 Low Noise Amplifier

5 Broadband Amplifiers

6 Power Amplifiers

7 Microwave Network Analysis (Appendix)

5.1 Applications
5.2 Performance Indicators
5.3 Wideband Matching
5.4 Wideband Intermediate Frequency (IF) Techniques
5.5 Travelling-Wave Amplifier

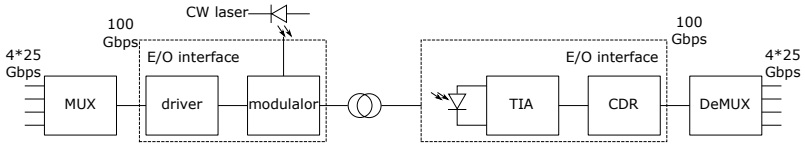
Applications

5.1

Optical Communication System

analog/mixed signal frontend topology

- TX: modulator driver
- RX: transimpedance amplifier transimpedance amplifier (TIA), clock-and-data recovery CDR



Optical Communication System

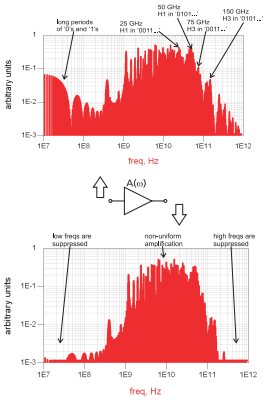


Figure 81: Frequency Spectrum of a 100 Gbps NRZ bit sequence.

Optical Communication System

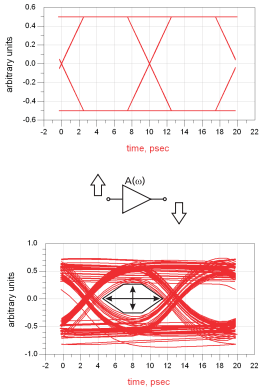


Figure 82: Eye diagrams of a 100 Gbps NRZ bit sequence.

Wideband IF Amplifier

Ku-band satellite receiver frontend

- RF carrier frequency within Ku-band
- IF frequency range: e.g. 950 - 2150 MHz
- IF bandwidth: 1200 MHz
- IF fractional bandwidth: 77%

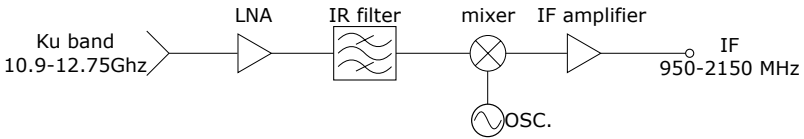


Figure 83: Ku-band receiver.

5.1 Applications
5.2 Performance Indicators
5.3 Wideband Matching
5.4 Wideband IF Techniques
5.5 Travelling-Wave Amplifier

Performance Indicators

5.2

Bandwidth

absolute and fractional bandwidth

- f_c operating/center frequency
- f_l, f_u lower, upper corner frequency
- absolute bandwidth BW:

$$BW = f_\mathrm{u} - f_\mathrm{l} \tag{84}$$

- fractional bandwidth FBW:

$$FBW = \frac{f_\mathrm{u} - f_\mathrm{l}}{f_\mathrm{c}} \tag{85}$$

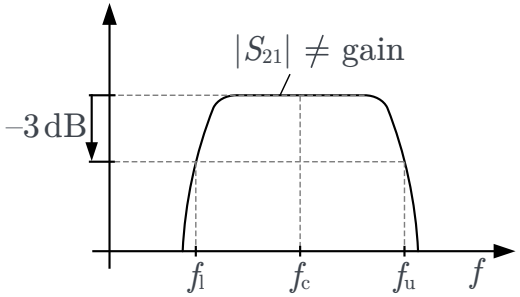


Figure 85: 3-dB gain bandwidth.

Bandwidth

bandwidth definitions

- 3-dB or 1-dB bandwidth
- defined by gain, matching, ...
- stringent: frequency range, in which all specifications are met

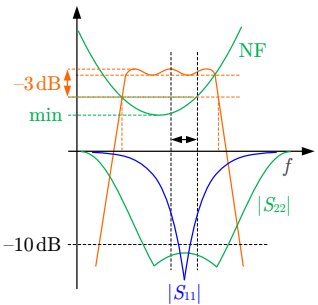


Figure 84: Visualization of the specifications.

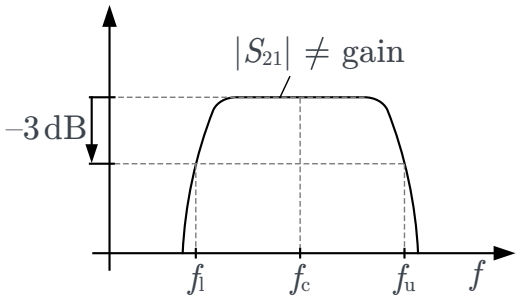


Figure 85: 3-dB gain bandwidth.

Bandwidth

frequent terms related to amplifier bandwidth

- "broadband:" $FBW > 25\%$ e.g. cover entire W-band (70-110 GHz) $FBW = \frac{40\,\text{GHz}}{90\,\text{GHz}} = 44\%$
- "ultra-broadband": f_l few MHz, f_u several GHz. e.g. optical communication (12.5/40/100 Gbps)

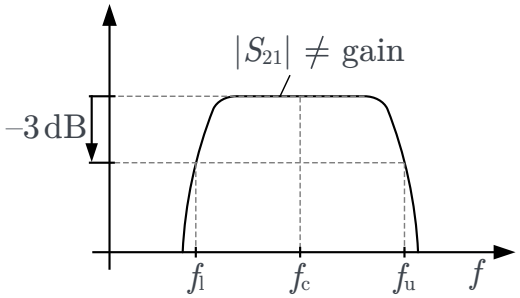


Figure 85: 3-dB gain bandwidth.

Group Delay Ripple

- transfer function of a distortion-less system

$$H(\omega) = Ce^{-j\omega\tau_0} \tag{86}$$

- group delay τ

$$\tau = -\frac{d\Phi}{d\omega} = \text{const.} = \tau_0 \tag{87}$$

- interpretation:
  - constant amplitude characteristics in frequency range of interest
  - linear phase characteristics in frequency range of interest

Group Delay Ripple

- transfer function of a distortion-less system

$$H(\omega) = Ce^{-j\omega\tau_0} \tag{86}$$

- group delay τ

$$\tau = -\frac{d\Phi}{d\omega} = \text{const.} = \tau_0 \tag{87}$$

- interpretation:
  - constant amplitude characteristics in frequency range of interest
  - linear phase characteristics in frequency range of interest

Group Delay Ripple

- transfer function of a distortion-less system

$$H(\omega) = Ce^{-j\omega\tau_0} \tag{86}$$

- group delay τ

$$\tau = -\frac{d\Phi}{d\omega} = \text{const.} = \tau_0 \tag{87}$$

- interpretation:
  - constant amplitude characteristics in frequency range of interest
  - linear phase characteristics in frequency range of interest

Group Delay Ripple

- transfer function of a distortion-less system

$$H(\omega) = Ce^{-j\omega\tau_0} \tag{86}$$

- group delay τ

$$\tau = -\frac{d\Phi}{d\omega} = \text{const.} = \tau_0 \tag{87}$$

- interpretation:
  - constant amplitude characteristics in frequency range of interest
  - linear phase characteristics in frequency range of interest

Group Delay Ripple

- transfer function of a distortion-less system

$$H(\omega) = Ce^{-j\omega\tau_0} \tag{86}$$

- group delay τ

$$\tau = -\frac{d\Phi}{d\omega} = \text{const.} = \tau_0 \tag{87}$$

- interpretation:
  - constant amplitude characteristics in frequency range of interest
  - linear phase characteristics in frequency range of interest

5.1 Applications
5.2 Performance Indicators
5.3 Wideband Matching
5.4 Wideband IF Techniques
5.5 Travelling-Wave Amplifier

Wideband Matching

5.3

Ways to Improve Bandwidth: Interstage Matching

interstage matching network

- improve bandwidth by matching to different center frequencies in the interstage matching networks

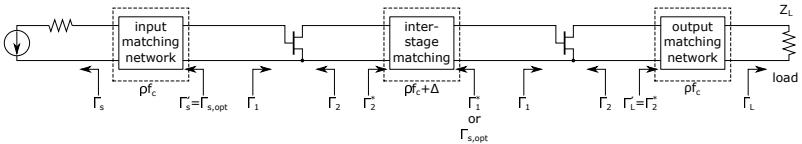


Figure 86: Interstage matching.

Wideband Matching

best possible match for parallel RC load (Fano Criterion)

$$|\Gamma_{\text{min}}| = e^{-\pi \frac{Q_2}{Q_1}} \tag{88}$$

- Γ_{min} ... minimum achievable reflection coefficient
- $Q_1 = RC\omega_0$
- $Q_2 = \frac{f_0}{\Delta f}$ ... RC quality factor
- ω_0 ... center frequency
- Δf ... bandwidth

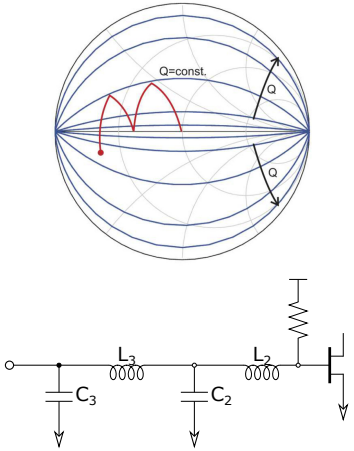


Figure 87: Impedance transformation path for multiple matching elements.

Wideband Matching

interpretation

- low-Q load can be matched in a broad bandwidth
- high-Q load has large reactance change with frequency $\rightarrow$ poor bandwidth

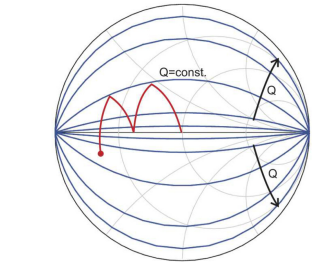


Figure 87: Impedance transformation path for multiple matching elements.

Wideband Matching

consider the extreme cases

- resistive load ($Q = 0$) can be matched in an infinite bandwidth
- ideal reactive load ($Q = \infty$) can be matched only at one frequency, the resonance frequency

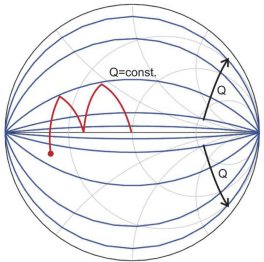


Figure 87: Impedance transformation path for multiple matching elements.

Wideband Matching

constant Q contour in Smith chart

$$\left|\frac{Im(Z)}{Re(Z)}\right| = \text{const.} \tag{89}$$

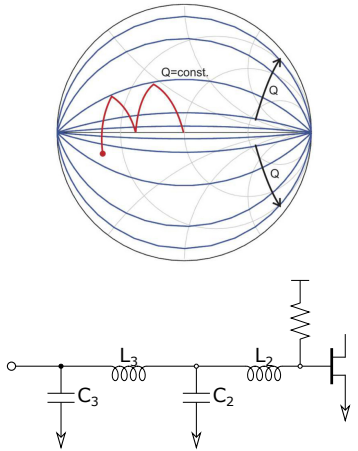


Figure 87: Impedance transformation path for multiple matching elements.

Wideband Matching

wideband matching

- use multiple reactive elements and stay within low Q regime
- do not increase the quality factor beyond its initial value along the transformation path

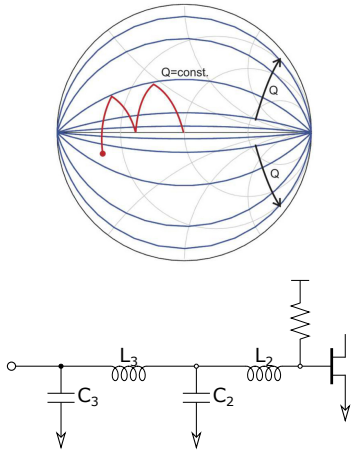


Figure 87: Impedance transformation path for multiple matching elements.

5.1 Applications
5.2 Performance Indicators
5.3 Wideband Matching
5.4 Wideband IF Techniques
5.5 Travelling-Wave Amplifier

Wideband IF Techniques

5.4

The Bandwidth Limitation of Cascaded Stages

- consider transadmittance of two cascaded common-source stages:

$$Y_t = \frac{i_2}{v_1} = g_m v_2 / v_1 = -g_m g_m Z_L = -g_m^2 \frac{R_L}{1 + j\omega C_{gs} R_L} \tag{90}$$

- single-pole roll-off
- 3-dB corner frequency

$$\omega_c = \frac{1}{C_{gs} R_L} \tag{91}$$

- with $A_v = g_m R_L$ and $\omega_T = \frac{g_m}{C_{gs}}$:

$$\omega_c A_v = \omega_T \tag{92}$$

- constant gain-bandwidth product

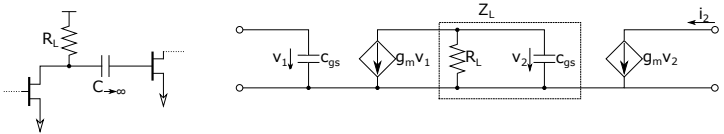


Figure 88: Bandwidth limitation of cascaded stages.

The Bandwidth Limitation of Cascaded Stages

- consider transadmittance of two cascaded common-source stages:

$$Y_t = \frac{i_2}{v_1} = g_m v_2 / v_1 = -g_m g_m Z_L = -g_m^2 \frac{R_L}{1 + j\omega C_{gs} R_L} \tag{90}$$

- single-pole roll-off
- 3-dB corner frequency

$$\omega_c = \frac{1}{C_{gs} R_L} \tag{91}$$

- with $A_v = g_m R_L$ and $\omega_T = \frac{g_m}{C_{gs}}$:

$$\omega_c A_v = \omega_T \tag{92}$$

- constant gain-bandwidth product

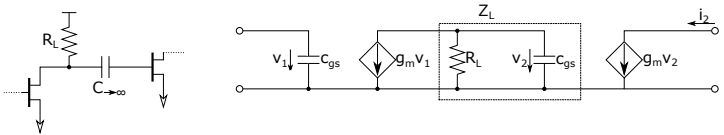


Figure 88: Bandwidth limitation of cascaded stages.

The Bandwidth Limitation of Cascaded Stages

- consider transadmittance of two cascaded common-source stages:

$$Y_t = \frac{i_2}{v_1} = g_m v_2 / v_1 = -g_m g_m Z_L = -g_m^2 \frac{R_L}{1 + j\omega C_{gs} R_L} \tag{90}$$

- single-pole roll-off
- 3-dB corner frequency

$$\omega_c = \frac{1}{C_{gs} R_L} \tag{91}$$

- with $A_v = g_m R_L$ and $\omega_T = \frac{g_m}{C_{gs}}$:

$$\omega_c A_v = \omega_T \tag{92}$$

- constant gain-bandwidth product

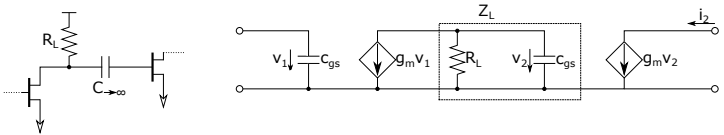


Figure 88: Bandwidth limitation of cascaded stages.

The Bandwidth Limitation of Cascaded Stages

- consider transadmittance of two cascaded common-source stages:

$$Y_t = \frac{i_2}{v_1} = g_m v_2 / v_1 = -g_m g_m Z_L = -g_m^2 \frac{R_L}{1 + j\omega C_{gs} R_L} \tag{90}$$

- single-pole roll-off
- 3-dB corner frequency

$$\omega_c = \frac{1}{C_{gs} R_L} \tag{91}$$

- with $A_v = g_m R_L$ and $\omega_T = \frac{g_m}{C_{gs}}$:

$$\omega_c A_v = \omega_T \tag{92}$$

- constant gain-bandwidth product

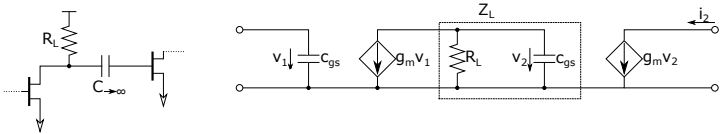


Figure 88: Bandwidth limitation of cascaded stages.

The Bandwidth Limitation of Cascaded Stages

- consider transadmittance of two cascaded common-source stages:

$$Y_t = \frac{i_2}{v_1} = g_m v_2 / v_1 = -g_m g_m Z_L = -g_m^2 \frac{R_L}{1 + j\omega C_{gs} R_L} \tag{90}$$

- single-pole roll-off
- 3-dB corner frequency

$$\omega_c = \frac{1}{C_{gs} R_L} \tag{91}$$

- with $A_v = g_m R_L$ and $\omega_T = \frac{g_m}{C_{gs}}$:

$$\omega_c A_v = \omega_T \tag{92}$$

- constant gain-bandwidth product

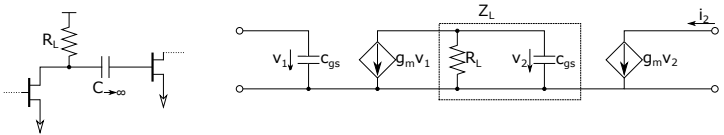


Figure 88: Bandwidth limitation of cascaded stages.

Ways to Improve IF Bandwidth: Inductive Peaking Fig. 89

- inductor L to compensate for load capacitance.
- effect of inductive (shunt) peaking:

$$Y_t = -g_m g_m Z_L = -g_m^2 \frac{R_L\,(1 + j\omega L/R_L)}{1 - \omega^2 C_{gs} L + j\omega R_L C_{gs}} \tag{93}$$

- with $\tau_C = R_L C_{gs} = \frac{1}{\omega_c}$ and $\tau_L = L/R_L$:

$$Y_t = -g_m^2 \frac{R_L\,(1 + j\omega\tau_L)}{1 - \omega^2 \tau_C \tau_L + j\omega\tau_C} \tag{94}$$

- 3-dB corner freq:

$$\omega_{c,I} = \omega_c \sqrt{\left(-\frac{\left(\frac{\tau_C}{\tau_L}\right)^2}{2} + \frac{\tau_C}{\tau_L} + 1\right) + \sqrt{\left(-\frac{\left(\frac{\tau_C}{\tau_L}\right)^2}{2} + \frac{\tau_C}{\tau_L} + 1\right)^2 + \left(\frac{\tau_C}{\tau_L}\right)^2}} \tag{95}$$

- has a maximum for $\frac{\tau_C}{\tau_L} = \sqrt{2}$

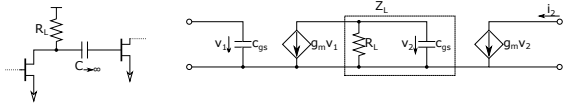


Figure 89: Inductive peaking.

Ways to Improve IF Bandwidth: Inductive Peaking Fig. 89

- inductor L to compensate for load capacitance.
- effect of inductive (shunt) peaking:

$$Y_t = -g_m g_m Z_L = -g_m^2 \frac{R_L\,(1 + j\omega L/R_L)}{1 - \omega^2 C_{gs} L + j\omega R_L C_{gs}} \tag{93}$$

- with $\tau_C = R_L C_{gs} = \frac{1}{\omega_c}$ and $\tau_L = L/R_L$:

$$Y_t = -g_m^2 \frac{R_L\,(1 + j\omega\tau_L)}{1 - \omega^2 \tau_C \tau_L + j\omega\tau_C} \tag{94}$$

- 3-dB corner freq:

$$\omega_{c,I} = \omega_c \sqrt{\left(-\frac{\left(\frac{\tau_C}{\tau_L}\right)^2}{2} + \frac{\tau_C}{\tau_L} + 1\right) + \sqrt{\left(-\frac{\left(\frac{\tau_C}{\tau_L}\right)^2}{2} + \frac{\tau_C}{\tau_L} + 1\right)^2 + \left(\frac{\tau_C}{\tau_L}\right)^2}} \tag{95}$$

- has a maximum for $\frac{\tau_C}{\tau_L} = \sqrt{2}$

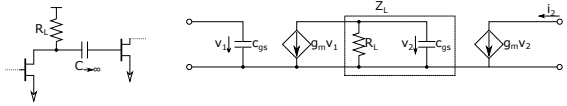


Figure 89: Inductive peaking.

Ways to Improve IF Bandwidth: Inductive Peaking Fig. 89

- inductor L to compensate for load capacitance.
- effect of inductive (shunt) peaking:

$$Y_t = -g_m g_m Z_L = -g_m^2 \frac{R_L\left(1 + j\omega L/R_L\right)}{1 - \omega^2 C_{gs} L + j\omega R_L C_{gs}} \tag{93}$$

- with $\tau_C = R_L C_{gs} = \frac{1}{\omega_c}$ and $\tau_L = L/R_L$:

$$Y_t = -g_m^2 \frac{R_L\left(1 + j\omega\tau_L\right)}{1 - \omega^2 \tau_C \tau_L + j\omega\tau_C} \tag{94}$$

- 3-dB corner freq:

$$\omega_{c,I} = \omega_c \sqrt{\left(-\frac{\left(\frac{\tau_C}{\tau_L}\right)^2}{2} + \frac{\tau_C}{\tau_L} + 1\right) + \sqrt{\left(-\frac{\left(\frac{\tau_C}{\tau_L}\right)^2}{2} + \frac{\tau_C}{\tau_L} + 1\right)^2 + \left(\frac{\tau_C}{\tau_L}\right)^2}} \tag{95}$$

- has a maximum for $\frac{\tau_C}{\tau_L} = \sqrt{2}$

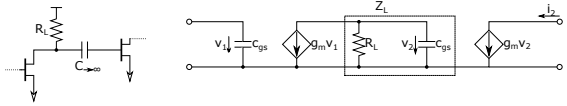


Figure 89: Inductive peaking.

Ways to Improve IF Bandwidth: Inductive Peaking Fig. 89

- inductor L to compensate for load capacitance.
- effect of inductive (shunt) peaking:

$$Y_t = -g_m g_m Z_L = -g_m^2 \frac{R_L\,(1 + j\omega L/R_L)}{1 - \omega^2 C_{gs} L + j\omega R_L C_{gs}} \tag{93}$$

- with $\tau_C = R_L C_{gs} = \frac{1}{\omega_c}$ and $\tau_L = L/R_L$:

$$Y_t = -g_m^2 \frac{R_L\,(1 + j\omega\tau_L)}{1 - \omega^2 \tau_C \tau_L + j\omega\tau_C} \tag{94}$$

- 3-dB corner freq:

$$\omega_{c,l} = \omega_c \sqrt{\left(-\frac{\left(\frac{\tau_C}{\tau_L}\right)^2}{2} + \frac{\tau_C}{\tau_L} + 1\right) + \sqrt{\left(-\frac{\left(\frac{\tau_C}{\tau_L}\right)^2}{2} + \frac{\tau_C}{\tau_L} + 1\right)^2 + \left(\frac{\tau_C}{\tau_L}\right)^2}} \tag{95}$$

- has a maximum for $\frac{\tau_C}{\tau_L} = \sqrt{2}$

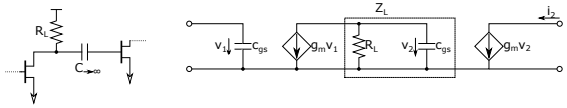


Figure 89: Inductive peaking.

Ways to Improve IF Bandwidth: Inductive Peaking Fig. 89

- inductor L to compensate for load capacitance.
- effect of inductive (shunt) peaking:

$$Y_t = -g_m g_m Z_L = -g_m^2 \frac{R_L\,(1 + j\omega L/R_L)}{1 - \omega^2 C_{gs} L + j\omega R_L C_{gs}} \tag{93}$$

- with $\tau_C = R_L C_{gs} = \frac{1}{\omega_c}$ and $\tau_L = L/R_L$:

$$Y_t = -g_m^2 \frac{R_L\,(1 + j\omega\tau_L)}{1 - \omega^2 \tau_C \tau_L + j\omega\tau_C} \tag{94}$$

- 3-dB corner freq:

$$\omega_{c,l} = \omega_c \sqrt{\left(-\frac{\left(\frac{\tau_C}{\tau_L}\right)^2}{2} + \frac{\tau_C}{\tau_L} + 1\right) + \sqrt{\left(-\frac{\left(\frac{\tau_C}{\tau_L}\right)^2}{2} + \frac{\tau_C}{\tau_L} + 1\right)^2 + \left(\frac{\tau_C}{\tau_L}\right)^2}} \tag{95}$$

- has a maximum for $\frac{\tau_C}{\tau_L} = \sqrt{2}$

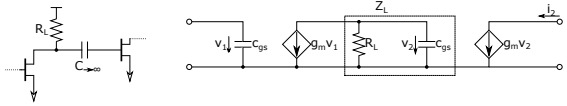


Figure 89: Inductive peaking.

Ways to Improve IF Bandwidth: Inductive Peaking

- normalized frequency: $\nu = \omega/\omega_c$
- normalized transconductance (Fig. 90):

$$Y_t/Y_{t0} = \frac{1 + j\nu c}{1 - \nu^2 c + j\nu} \qquad (96)$$

  with $c = \omega_c \tau_L$
- maximum bandwidth for $c = 1/\sqrt{2}$, BW enhancement ≈ 1.8-fold
- maximum flatness for $c = 1/\left(1 + \sqrt{2}\right)$

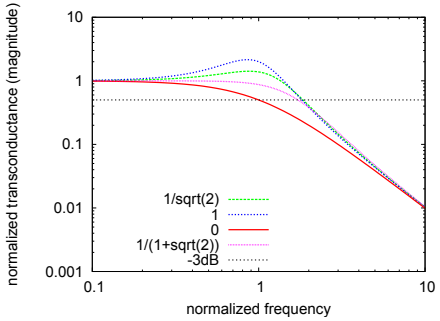


Figure 90: Transconductance with inductive peaking effect.

R-C Bandwidth Limitation

- Consider transimpedance $A_Z = \frac{v_2}{i_1}$ of a general amplifier with an R-C input characteristic (Fig. 91):

$$A_Z = -\frac{R_L g_m R_{in}}{1 + j\omega R_{in} C_{gs}} = -\frac{A_0 R_{in}}{1 + j\frac{\omega}{\omega_c}} \tag{97}$$

- low-frequency voltage gain A_0 (inverting amplifier):

$$A_0 = R_L g_m \tag{98}$$

- single-pole role-off with 3-dB cutoff freq:

$$f_c = \frac{1}{R_{in} C_{gs}} \tag{99}$$

- ways to improve bandwidth:
  - reduce input capacitance: series-series feedback, cascode, travelling-wave amplifier
  - reduce input resistance: $\Rightarrow$ would increase noise $\Rightarrow$ shunt-shunt feedback

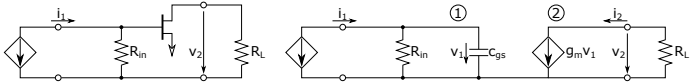


Figure 91: RC bandwidth limitation of a gain cell.

R-C Bandwidth Limitation

- Consider transimpedance $A_Z = \frac{v_2}{i_1}$ of a general amplifier with an R-C input characteristic (Fig. 91):

$$A_Z = -\frac{R_L g_m R_{in}}{1 + j\omega R_{in} C_{gs}} = -\frac{A_0 R_{in}}{1 + j\frac{\omega}{\omega_c}} \tag{97}$$

- low-frequency voltage gain A_0 (inverting amplifier):

$$A_0 = R_L g_m \tag{98}$$

- single-pole role-off with 3-dB cutoff freq:

$$f_c = \frac{1}{R_{in} C_{gs}} \tag{99}$$

- ways to improve bandwidth:
  - reduce input capacitance: series-series feedback, cascode, travelling-wave amplifier
  - reduce input resistance: -> would increase noise -> shunt-shunt feedback

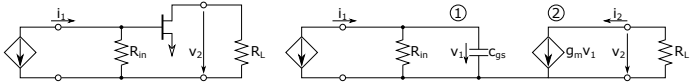


Figure 91: RC bandwidth limitation of a gain cell.

R-C Bandwidth Limitation

- Consider transimpedance $A_Z = \frac{v_2}{i_1}$ of a general amplifier with an R-C input characteristic (Fig. 91):

$$A_Z = -\frac{R_L g_m R_{in}}{1 + j\omega R_{in} C_{gs}} = -\frac{A_0 R_{in}}{1 + j\frac{\omega}{\omega_c}} \tag{97}$$

- low-frequency voltage gain A_0 (inverting amplifier):

$$A_0 = R_L g_m \tag{98}$$

- single-pole role-off with 3-dB cutoff freq:

$$f_c = \frac{1}{R_{in} C_{gs}} \tag{99}$$

- ways to improve bandwidth:
  - reduce input capacitance: series-series feedback, cascode, travelling-wave amplifier
  - reduce input resistance: -> would increase noise -> shunt-shunt feedback

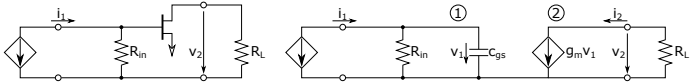


Figure 91: RC bandwidth limitation of a gain cell.

R-C Bandwidth Limitation

- Consider transimpedance $A_Z = \frac{v_2}{i_1}$ of a general amplifier with an R-C input characteristic (Fig. 91):

$$A_Z = -\frac{R_L g_m R_{in}}{1 + j\omega R_{in} C_{gs}} = -\frac{A_0 R_{in}}{1 + j\frac{\omega}{\omega_c}} \tag{97}$$

- low-frequency voltage gain A_0 (inverting amplifier):

$$A_0 = R_L g_m \tag{98}$$

- single-pole role-off with 3-dB cutoff freq:

$$f_c = \frac{1}{R_{in} C_{gs}} \tag{99}$$

- ways to improve bandwidth:
  - reduce input capacitance: series-series feedback, cascode, travelling-wave amplifier
  - reduce input resistance: → would increase noise → shunt-shunt feedback

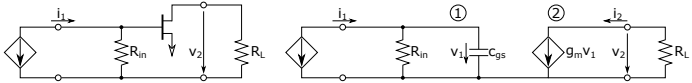


Figure 91: RC bandwidth limitation of a gain cell.

R-C Bandwidth Limitation

- Consider transimpedance $A_Z = \frac{v_2}{i_1}$ of a general amplifier with an R-C input characteristic (Fig. 91):

$$A_Z = -\frac{R_L g_m R_{in}}{1 + j\omega R_{in} C_{gs}} = -\frac{A_0 R_{in}}{1 + j\frac{\omega}{\omega_c}} \tag{97}$$

- low-frequency voltage gain A_0 (inverting amplifier):

$$A_0 = R_L g_m \tag{98}$$

- single-pole role-off with 3-dB cutoff freq:

$$f_c = \frac{1}{R_{in} C_{gs}} \tag{99}$$

- ways to improve bandwidth:
  - reduce input capacitance: series-series feedback, cascode, travelling-wave amplifier
  - reduce input resistance: → would increase noise → shunt-shunt feedback

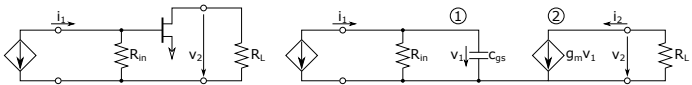


Figure 91: RC bandwidth limitation of a gain cell.

R-C Bandwidth Limitation

- Consider transimpedance $A_Z = \frac{v_2}{i_1}$ of a general amplifier with an R-C input characteristic (Fig. 91):

$$A_Z = -\frac{R_L g_m R_{in}}{1 + j\omega R_{in} C_{gs}} = -\frac{A_0 R_{in}}{1 + j\frac{\omega}{\omega_c}} \tag{97}$$

- low-frequency voltage gain A_0 (inverting amplifier):

$$A_0 = R_L g_m \tag{98}$$

- single-pole role-off with 3-dB cutoff freq:

$$f_c = \frac{1}{R_{in} C_{gs}} \tag{99}$$

- ways to improve bandwidth:
  - reduce input capacitance: series-series feedback, cascode, travelling-wave amplifier
  - reduce input resistance: $\rightarrow$ would increase noise $\rightarrow$ shunt-shunt feedback

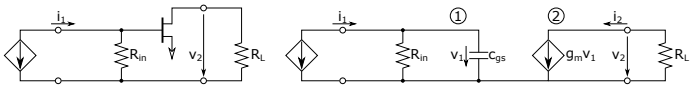


Figure 91: RC bandwidth limitation of a gain cell.

Shunt-Shunt Feedback

- transimpedance $A_Z = \frac{v_2}{i_1}$ including shunt-shunt feedback (Fig. 92):

$$A_Z = -\frac{R_F A_0 \left(1 - \frac{R_L}{R_F A_0}\right)}{1 + A_0} \cdot \frac{1}{1 + j\omega \frac{C_{gs} R_F \left(1 + \frac{R_L}{R_F}\right)}{1 + A_0}} \tag{100}$$

  - increased 3-dB cutoff freq.

$$\omega_c' = \frac{1 + A_0}{C_{gs} R_F \left(1 + \frac{R_L}{R_F}\right)} \gg \omega_c = \frac{1}{R_{in} C_{gs}} \tag{101}$$

  - reduced low-freq. transimpedance:

$$A_{Z,0}' \approx \frac{R_F A_0}{1 + A_0} \approx R_F|_{A_0 \gg 1} \tag{102}$$

- input impedance $Z_{in} = \frac{v_1}{i_1}$:

$$Z_{in} = \frac{R_F}{1 + \frac{A_0 - \frac{R_L}{R_F}}{1 + \frac{R_L}{R_F}} + j\omega R_F C_{gs}} \approx \frac{R_F}{1 + A_0}\Big|_{A_0 \gg 1,\, f \to 0} \tag{103}$$

  - can be made very low, while maintaining low noise contribution from R_F!

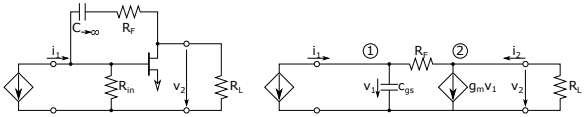


Figure 92: Shunt-shunt feedback.

Shunt-Shunt Feedback

- transimpedance $A_Z = \frac{v_2}{i_1}$ including shunt-shunt feedback (Fig. 92):

$$A_Z = -\frac{R_F A_0 \left(1 - \frac{R_L}{R_F A_0}\right)}{1 + A_0} \cdot \frac{1}{1 + j\omega \frac{C_{gs} R_F \left(1 + \frac{R_L}{R_F}\right)}{1 + A_0}} \tag{100}$$

  - increased 3-dB cutoff freq.

$$\omega_c' = \frac{1 + A_0}{C_{gs} R_F \left(1 + \frac{R_L}{R_F}\right)} \gg \omega_c = \frac{1}{R_{in} C_{gs}} \tag{101}$$

  - reduced low-freq. transimpedance:

$$A_{Z,0}' \approx \frac{R_F A_0}{1 + A_0} \approx R_F|_{A_0 \gg 1} \tag{102}$$

- input impedance $Z_{in} = \frac{v_1}{i_1}$:

$$Z_{in} = \frac{R_F}{1 + \frac{A_0 - \frac{R_L}{R_F}}{1 + \frac{R_L}{R_F}} + j\omega R_F C_{gs}} \approx \frac{R_F}{1 + A_0}\Big|_{A_0 \gg 1,\, f \to 0} \tag{103}$$

  - can be made very low, while maintaining low noise contribution (see 8b)

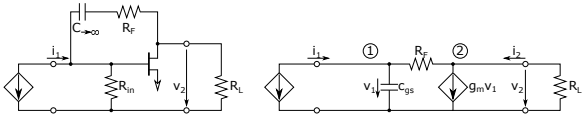


Figure 92: Shunt-shunt feedback.

Shunt-Shunt Feedback

- transimpedance $A_Z = \frac{v_2}{i_1}$ including shunt-shunt feedback (Fig. 92):

$$A_Z = -\frac{R_F A_0 \left(1 - \frac{R_L}{R_F A_0}\right)}{1 + A_0} \cdot \frac{1}{1 + j\omega \frac{C_{gs} R_F \left(1 + \frac{R_L}{R_F}\right)}{1 + A_0}} \tag{100}$$

  - increased 3-dB cutoff freq.

$$\omega_c' = \frac{1 + A_0}{C_{gs} R_F \left(1 + \frac{R_L}{R_F}\right)} \gg \omega_c = \frac{1}{R_{in} C_{gs}} \tag{101}$$

  - reduced low-freq. transimpedance:

$$A_{Z,0}' \approx \frac{R_F A_0}{1 + A_0} \approx R_F|_{A_0 \gg 1} \tag{102}$$

- input impedance $Z_{in} = \frac{v_1}{i_1}$:

$$Z_{in} = \frac{R_F}{1 + \frac{A_0 - \frac{R_L}{R_F}}{1 + \frac{R_L}{R_F}} + j\omega R_F C_{gs}} \approx \frac{R_F}{1 + A_0}\Big|_{A_0 \gg 1,\, f \to 0} \tag{103}$$

  - can be made very low, easily maintaining low noise contribution from R_F

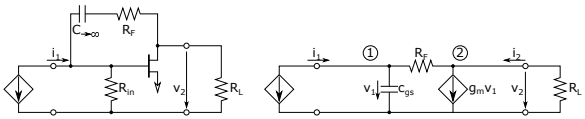


Figure 92: Shunt-shunt feedback.

Shunt-Shunt Feedback

- transimpedance $A_Z = \frac{v_2}{i_1}$ including shunt-shunt feedback (Fig. 92):

$$A_Z = -\frac{R_F A_0 \left(1 - \frac{R_L}{R_F A_0}\right)}{1 + A_0} \cdot \frac{1}{1 + j\omega \frac{C_{gs} R_F \left(1 + \frac{R_L}{R_F}\right)}{1 + A_0}} \tag{100}$$

  - increased 3-dB cutoff freq.

$$\omega_c' = \frac{1 + A_0}{C_{gs} R_F \left(1 + \frac{R_L}{R_F}\right)} \gg \omega_c = \frac{1}{R_{in} C_{gs}} \tag{101}$$

  - reduced low-freq. transimpedance:

$$A_{Z,0}' \approx \frac{R_F A_0}{1 + A_0} \approx R_F|_{A_0 \gg 1} \tag{102}$$

- input impedance $Z_{in} = \frac{v_1}{i_1}$:

$$Z_{in} = \frac{R_F}{1 + \frac{A_0 - \frac{R_L}{R_F}}{1 + \frac{R_L}{R_F}} + j\omega R_F C_{gs}} \approx \frac{R_F}{1 + A_0}\Big|_{A_0 \gg 1,\, f \to 0} \tag{103}$$

  - can be made very low, while maintaining low noise contribution from R_F

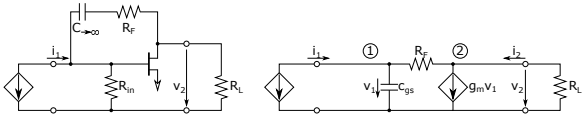


Figure 92: Shunt-shunt feedback.

Shunt-Shunt Feedback

- transimpedance $A_Z = \frac{v_2}{i_1}$ including shunt-shunt feedback (Fig. 92):

$$A_Z = -\frac{R_F A_0 \left(1 - \frac{R_L}{R_F A_0}\right)}{1 + A_0} \cdot \frac{1}{1 + j\omega \frac{C_{gs} R_F \left(1 + \frac{R_L}{R_F}\right)}{1 + A_0}} \tag{100}$$

  - increased 3-dB cutoff freq.

$$\omega_c^{'} = \frac{1 + A_0}{C_{gs} R_F \left(1 + \frac{R_L}{R_F}\right)} \gg \omega_c = \frac{1}{R_{in} C_{gs}} \tag{101}$$

  - reduced low-freq. transimpedance:

$$A_{Z,0}^{'} \approx \frac{R_F A_0}{1 + A_0} \approx R_F|_{A_0 \gg 1} \tag{102}$$

- input impedance $Z_{in} = \frac{v_1}{i_1}$:

$$Z_{in} = \frac{R_F}{1 + \frac{A_0 - \frac{R_L}{R_F}}{1 + \frac{R_L}{R_F}} + j\omega R_F C_{gs}} \approx \frac{R_F}{1 + A_0}\Big|_{A_0 \gg 1,\, f \to 0} \tag{103}$$

  - can be made very low, while maintaining low noise contribution from R_F

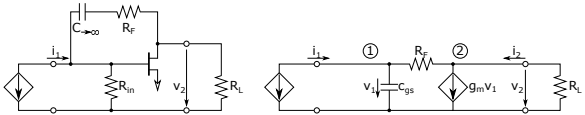


Figure 92: Shunt-shunt feedback.

DC-Coupled Transimpedance Amplifier (Fig. (93))

- DC-coupling to avoid DC decoupling capacitor in series to R_F, which limits the lower cutoff frequency in case of direct drain to gate feedback
- uses DC level shifter diodes (forward-biased) and source follower (common-drain stage)

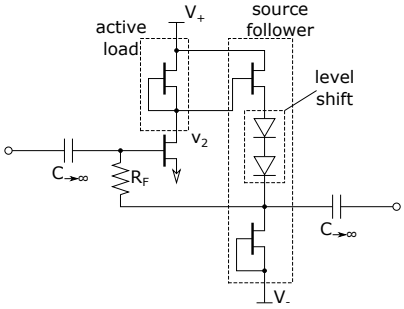


Figure 93: Transimpedance amplifier.

5.1 Applications
5.2 Performance Indicators
5.3 Wideband Matching
5.4 Wideband IF Techniques
5.5 Travelling-Wave Amplifier

Travelling-Wave Amplifier

5.5

Travelling-Wave Amplifier Concept (Fig. (94))

- overcome gain-bandwidth limitation by compensation of the amplifier's in- and output capacitance
- embedding of the amplifier into inductive transmission lines ($Z_L^{'} > Z_0 = 50\Omega$) to form "artificial" transmission lines
- n amplification stages and constructive signal summation (hence also called "distributed amplifier")

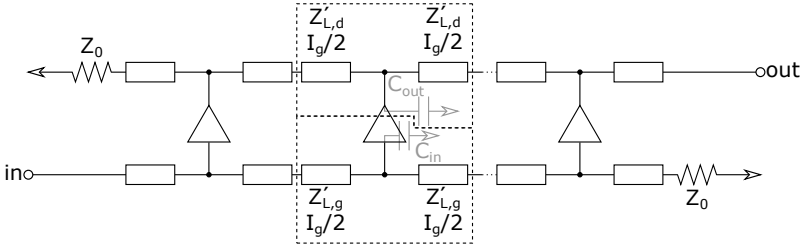


Figure 94: Travelling-wave amplifier.

Travelling-Wave Amplifier Concept (Fig. (94))

- overcome gain-bandwidth limitation by compensation of the amplifier's in- and output capacitance
- embedding of the amplifier into inductive transmission lines ($Z_L^{'} > Z_0 = 50\Omega$) to form "artificial" transmission lines
- n amplification stages and constructive signal summation (hence also called "distributed amplifier")

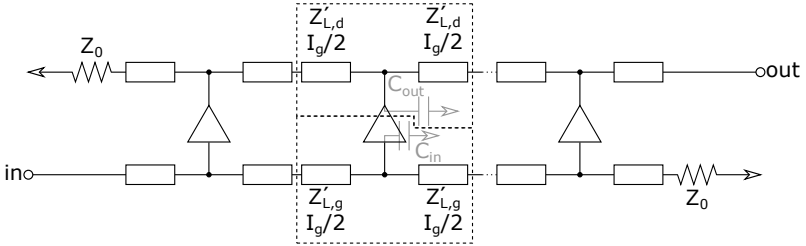


Figure 94: Travelling-wave amplifier.

Travelling-Wave Amplifier Concept (Fig. (94))

- overcome gain-bandwidth limitation by compensation of the amplifier's in- and output capacitance
- embedding of the amplifier into inductive transmission lines ($Z_L^{'} > Z_0 = 50\Omega$) to form "artificial" transmission lines
- n amplification stages and constructive signal summation (hence also called "distributed amplifier")

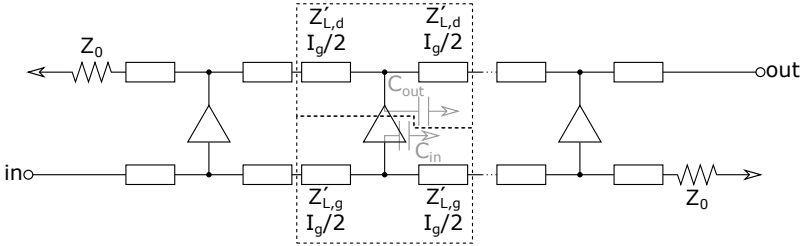


Figure 94: Travelling-wave amplifier.

TWA: Artificial Transmission Line (Fig. (95))

characteristic impedance of the artificial TRL

$$Z_L = \sqrt{\frac{L'}{C' + C_{\text{amp}}/l}} = \frac{Z_L'}{\sqrt{1 + \frac{C_{\text{amp}}}{C' \cdot l}}} \qquad (104)$$

- l ...length of transmission line section
- $Z_L' = \sqrt{\frac{L'}{C'}}$ ...impedance of (inductive) transmission line

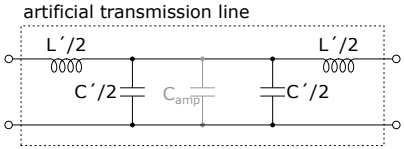


Figure 95: One section of a Travelling-Wave Amplifier (TWA) "artifical" transmission line.

TWA: Artificial Transmission Line (Fig. (95))

phase constant β of the artificial TRL

$$\beta = \omega\sqrt{L'\left(C' + C_{\mathrm{amp}}/l\right)} = \beta'\sqrt{1 + \frac{C_{\mathrm{amp}}}{C' \cdot l}} \tag{105}$$

$$\beta' = \omega\sqrt{L'C'} \tag{106}$$

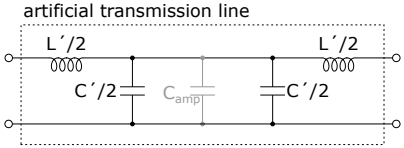


Figure 95: One section of a TWA "artifical" transmission line.

TWA: Artificial Transmission Line (Fig. (95))

design equations for input and output impedance match

$$Z_{L,g} = Z_{L,d} = 50\Omega \tag{107}$$

$$Z_{L,g} = \sqrt{\frac{L'_g}{C'_g + C_{\text{in}}/l_g}} = \frac{Z'_{L,g}}{\sqrt{1 + \frac{C_{\text{in}}}{C'_g \cdot l_g}}} = 50\Omega \tag{108}$$

$$Z_{L,d} = \sqrt{\frac{L'_d}{C'_d + C_{\text{out}}/l_d}} = \frac{Z'_{L,d}}{\sqrt{1 + \frac{C_{\text{out}}}{C'_d \cdot l_d}}} = 50\Omega \tag{109}$$

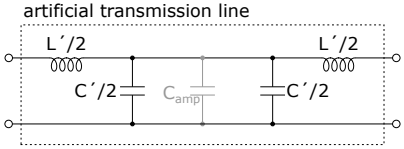


Figure 95: One section of a TWA "artifical" transmission line.

TWA: Artificial Transmission Line (Fig. (95))

design equations for constructive wave interference at output

$$\beta_g l_g = \beta_d l_d \tag{110}$$

$$\beta_g = \omega\sqrt{L'_g\left(C'_g + C_{\text{in}}/l_g\right)} = \beta'_g\sqrt{1 + \frac{C_{\text{in}}}{C'_g \cdot l_g}} \tag{111}$$

$$\beta_d = \omega\sqrt{L'_d\left(C'_d + C_{\text{out}}/l_d\right)} = \beta'_d\sqrt{1 + \frac{C_{\text{out}}}{C'_d \cdot l_d}} \tag{112}$$

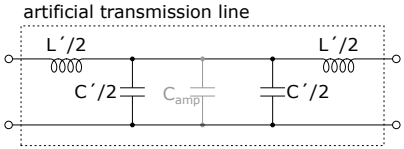


Figure 95: One section of a TWA "artifical" transmission line.

TWA: Artificial Transmission Line (Fig. (95))

- given C_{in} and C_{out}, the equations can be solved iteratively for $Z'_{L,g/d}$ and $l_{g/d}$.

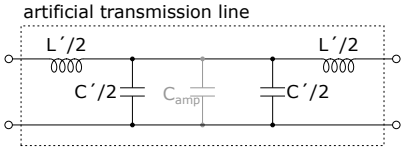


Figure 95: One section of a TWA "artifical" transmission line.

TWA Example (Fig. 96)

- 6-stage TWA in coplanar environment
- note different length and impedance of gate- and drain lines, due to $C_{\text{in}} \neq C_{\text{out}}$

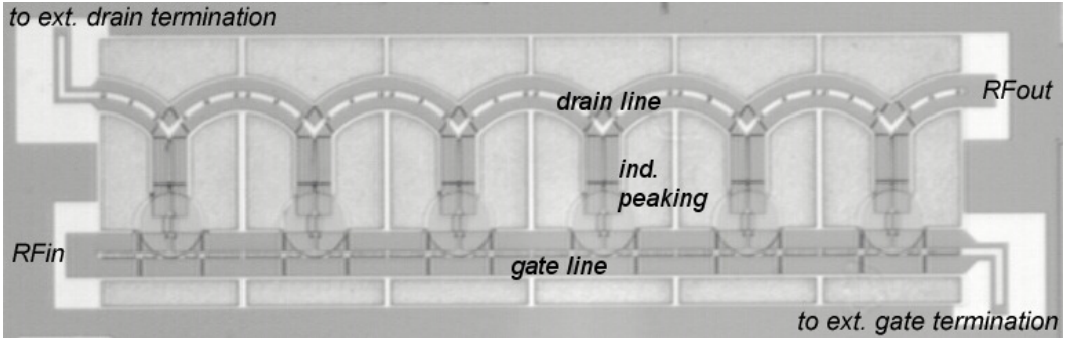


Figure 96: 6-stage uniform TWA.

TWA Example (Fig. 96)

- 6-stage TWA in coplanar environment
- note different length and impedance of gate- and drain lines, due to $C_{\text{in}} \neq C_{\text{out}}$

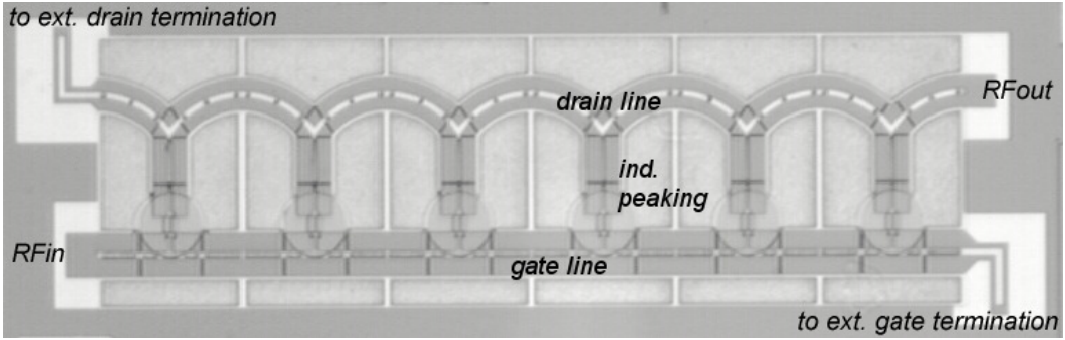


Figure 96: 6-stage uniform TWA.

TWA Gain vs. Number of Stages

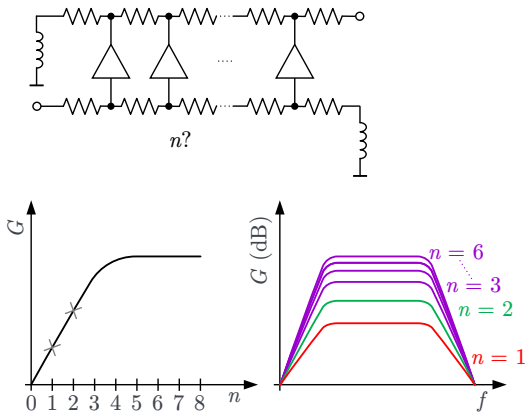


Figure 97: Influence of the number of stages.

Input Capacitive Voltage Divider (Fig. 98)

- capacitive voltage-divider on input line
  - reduced gain
  - Equalize C_{in} and C_{out}
  - popular option for power amplifiers: equalize input power for all stages by $C_{g1} < C_{g2} < .. < C_{gn}$ (non-uniform TWA)

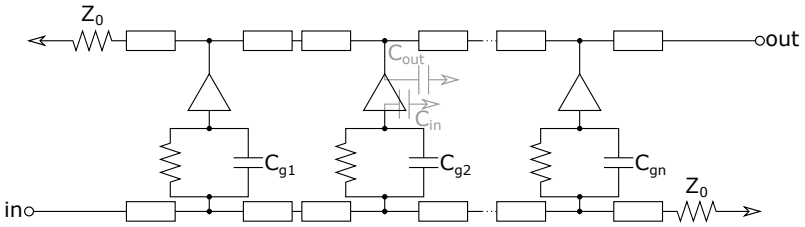


Figure 98: Input capacitive voltage divider.

Input Capacitive Voltage Divider (Fig. 98)

- capacitive voltage-divider on input line
  - reduced gain
  - Equalize C_{in} and C_{out}
  - popular option for power amplifiers: equalize input power for all stages by $C_{g1} < C_{g2} < .. < C_{gn}$ (non-uniform TWA)

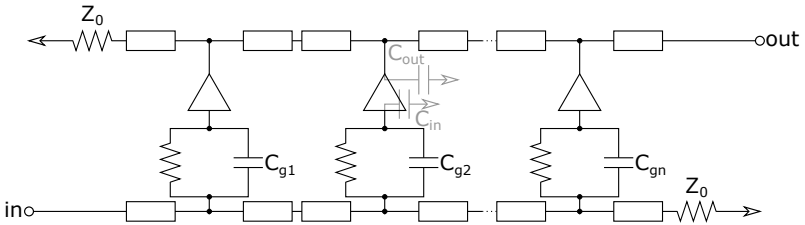


Figure 98: Input capacitive voltage divider.

Input Capacitive Voltage Divider (Fig. 98)

- capacitive voltage-divider on input line
  - reduced gain
  - Equalize C_{in} and C_{out}
  - popular option for power amplifiers: equalize input power for all stages by $C_{g1} < C_{g2} < .. < C_{gn}$ (non-uniform TWA)

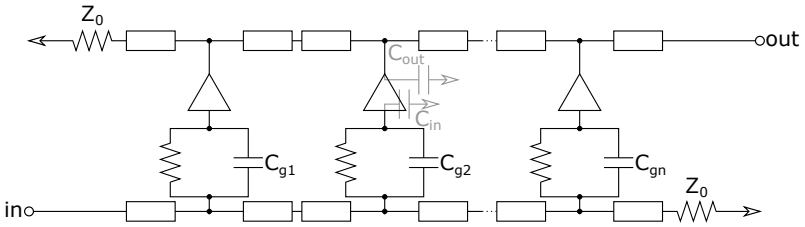


Figure 98: Input capacitive voltage divider.

Input Capacitive Voltage Divider (Fig. 98)

- capacitive voltage-divider on input line
  - reduced gain
  - Equalize C_{in} and C_{out}
  - popular option for power amplifiers: equalize input power for all stages by $C_{g1} < C_{g2} < .. < C_{gn}$ (non-uniform TWA)

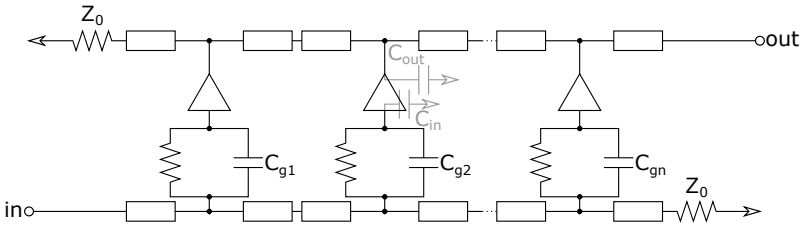


Figure 98: Input capacitive voltage divider.

Providing Bias to a TWA (Fig. 99)

gate bias

- provided by large resistor

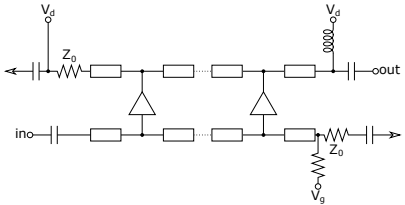


Figure 99: Biasing a TWA.

Providing Bias to a TWA (Fig. 99)

drain bias

- provided by (external) bias-T
- provided via drain terminating resistor (power loss)

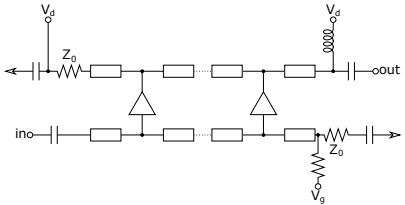


Figure 99: Biasing a TWA.

Providing Bias to a TWA (Fig. 99)

low frequency termination

- shunting and coupling C's need to be large for the TWA to work at low frequencies (kHz-MHz)
- difficult on-wafer (resonance, wafer space)

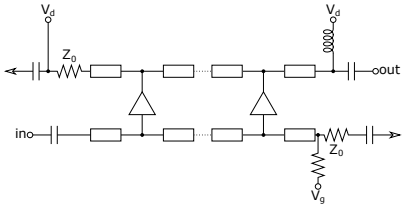


Figure 99: Biasing a TWA.

High-Pass TWA (Fig. 100)

- include amplifier capacitance into high-pass TRL
- inherent bias circuit: supplied via gate and drain inductors

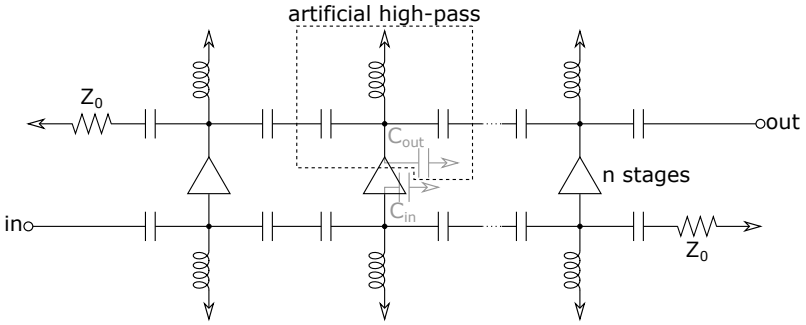


Figure 100: High-pass transmission line TWA.

High-Pass TWA (Fig. 100)

- include amplifier capacitance into high-pass TRL
- inherent bias circuit: supplied via gate and drain inductors

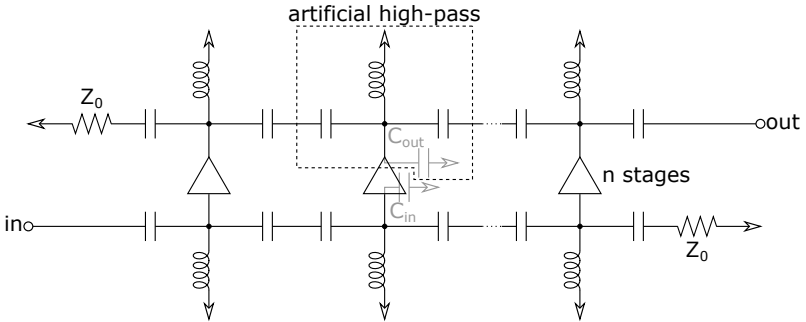


Figure 100: High-pass transmission line TWA.

Loss in TWA (Fig. 101)

loss in input line

$$Y_{in} = \frac{1}{R_{in} + \frac{1}{j\omega C_{in}}} = \frac{\omega^2 R_{in} C_{in}^2}{1 + \omega^2 R_{in}^2 C_{in}^2} + j\omega \frac{C_{in}}{1 + \omega^2 R_{in}^2 C_{in}^2} \tag{113}$$

- real part increases with $\approx \omega^2$

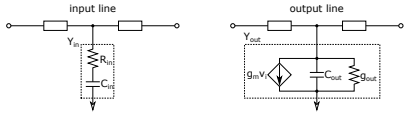


Figure 101: Input and output loss of a TWA gain cell.

Loss in TWA (Fig. 101)

loss in output line

$$Y_{out} = g_{out} + j\omega C_{out} \tag{114}$$

- frequency independent loss due to finite output conductance

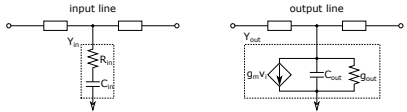


Figure 101: Input and output loss of a TWA gain cell.

Loss in TWA (Fig. 101)

transmission line loss

- skin effect, dielectric loss etc.

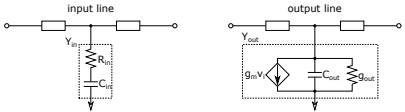


Figure 101: Input and output loss of a TWA gain cell.

Broadband vs. Narrowband Amplifiers I

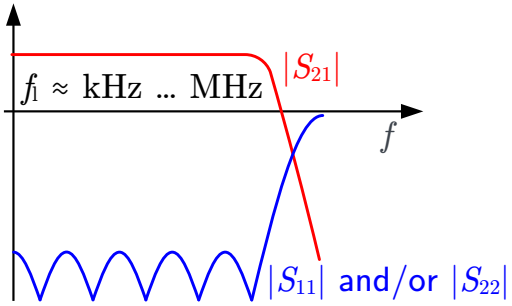


Figure 102: Idealized S-parameters of a TWA.

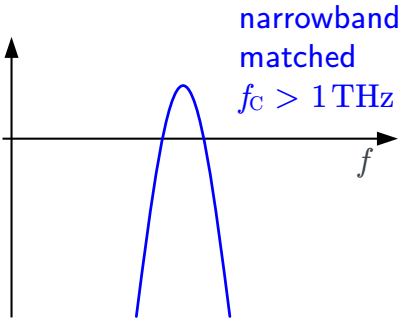


Figure 103: Narrowband match.

Amplification Stage: Common-Source FET(Fig. 104)

unilateral, $C_{gd} = 0$

$$\mathrm{Im}\,(\mathrm{Y_{in}}) = \mathrm{j}\omega C_{gs} \rightarrow C_{in} = C_{gs} \tag{115}$$

- idem:

$$C_{out} = C_{ds} \tag{116}$$

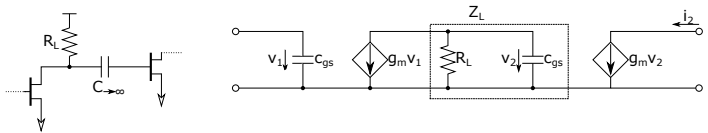


Figure 104: Low frequency equivalent circuit of a TWA gain cell with common-source transistor.

Amplification Stage: Common-Source FET(Fig. 104)

input admittance Y_{in}, if C_{gd} is considered$^{(*)}$

$$i_{in} = v_{in} j\omega C_{gs} + (v_{in} - v_{out})\, j\omega C_{gd} \tag{117}$$

- with $R_l^* = r_{ds} \parallel 50\Omega \parallel R_l$:

$$v_{out} = R_l^* \left(-g_m v_{in} + (v_{in} - v_{out})\, j\omega C_{gd} - v_{out} j\omega C_{ds}\right) \tag{118}$$

$$Y_{in} = j\omega \left(C_{gs} + C_{gd}\left(1 + g_m R_l^* \frac{1 - j\omega C_{gd}/g_m R_l^*}{1 + j\omega R_l^* (C_{ds} + C_{gd})}\right)\right) \tag{119}$$

- with $A_0 = g_m R_l^*$ for $\omega R_l^* \left(C_{ds} + C_{gd}\right) \ll 1$:

$$Y_{in} \approx j\omega \left(C_{gs} + C_{gd}\,(1 + A_0)\right) \tag{120}$$

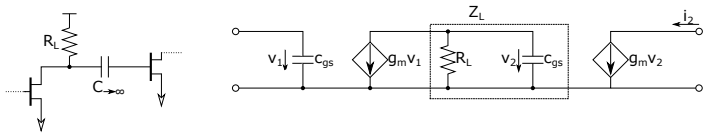


Figure 104: Low frequency equivalent circuit of a TWA gain cell with common-source transistor.

The Miller Effect (Fig. 105)

- input capacitance:

$$C_{in} \approx C_{gs} + C_{gd}\,(1 + A_0) \qquad (121)$$

  with $A_0 = g_m R_l^*$
- the gate-drain capacitance appears multiplied by the voltage amplification at the transistor input
- avoid Miller-effect $\rightarrow$ Cascode topology

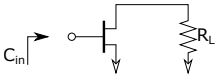


Figure 105: Miller-effect.

The Cascode (Fig. 106)

cascode topology

- cascade of common-source and common-gate transistor (without interstage matching)

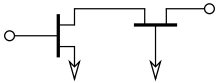


Figure 106: “Cascode” connection of a common-source and common-gate transistor.

The Cascode (Fig. 106)

advantages

- A_v of CS device $\approx 1 \rightarrow$ reduced Miller-effect
- significantly increased output resistance (cp. drain line loss in TWAs)
- significantly increased isolation ($-S_{12}$)
- increased gain

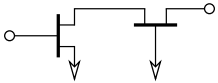


Figure 106: "Cascode" connection of a common-source and common-gate transistor.

The Cascode (Fig. 106)

disadvantages

- risk of instability
- doubles the power consumption → reduced efficiency

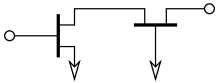


Figure 106: “Cascode” connection of a common-source and common-gate transistor.

Cascode Bias (Fig. 107)

bias conditions

$$I_{d1} = I_{d2} = I_d \tag{122}$$

- if class A:

$$V_{DS1} = V_{DS2} = V_{DS,sat} \tag{123}$$

- then, if $W_{g1} = W_{g2}$ (symmetrical cascode):

$$V_{gs1} \approx V_{gs2} \tag{124}$$

$$V_{D1} = V_{G2} - V_{G1} \tag{125}$$

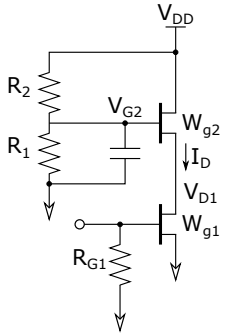


Figure 107: Cascode biasing.

Cascode Bias (Fig. 107)

CG gate bias V_{G2}

- supplied directly or
- via resistive voltage divider from drain supply

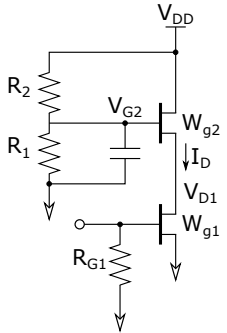


Figure 107: Cascode biasing.

Cascode Gain (Fig. 108)

reduced Miller-effect

- for $r_{ds} > R_l$:

$$\frac{v_i}{v_1} \approx -\left(1 + R_l g_{ds}\right) \approx -1 \tag{126}$$

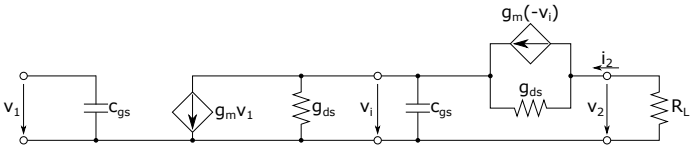


Figure 108: Equivalent circuit for cascode gain analysis.

Cascode Gain (Fig. 108)

cascode gain

- intrinsic gain:

$$A_v = \left.\frac{v_2}{v_1}\right|_{R_l=\infty} = -\frac{g_m\left(g_m + g_{ds}\right)}{g_{ds}\left(g_{ds} + j\omega C_{gs}\right)} \approx \frac{g_m^2}{g_{ds}^2} \tag{127}$$

- including load:

$$A_v = -\frac{g_m R_l}{1 + \left(g_{ds}R_l + j\omega C_{gs}R_l\right)\left(\frac{g_{ds}+\frac{1}{R_l}}{g_{ds}+g_m}\right)} \tag{128}$$

- compare to CS:

$$A_{v,CS} = -\frac{g_m R_l}{1 + g_{ds}R_l} \tag{129}$$

- $|A_{v,Casc}| > |A_{v,CS}|$

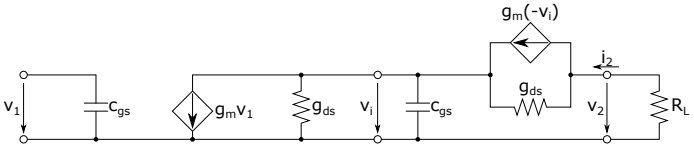


Figure 108: Equivalent circuit for cascode gain analysis.

Cascode Output Impedance (Fig. 109)

cascode output impedance

$$Z_{out} = \left.\frac{v_2}{i_2}\right|_{v_1=0} = \left(\frac{g_{ds}}{1+\frac{g_m+g_{ds}}{g_{ds}+j\omega C_{gs}}}\right)^{-1} \gg (g_{ds})^{-1} \tag{130}$$

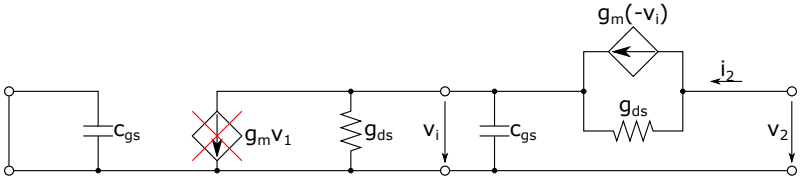


Figure 109: Equivalent circuit for cascode output impedance analysis.

Cascode Isolation (Fig. 110)

- e.g. reverse short-circuit transadmittance $\left.\frac{i_1}{v_2}\right|_{v_1=0}$
  - CS:

$$\left.\frac{i_1}{v_2}\right|_{v_1=0} = -j\omega C_{gd} \tag{131}$$

  - cascode:

$$\left.\frac{i_1}{v_2}\right|_{v_1=0} = -j\omega C_{gd}\left(\frac{1}{2+\frac{g_m+j\omega C_{gs}}{g_{ds}}}\right) \tag{132}$$

  $\rightarrow$ effectively diminished C_{gd}

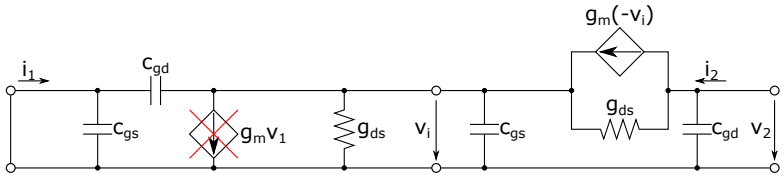


Figure 110: Equivalent circuit for cascode isolation analysis.

Cascode Output Impedance (Fig. 111)

- negative real part → requires stabilization

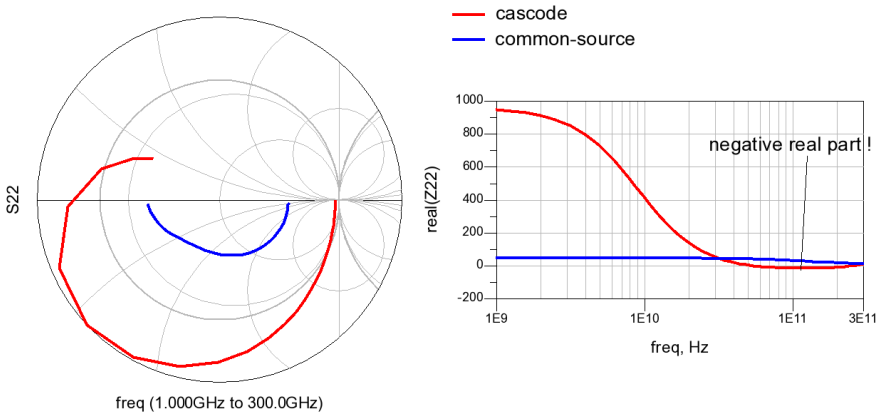


Figure 111: Cascode output conductance. Simulation using 2x30µm IAF 100 nm MHEMTs.

Cascode Isolation (Fig. 112)

- signficantly increased isolation at low frequencies, but reduced effect at high frequencies

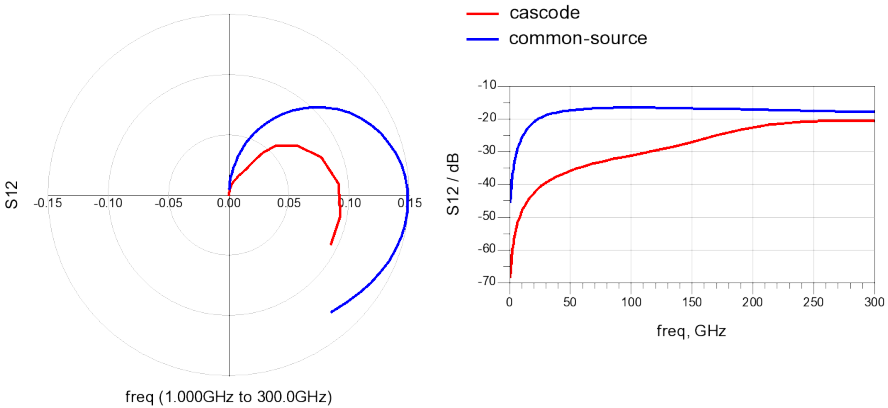


Figure 112: Cascode isolation. Simulation using 2x30µm IAF 100 nm MHEMTs.

Cascode Gain (Fig. 113)

- MSG / MAG in $50\,\Omega$ load larger in cascode, but not double
- close to $f_{\max}$ the cascode gain decreases faster than in CS

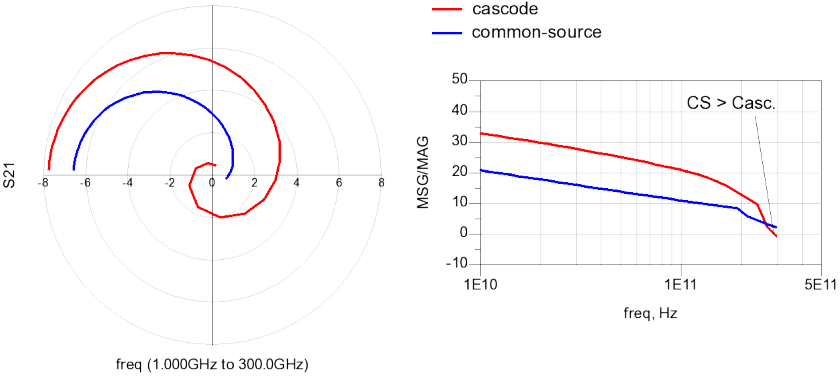


Figure 113: Cascode gain. Simulation using 2x30µm IAF 100 nm MHEMTs.

Cascode Stabilization

resistive gate feedback in CG (Fig. 114)

- avoid neative real part of output impedance by proper choice of R_g and C_g
- but: reduced gain (Fig. 115)

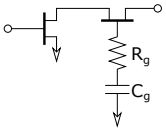


Figure 114: Cascode stabilization via resistive gate feedback in CG.

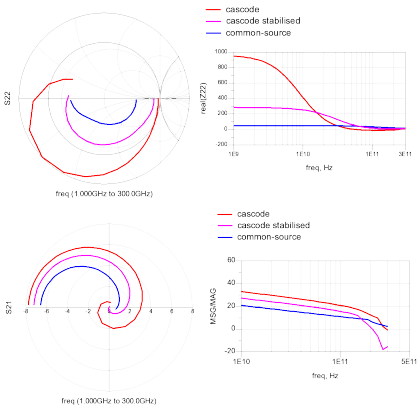


Figure 115: Cacode stabilization - S-parameters.

Cascode Stabilization

other possibilities

- inductive source feedback in CS or in CG device

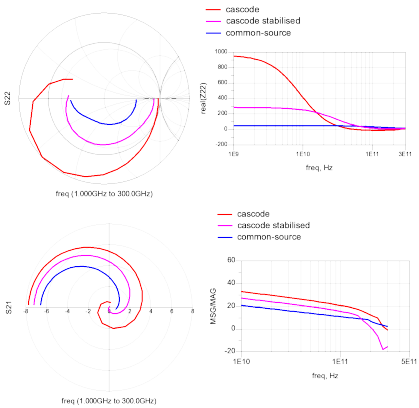


Figure 115: Cacode stabilization - S-parameters.

TWA Design Sequence

- gain cell
  - stable cascode
  - equalize C_{in} and C_{out}?
- design of gate and drain lines: $Z^*_{L,g/d}$ and $l_{g/d}$
  - use CAD optimizer with starting values from theory
- decide on number of stages
  - required gain
  - benötigte Ausgangsleistung
- design bias networks and low-frequency terminations
- possible modifications
  - group delay improvement by unequal TRL length of individual stages (non-uniform TWA)

TWA Design Sequence

- gain cell
  - stable cascode
  - equalize C_{in} and C_{out}?
- design of gate and drain lines: $Z^*_{L,g/d}$ and $l_{g/d}$
  - use CAD optimizer with starting values from theory
- decide on number of stages
  - required gain
  - benötigte Ausgangsleistung
- design bias networks and low-frequency terminations
- possible modifications
  - group delay improvement by unequal TRL length of individual stages (non-uniform TWA)

TWA Design Sequence

- gain cell
  - stable cascode
  - equalize C_{in} and C_{out}?
- design of gate and drain lines: $Z^*_{L,g/d}$ and $l_{g/d}$
  - use CAD optimizer with starting values from theory
- decide on number of stages
  - required gain
  - benötigte Ausgangsleistung
- design bias networks and low-frequency terminations
- possible modifications
  - group delay improvement by unequal TRL length of individual stages (non-uniform TWA)

TWA Design Sequence

- gain cell
  - stable cascode
  - equalize C_{in} and C_{out}?
- design of gate and drain lines: $Z^*_{L,g/d}$ and $l_{g/d}$
  - use CAD optimizer with starting values from theory
- decide on number of stages
  - required gain
  - benötigte Ausgangsleistung
- design bias networks and low-frequency terminations
- possible modifications
  - group delay improvement by unequal TRL length of individual stages (non-uniform TWA)

TWA Design Sequence

- gain cell
  - stable cascode
  - equalize C_{in} and C_{out}?
- design of gate and drain lines: $Z^*_{L,g/d}$ and $l_{g/d}$
  - use CAD optimizer with starting values from theory
- decide on number of stages
  - required gain
  - benötigte Ausgangsleistung
- design bias networks and low-frequency terminations
- possible modifications
  - group delay improvement by unequal TRL length of individual stages (non-uniform TWA)

TWA Design Sequence

- gain cell
  - stable cascode
  - equalize C_{in} and C_{out}?
- design of gate and drain lines: $Z^*_{L,g/d}$ and $l_{g/d}$
  - use CAD optimizer with starting values from theory
- decide on number of stages
  - required gain
  - benötigte Ausgangsleistung
- design bias networks and low-frequency terminations
- possible modifications
  - group delay improvement by unequal TRL length of individual stages (non-uniform TWA)

TWA Design Sequence

- gain cell
  - stable cascode
  - equalize C_{in} and C_{out}?
- design of gate and drain lines: $Z^*_{L,g/d}$ and $l_{g/d}$
  - use CAD optimizer with starting values from theory
- decide on number of stages
  - required gain
  - benötigte Ausgangsleistung
- design bias networks and low-frequency terminations
- possible modifications
  - group delay improvement by unequal TRL length of individual stages (non-uniform TWA)

TWA Design Sequence

- gain cell
  - stable cascode
  - equalize C_{in} and C_{out}?
- design of gate and drain lines: $Z^*_{L,g/d}$ and $l_{g/d}$
  - use CAD optimizer with starting values from theory
- decide on number of stages
  - required gain
  - benötigte Ausgangsleistung
- design bias networks and low-frequency terminations
- possible modifications
  - group delay improvement by unequal TRL length of individual stages (non-uniform TWA)

TWA Design Sequence

- gain cell
  - stable cascode
  - equalize C_{in} and C_{out}?
- design of gate and drain lines: $Z^*_{L,g/d}$ and $l_{g/d}$
  - use CAD optimizer with starting values from theory
- decide on number of stages
  - required gain
  - benötigte Ausgangsleistung
- design bias networks and low-frequency terminations
- possible modifications
  - group delay improvement by unequal TRL length of individual stages (non-uniform TWA)

TWA Design Sequence

- gain cell
  - stable cascode
  - equalize C_{in} and C_{out}?
- design of gate and drain lines: $Z^*_{L,g/d}$ and $l_{g/d}$
  - use CAD optimizer with starting values from theory
- decide on number of stages
  - required gain
  - benötigte Ausgangsleistung
- design bias networks and low-frequency terminations
- possible modifications
  - group delay improvement by unequal TRL length of individual stages (non-uniform TWA)

TWA Design Sequence

- gain cell
  - stable cascode
  - equalize C_{in} and C_{out}?
- design of gate and drain lines: $Z^*_{L,g/d}$ and $l_{g/d}$
  - use CAD optimizer with starting values from theory
- decide on number of stages
  - required gain
  - benötigte Ausgangsleistung
- design bias networks and low-frequency terminations
- possible modifications
  - group delay improvement by unequal TRL length of individual stages (non-uniform TWA)

University of Stuttgart
Germany

Institute of
Robust Power
Semiconductor Systems

Power Amplifiers

6

1 Course Outline

2 The Millimeterwave Frequency Spectrum: Applicatio
and Technologies

3 Functional Buildung Blocks of MMICs

4 Low Noise Amplifier

5 Broadband Amplifiers

6 Power Amplifiers

7 Microwave Network Analysis (Appendix)

6.1 Applications
6.2 Performance Indicators
6.3 Design Methods

Applications

6.1

Radar (Fig. 116)

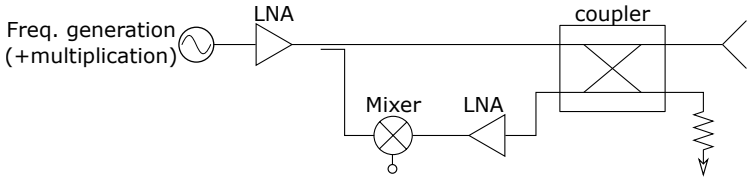


Figure 116: Generic monostatic FMCW radar frontend.

(saturated) output power

- the output power of the PA stage determines the detection range.

further purposes

- provide LO power to mixer
- enhance output power of frequency multipliers and multiplier chains

solid state power amplifier (SSPA)

- in hybrid integrated transceivers, where the PA is a stand-alone components, such as in satellite payloads, an SSPA designates a PA built with semiconductor technology

Communication (Fig. 117)

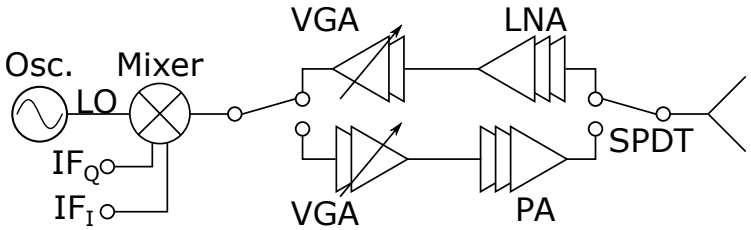


Figure 117: Generic wireless transceiver analog frontend.

linear output power

- modulation format dictates linearity requirements (backoff from power compression)
- avoid signal distortion, intermodulation and harmonic frequencies

efficiency

- in hand-held/portable/battery-powered devices
- reduce cooling (e.g. in base-stations)

6.1 Applications
6.2 Performance Indicators
6.3 Design Methods

Performance Indicators

6.2

Gain Compression

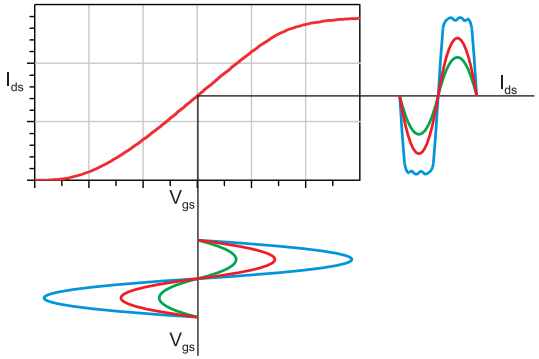


Figure 118: Gain compression due to nonlinear transfer characteristics.

gain compression (Fig. 118)

- signal clipping at $V_{\mathrm{gs}}{=}V_{\mathrm{th}}$, $V_{\mathrm{gs}}{=}V_{\mathrm{d}}$ and $V_{\mathrm{ds}}{=}V_{knee}$
- gain reduces at high power levels
- signal distortion and harmonic frequencies appear in the output spectrum

Gain Compression

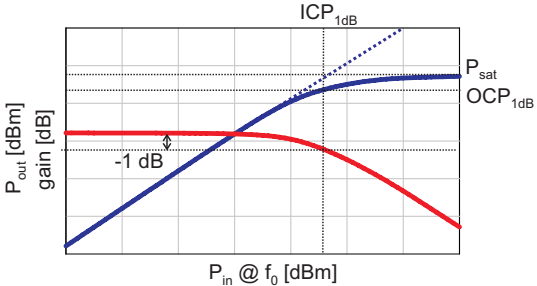


Figure 119: 1-tone power sweep.

1-tone power sweep (Fig. 119)

- saturated output power P_{sat}
- 1-dB compression point
  - output-related OCP_{1dB}
  - input-related ICP_{1dB}

Generation of Harmonics

- due to non-linearity energy is converted into higher order harmonics
- under weak non-linearity, harmonics rise with a constant slope:

$$\Delta H1 \approx 3\,\mathrm{dB/octave}|_{\text{weak non-linearity}} \quad (133)$$

$$\Delta H2 \approx 6\,\mathrm{dB/octave}|_{\text{weak non-linearity}} \quad (134)$$

$$\Delta H3 \approx 9\,\mathrm{dB/octave}|_{\text{weak non-linearity}} \quad (135)$$

- note: this is an approximation, since higher-order non-linearities affect also the lower-order harmonics (cp. analysis)

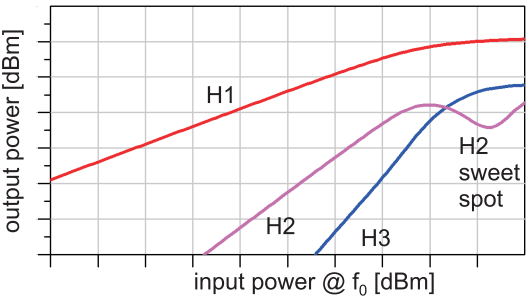


Figure 120: Harmonic generation.

Generation of Harmonics

- under strong non-linearity, there may occur “sweet spots” for the suppression of individual harmonics
- e.g. suppression of even-order harmonics under symmtrical compression of a class-A amplifier
- these sweet spots are functions of input power, bias and frequency

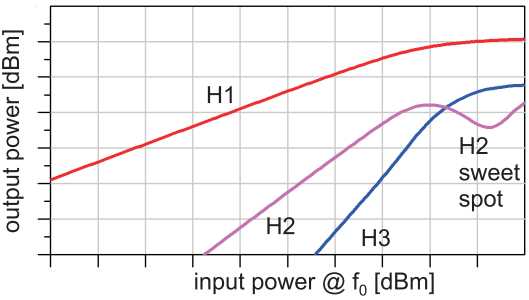


Figure 120: Harmonic generation.

- Taylor series expansion of a nonlinear transfer function
- example: nonlinear current-voltage characteristics

$$\begin{aligned} I(V) &= \sum_{i=0}^{i=\infty} \frac{I^{(i)}(V_0)}{i!} (V - V_0)^i \\ &= I(V_0) + \frac{dI(V_0)}{dV} (V - V_0) + \frac{1}{2!} \frac{d^2 I(V_0)}{dV^2} (V - V_0)^2 + \frac{1}{3!} \frac{d^3 I(V_0)}{dV^3} (V - V_0)^3 \ldots \end{aligned} \tag{136}$$

- V_0, $I_0 = I(V_0)$ ... quiescent point
- for the sake of simplicity, consider $V_0 = 0$ in the following analysis
- 1-tone power sweep: $V = v_1 \sin \omega_1 t$
-

$$I(V) = I_0 + \frac{dI}{dV} v_1 \sin \omega_1 t + \frac{1}{2} \frac{d^2 I}{dV^2} (v_1 \sin \omega_1 t)^2 + \frac{1}{6} \frac{d^3 I}{dV^3} (v_1 \sin \omega_1 t)^3 \ldots \tag{137}$$

- Trigonometric identities - power reduction

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x) \tag{138}$$

$$\sin^3 x = \frac{1}{4} (3 \sin x - \sin 3x) \tag{139}$$

- Taylor series expansion of a nonlinear transfer function
- example: nonlinear current-voltage characteristics

$$\begin{aligned} I(V) &= \sum_{i=0}^{i=\infty} \frac{I^{(i)}(V_0)}{i!} (V - V_0)^i \\ &= I(V_0) + \frac{dI(V_0)}{dV} (V - V_0) + \frac{1}{2!} \frac{d^2 I(V_0)}{dV^2} (V - V_0)^2 + \frac{1}{3!} \frac{d^3 I(V_0)}{dV^3} (V - V_0)^3 \ldots \end{aligned} \tag{136}$$

- V_0, $I_0 = I(V_0)$ ... quiescent point
- for the sake of simplicity, consider $V_0 = 0$ in the following analysis
- 1-tone power sweep: $V = v_1 \sin \omega_1 t$
-

$$I(V) = I_0 + \frac{dI}{dV} v_1 \sin \omega_1 t + \frac{1}{2} \frac{d^2 I}{dV^2} (v_1 \sin \omega_1 t)^2 + \frac{1}{6} \frac{d^3 I}{dV^3} (v_1 \sin \omega_1 t)^3 \ldots \tag{137}$$

- Trigonometric identities - power reduction

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x) \tag{138}$$

$$\sin^3 x = \frac{1}{4} (3 \sin x - \sin 3x) \tag{139}$$

- Taylor series expansion of a nonlinear transfer function
- example: nonlinear current-voltage characteristics

$$\begin{aligned} I(V) &= \sum_{i=0}^{i=\infty} \frac{I^{(i)}(V_0)}{i!} (V - V_0)^i \\ &= I(V_0) + \frac{dI(V_0)}{dV} (V - V_0) + \frac{1}{2!} \frac{d^2 I(V_0)}{dV^2} (V - V_0)^2 + \frac{1}{3!} \frac{d^3 I(V_0)}{dV^3} (V - V_0)^3 \ldots \end{aligned} \tag{136}$$

- V_0, $I_0 = I(V_0)$ ... quiescent point
- for the sake of simplicity, consider $V_0 = 0$ in the following analysis
- 1-tone power sweep: $V = v_1 \sin \omega_1 t$
-

$$I(V) = I_0 + \frac{dI}{dV} v_1 \sin \omega_1 t + \frac{1}{2} \frac{d^2 I}{dV^2} (v_1 \sin \omega_1 t)^2 + \frac{1}{6} \frac{d^3 I}{dV^3} (v_1 \sin \omega_1 t)^3 \ldots \tag{137}$$

- Trigonometric identities - power reduction

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x) \tag{138}$$

$$\sin^3 x = \frac{1}{4} (3 \sin x - \sin 3x) \tag{139}$$

- Taylor series expansion of a nonlinear transfer function
- example: nonlinear current-voltage characteristics

$$\begin{aligned} I(V) &= \sum_{i=0}^{i=\infty} \frac{I^{(i)}(V_0)}{i!} (V - V_0)^i \\ &= I(V_0) + \frac{dI(V_0)}{dV} (V - V_0) + \frac{1}{2!} \frac{d^2 I(V_0)}{dV^2} (V - V_0)^2 + \frac{1}{3!} \frac{d^3 I(V_0)}{dV^3} (V - V_0)^3 \ldots \end{aligned} \tag{136}$$

- V_0, $I_0 = I(V_0)$ ... quiescent point
- for the sake of simplicity, consider $V_0 = 0$ in the following analysis
- 1-tone power sweep: $V = v_1 \sin \omega_1 t$
-

$$I(V) = I_0 + \frac{dI}{dV} v_1 \sin \omega_1 t + \frac{1}{2} \frac{d^2 I}{dV^2} (v_1 \sin \omega_1 t)^2 + \frac{1}{6} \frac{d^3 I}{dV^3} (v_1 \sin \omega_1 t)^3 \ldots \tag{137}$$

- Trigonometric identities - power reduction

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x) \tag{138}$$

$$\sin^3 x = \frac{1}{4} (3 \sin x - \sin 3x) \tag{139}$$

- Taylor series expansion of a nonlinear transfer function
- example: nonlinear current-voltage characteristics

$$\begin{aligned} I(V) &= \sum_{i=0}^{i=\infty} \frac{I^{(i)}(V_0)}{i!} (V - V_0)^i \\ &= I(V_0) + \frac{dI(V_0)}{dV} (V - V_0) + \frac{1}{2!} \frac{d^2 I(V_0)}{dV^2} (V - V_0)^2 + \frac{1}{3!} \frac{d^3 I(V_0)}{dV^3} (V - V_0)^3 \ldots \end{aligned} \tag{136}$$

- V_0, $I_0 = I(V_0)$ ... quiescent point
- for the sake of simplicity, consider $V_0 = 0$ in the following analysis
- 1-tone power sweep: $V = v_1 \sin \omega_1 t$
-

$$I(V) = I_0 + \frac{dI}{dV} v_1 \sin \omega_1 t + \frac{1}{2} \frac{d^2 I}{dV^2} (v_1 \sin \omega_1 t)^2 + \frac{1}{6} \frac{d^3 I}{dV^3} (v_1 \sin \omega_1 t)^3 \ldots \tag{137}$$

- Trigonometric identities - power reduction

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x) \tag{138}$$

$$\sin^3 x = \frac{1}{4} (3 \sin x - \sin 3x) \tag{139}$$

- Taylor series expansion of a nonlinear transfer function
- example: nonlinear current-voltage characteristics

$$\begin{aligned} I(V) &= \sum_{i=0}^{i=\infty} \frac{I^{(i)}(V_0)}{i!} (V - V_0)^i \\ &= I(V_0) + \frac{dI(V_0)}{dV} (V - V_0) + \frac{1}{2!} \frac{d^2 I(V_0)}{dV^2} (V - V_0)^2 + \frac{1}{3!} \frac{d^3 I(V_0)}{dV^3} (V - V_0)^3 \ldots \end{aligned} \tag{136}$$

- V_0, $I_0 = I(V_0)$ ... quiescent point
- for the sake of simplicity, consider $V_0 = 0$ in the following analysis
- 1-tone power sweep: $V = v_1 \sin \omega_1 t$
-

$$I(V) = I_0 + \frac{dI}{dV} v_1 \sin \omega_1 t + \frac{1}{2} \frac{d^2 I}{dV^2} (v_1 \sin \omega_1 t)^2 + \frac{1}{6} \frac{d^3 I}{dV^3} (v_1 \sin \omega_1 t)^3 \ldots \tag{137}$$

- Trigonometric identities - power reduction

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x) \tag{138}$$

$$\sin^3 x = \frac{1}{4} (3 \sin x - \sin 3x) \tag{139}$$

- Taylor series expansion of a nonlinear transfer function
- example: nonlinear current-voltage characteristics

$$\begin{aligned} I(V) &= \sum_{i=0}^{i=\infty} \frac{I^{(i)}(V_0)}{i!} (V - V_0)^i \\ &= I(V_0) + \frac{dI(V_0)}{dV} (V - V_0) + \frac{1}{2!} \frac{d^2 I(V_0)}{dV^2} (V - V_0)^2 + \frac{1}{3!} \frac{d^3 I(V_0)}{dV^3} (V - V_0)^3 \ldots \end{aligned} \tag{136}$$

- V_0, $I_0 = I(V_0)$ ... quiescent point
- for the sake of simplicity, consider $V_0 = 0$ in the following analysis
- 1-tone power sweep: $V = v_1 \sin \omega_1 t$
-

$$I(V) = I_0 + \frac{dI}{dV} v_1 \sin \omega_1 t + \frac{1}{2} \frac{d^2 I}{dV^2} (v_1 \sin \omega_1 t)^2 + \frac{1}{6} \frac{d^3 I}{dV^3} (v_1 \sin \omega_1 t)^3 \ldots \tag{137}$$

- Trigonometric identities - power reduction

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x) \tag{138}$$

$$\sin^3 x = \frac{1}{4} (3 \sin x - \sin 3x) \tag{139}$$

- after power reduction:

$$I = I_0 + \frac{dI}{dV} v_1 \sin\omega_1 t + \frac{1}{2}\frac{d^2I}{dV^2} v_1^2 \frac{1}{2}\left(1 - \cos 2\omega_1 t\right) + \frac{1}{6}\frac{d^3I}{dV^3} v_1^3 \frac{1}{4}\left(3\sin\omega_1 t - \sin 3\omega_1 t\right) \ldots \tag{140}$$

- arranged for frequency components:

$$I = I_0 + \frac{v_1^2}{4}\frac{d^2I}{dV^2} + \left(v_1 \frac{dI}{dV} + \frac{v_1^3}{8}\frac{d^3I}{dV^3}\right)\sin\omega_1 t - \frac{v_1^2}{4}\frac{d^2I}{dV^2}\cos 2\omega_1 t - \frac{v_1^3}{24}\frac{d^3I}{dV^3}\sin 3\omega_1 t \ldots \tag{141}$$

- after power reduction:

$$I = I_0 + \frac{dI}{dV} v_1 \sin\omega_1 t + \frac{1}{2}\frac{d^2I}{dV^2} v_1^2 \frac{1}{2}\left(1 - \cos 2\omega_1 t\right) + \frac{1}{6}\frac{d^3I}{dV^3} v_1^3 \frac{1}{4}\left(3\sin\omega_1 t - \sin 3\omega_1 t\right) \ldots \tag{140}$$

- arranged for frequency components:

$$I = I_0 + \frac{v_1^2}{4}\frac{d^2I}{dV^2} + \left(v_1 \frac{dI}{dV} + \frac{v_1^3}{8}\frac{d^3I}{dV^3}\right)\sin\omega_1 t - \frac{v_1^2}{4}\frac{d^2I}{dV^2}\cos 2\omega_1 t - \frac{v_1^3}{24}\frac{d^3I}{dV^3}\sin 3\omega_1 t \ldots \tag{141}$$

analysis: DC amplitude

$$|I|_{\mathrm{DC}} = I_0 + \frac{v_1^2}{4}\frac{d^2 I}{dV^2} \tag{142}$$

- the second-order non-linearity of the transfer characteristic affects the DC operating point
- the effect increases with the square of the input tone amplitude

analysis: fundamental frequency

$$|I|_{\omega_0} = v_1 \frac{dI}{dV} + \frac{v_1^3}{8}\frac{d^3 I}{dV^3} \tag{143}$$

- the third-order non-linearity of the transfer characteristic affects the fundamental power
- the effect increases with the cube of the input tone amplitude

analysis: DC amplitude

$$|I|_{\mathrm{DC}} = I_0 + \frac{v_1^2}{4}\frac{d^2 I}{dV^2} \tag{142}$$

- the second-order non-linearity of the transfer characteristic affects the DC operating point
- the effect increases with the square of the input tone amplitude

analysis: fundamental frequency

$$|I|_{\omega_0} = v_1 \frac{dI}{dV} + \frac{v_1^3}{8}\frac{d^3 I}{dV^3} \tag{143}$$

- the third-order non-linearity of the transfer characteristic affects the fundamental power
- the effect increases with the cube of the input tone amplitude

analysis: DC amplitude

$$|I|_{\mathrm{DC}} = I_0 + \frac{v_1^2}{4}\frac{d^2 I}{dV^2} \tag{142}$$

- the second-order non-linearity of the transfer characteristic affects the DC operating point
- the effect increases with the square of the input tone amplitude

analysis: fundamental frequency

$$|I|_{\omega_0} = v_1 \frac{dI}{dV} + \frac{v_1^3}{8}\frac{d^3 I}{dV^3} \tag{143}$$

- the third-order non-linearity of the transfer characteristic affects the fundamental power
- the effect increases with the cube of the input tone amplitude

analysis: DC amplitude

$$|I|_{\mathrm{DC}} = I_0 + \frac{v_1^2}{4}\frac{d^2 I}{dV^2} \tag{142}$$

- the second-order non-linearity of the transfer characteristic affects the DC operating point
- the effect increases with the square of the input tone amplitude

analysis: fundamental frequency

$$|I|_{\omega_0} = v_1 \frac{dI}{dV} + \frac{v_1^3}{8}\frac{d^3 I}{dV^3} \tag{143}$$

- the third-order non-linearity of the transfer characteristic affects the fundamental power
- the effect increases with the cube of the input tone amplitude

analysis: 2^{nd} harmonic

$$|I|_{2\omega_0} = \frac{v_1^2}{4}\frac{d^2 I}{dV^2} \tag{14}$$

- the second-order non-linearity of the transfer characteristic introduces a spectral component at the second harmonic
- the effect increases with the square of the input tone amplitude

analysis: 3^{rd} harmonic

$$|I|_{3\omega_0} = \frac{v_1^3}{24}\frac{d^3 I}{dV^3} \tag{145}$$

- the third-order non-linearity of the transfer characteristic introduces a spectral component at the third harmonic
- the effect increases with the cube of the input tone amplitude

analysis: 2^{nd} harmonic

$$|I|_{2\omega_0} = \frac{v_1^2}{4}\frac{d^2 I}{dV^2} \tag{14}$$

- the second-order non-linearity of the transfer characteristic introduces a spectral component at the second harmonic
- the effect increases with the square of the input tone amplitude

analysis: 3^{rd} harmonic

$$|I|_{3\omega_0} = \frac{v_1^3}{24}\frac{d^3 I}{dV^3} \tag{145}$$

- the third-order non-linearity of the transfer characteristic introduces a spectral component at the third harmonic
- the effect increases with the cube of the input tone amplitude

analysis: 2^{nd} harmonic

$$|I|_{2\omega_0} = \frac{v_1^2}{4}\frac{d^2 I}{dV^2} \tag{14}$$

- the second-order non-linearity of the transfer characteristic introduces a spectral component at the second harmonic
- the effect increases with the square of the input tone amplitude

analysis: 3^{rd} harmonic

$$|I|_{3\omega_0} = \frac{v_1^3}{24}\frac{d^3 I}{dV^3} \tag{145}$$

- the third-order non-linearity of the transfer characteristic introduces a spectral component at the third harmonic
- the effect increases with the cube of the input tone amplitude

analysis: 2^{nd} harmonic

$$|I|_{2\omega_0} = \frac{v_1^2}{4}\frac{d^2 I}{dV^2} \tag{14}$$

- the second-order non-linearity of the transfer characteristic introduces a spectral component at the second harmonic
- the effect increases with the square of the input tone amplitude

analysis: 3^{rd} harmonic

$$|I|_{3\omega_0} = \frac{v_1^3}{24}\frac{d^3 I}{dV^3} \tag{145}$$

- the third-order non-linearity of the transfer characteristic introduces a spectral component at the third harmonic
- the effect increases with the cube of the input tone amplitude

Third-Order Intermodulation Intercept Point

third-order intermodulation products in two-tone measurement (Fig. 121)

- linear extrapolation →Third-order intercept point (IP3)
- input-related: IIP3
- output-related: OIP3

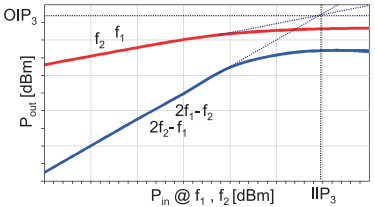


Figure 121: Two-tone power sweep.

Dynamic Range (Fig. 122)

- minimum detectable signal (MDS) at input

$$\mathrm{MDS_{in}} = kTB + 3\mathrm{dB} + NF \qquad (146)$$

- minimum detectable signal at output

$$\mathrm{MDS_{out}} = \mathrm{MDS_{in}} + G \qquad (147)$$

- thermal noise at 290 K:

$$kT|_{290\mathrm{K}} = -174\,\mathrm{dBm/Hz} = -84\,\mathrm{dBm/GHz} \qquad (148)$$

- dynamic range (DR)

$$\mathrm{DR} = \mathrm{OCP_{1dB}} - \mathrm{MDS_{out}} \qquad (149)$$

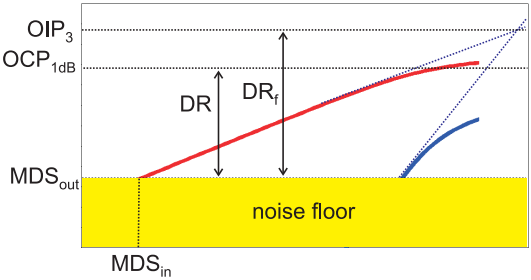


Figure 122: Dynamic range.

Dynamic Range (Fig. 122)

- minimum detectable signal (MDS) at input

$$\mathrm{MDS_{in}} = kTB + 3\mathrm{dB} + NF \qquad (146)$$

- minimum detectable signal at output

$$\mathrm{MDS_{out}} = \mathrm{MDS_{in}} + G \qquad (147)$$

- thermal noise at 290 K:

$$kT|_{290\mathrm{K}} = -174\,\mathrm{dBm/Hz} = -84\,\mathrm{dBm/GHz} \qquad (148)$$

- dynamic range (DR)

$$\mathrm{DR} = \mathrm{OCP_{1dB}} - \mathrm{MDS_{out}} \qquad (149)$$

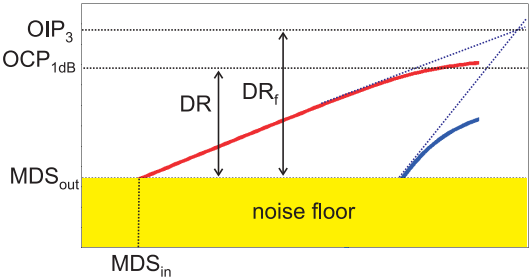


Figure 122: Dynamic range.

Dynamic Range (Fig. 122)

- minimum detectable signal (MDS) at input

$$\mathrm{MDS_{in}} = kTB + 3\mathrm{dB} + NF \qquad (146)$$

- minimum detectable signal at output

$$\mathrm{MDS_{out}} = \mathrm{MDS_{in}} + G \qquad (147)$$

- thermal noise at 290 K:

$$kT|_{290\mathrm{K}} = -174\,\mathrm{dBm/Hz} = -84\,\mathrm{dBm/GHz} \qquad (148)$$

- dynamic range (DR)

$$\mathrm{DR} = \mathrm{OCP_{1dB}} - \mathrm{MDS_{out}} \qquad (149)$$

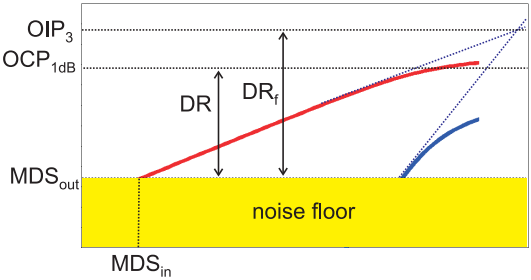


Figure 122: Dynamic range.

Dynamic Range (Fig. 122)

- minimum detectable signal (MDS) at input

$$\mathrm{MDS_{in}} = kTB + 3\mathrm{dB} + NF \tag{146}$$

- minimum detectable signal at output

$$\mathrm{MDS_{out}} = \mathrm{MDS_{in}} + G \tag{147}$$

- thermal noise at 290 K:

$$kT|_{290\mathrm{K}} = -174\,\mathrm{dBm/Hz} = -84\,\mathrm{dBm/GHz} \tag{148}$$

- dynamic range (DR)

$$\mathrm{DR} = \mathrm{OCP_{1dB}} - \mathrm{MDS_{out}} \tag{149}$$

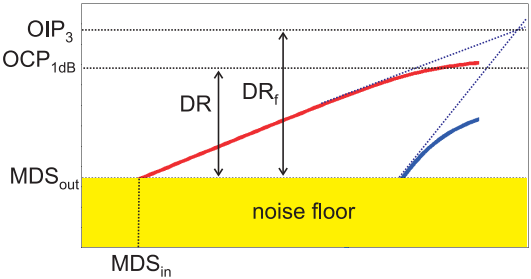


Figure 122: Dynamic range.

Spurious-free dynamic range (SFDR)

- definition via two-tone test:

$$\mathrm{DR_f} = \frac{2}{3}\left(\mathrm{OIP_3} - \mathrm{MDS_{out}}\right) \tag{150}$$

$$[\mathrm{DR_f}] = \mathrm{dB} \tag{151}$$

- general definition:

$$\mathrm{SFDR} = \frac{P_{\text{signal}}}{P_{\text{strongest spurious signal}}} \tag{152}$$

$$[\mathrm{SFDR}] = \mathrm{dBc} \tag{153}$$

Spurious-free dynamic range (SFDR)

- definition via two-tone test:

$$\mathrm{DR_f} = \frac{2}{3}\left(\mathrm{OIP_3} - \mathrm{MDS_{out}}\right) \tag{150}$$

$$[\mathrm{DR_f}] = \mathrm{dB} \tag{151}$$

- general definition:

$$\mathrm{SFDR} = \frac{P_{\text{signal}}}{P_{\text{strongest spurious signal}}} \tag{152}$$

$$[\mathrm{SFDR}] = \mathrm{dBc} \tag{153}$$

Efficiency

drain efficiency

$$\eta = \frac{P_{\mathrm{RF,out}}}{P_{\mathrm{DC}}} \tag{154}$$

power-added-efficiency (PAE)

$$PAE = \frac{P_{\mathrm{RF,out}} - P_{\mathrm{RF,in}}}{P_{\mathrm{DC}}} = \eta \left(1 - \frac{1}{G_{\mathrm{ass}}}\right) \tag{155}$$

- G_{ass} associated gain at $P_{\mathrm{RF,in}}$

Efficiency (Fig. 123)

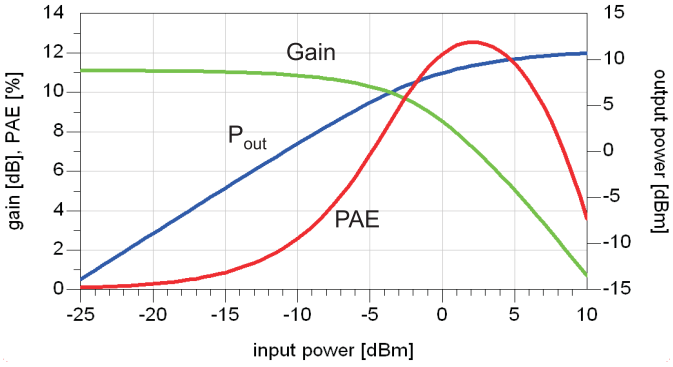


Figure 123: Power-added-efficiency.

1-tone power sweep

- input power for maximum PAE
- load impedance for maximum PAE → load-pull analysis

Self-Biasing (Fig. 124)

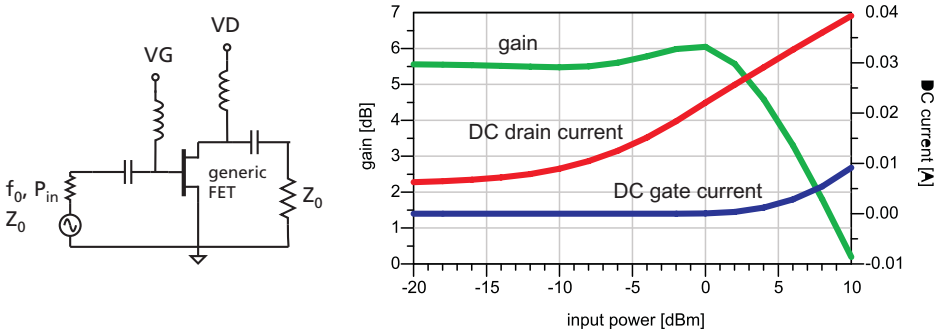


Figure 124: Self-biasing effect.

unsymmetrical power compression at high input power levels

- DC component in output spectrum
- dependent on input power, quiescent point, load impedance
- onset of gate current → possible device degradation or failure
- may lead to intermediate increase in gain (e.g. shift from class AB to effective class A)

Self-Biasing (2)

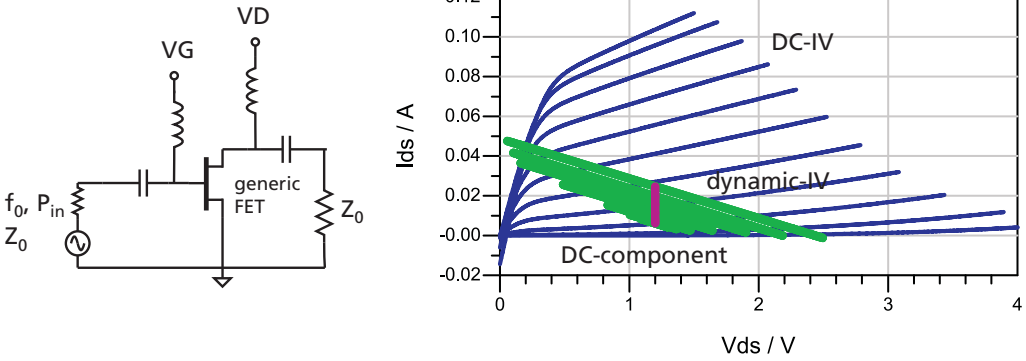


Figure 125: Self-biasing observed in dynamic IV traces.

dynamic IV (Fig. 125)

- shift in drain current observable with increasing RF power level
- the FET “opens itself”

6.1 Applications
6.2 Performance Indicators
6.3 Design Methods

Design Methods

6.3

Amplifier Classes A to C (Fig. 126)

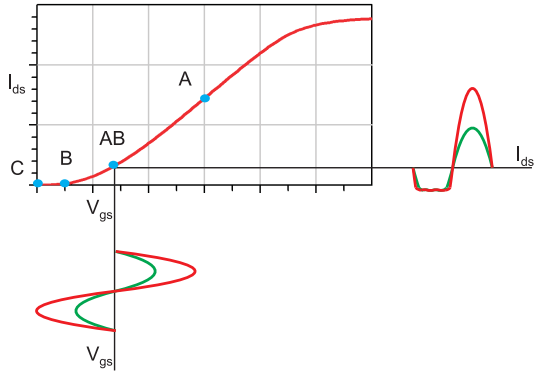


Figure 126: Power amplifier classes.

conduction angle η

- class A: $\eta = 2\pi$ ...device is conducting over the full input signal cycle
- class AB: $\pi < \eta < 2\pi$
- class B: $\eta = \pi$ ...device is conducting during one half cycle
- class C: $\eta < \pi$

Amplifier Classes A to C

class A-C considerations

- AB...C improves efficiency due to reduced DC current, but consider self-biasing effect under saturation
- AB...C leads to diminished linearity

Optimum Load Impedance for Output Power

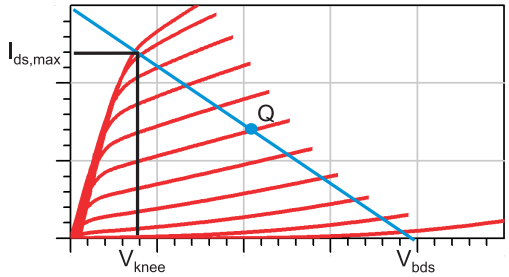


Figure 127: Cripps load.

Cripps load for maximum output power (Fig. 127)

- class A quiescent condition
- optimum load resistance:

$$R_{\mathrm{opt}} = \frac{V_{\mathrm{bds}} - V_{\mathrm{knee}}}{I_{\mathrm{ds,max}}} \tag{156}$$

- maximum output power:

$$P_{\mathrm{out,max}} = \frac{1}{\sqrt{2}}\frac{1}{2} I_{\mathrm{ds,max}} \cdot \frac{1}{\sqrt{2}}\frac{1}{2}\left(V_{\mathrm{bds}} - V_{\mathrm{knee}}\right) = \frac{1}{8} I_{\mathrm{ds,max}}\left(V_{\mathrm{bds}} - V_{\mathrm{knee}}\right) \tag{157}$$

- important: avoid both on- and off-state breakdown

Single-Stage Matching (Fig. 128)

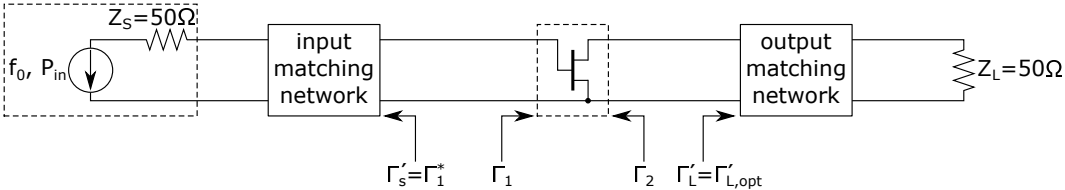


Figure 128: Single stage matching.

output

- transform external load Γ_L to ideal load for power or PAE $\Gamma^{'}_{\mathrm{L,opt}}$

input

- achieve power match ($\Gamma^{'}_{\mathrm{S}} = \Gamma_1^*$) under ideal output matching conditions ($S_{12} \neq 0$)

$$\Gamma_{in} = S_{11} + \frac{S_{12}S_{21}\Gamma_{l,opt}}{1 - S_{22}\Gamma_{l,opt}} = \Gamma_S^* \tag{158}$$

Dynamic IV

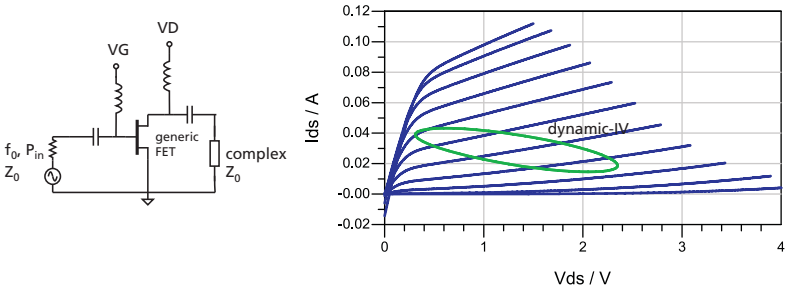


Figure 129: Dynamic IV traces.

dynamic IV traces (Fig. 129)

- due to the reactive component in Γ_2, $\Gamma_L^{'}$ will have a reactive component
- the load impedance $\Gamma_L^{'}$ has a reactive component
- the dynamic output IV curves become ellipses
- the ellipse widens with increasing frequency

goal

- present the ideal real load impedance (Cripps load) to the FET internal current source I_{ds}
- compensate for the FET output reactance in the output matching network

Reactive Matching

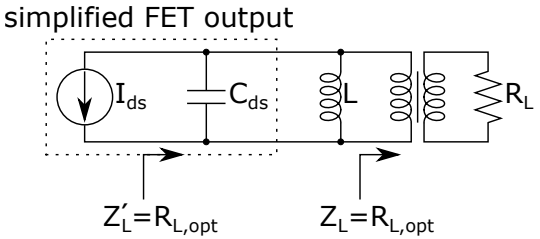


Figure 130: Reactive matching via inductive compensation of output capacitance.

first-order model (Fig. 130)

- neglects C_{gd}, R_s, ...
- valid up to $\sim f_{\mathrm{T}}/10$

reactive output matching

- inductive match to compensate for drain-source capacitance
- achieve $Z_{\mathrm{L}}^{'} = R_{\mathrm{L,opt}}$ for internal current source

Load-Pull Analysis (Fig. 131)

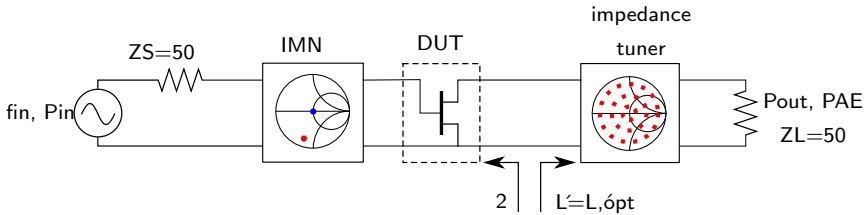


Figure 131: Load pull analysis.

tune load impedance

- input parameters: f_0, P_{in}
- measure Gain, P_{out}, PAE contours in $\Gamma_{\mathrm{L}}^{'}$ plane

pre-requisite

- accurate large-signal model

Load-Pull Contours (Fig. 132)

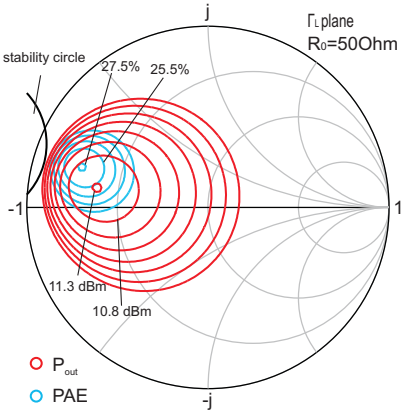


Figure 132: Load-pull contours[1].

load-pull example

- transistor: $4 \times 30\,\mu\text{m}$ IAF 100 nm mHEMT, class A
- $f_0 = 140\,\text{GHz}$, $P_{\text{in}} = 5.6\,\text{dBm}$

Load-Pull Example (Fig. 133)

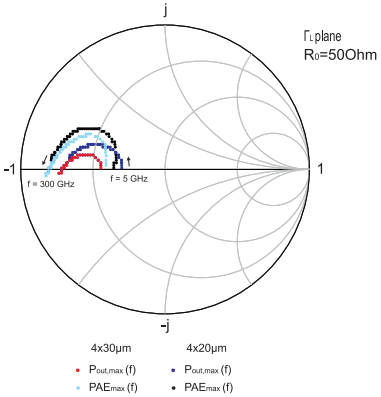


Figure 133: Load-pull impedances: frequency sweep.[1]

example: frequency sweep

- transistors: IAF 100 nm mHEMT, class A, $4 \times 30\,\mu\mathrm{m}$ and $4 \times 20\,\mu\mathrm{m}$
- $P_{\mathrm{in}} = 5.6\,\mathrm{dBm}$
- $\Gamma_{\mathrm{opt}}^{'}$ for max. PAE and max. P_{out}

Maximising Individual Transistor Size (Fig. 134)

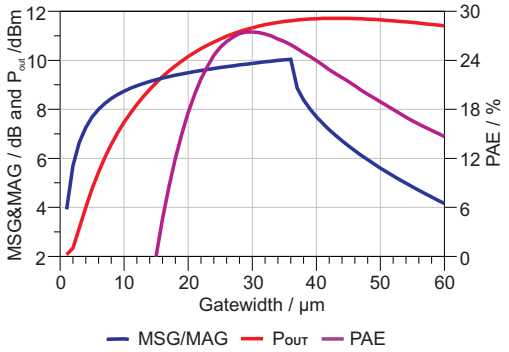


Figure 134: Sweep of transistor size.

sweep of transistor size

- example: IAF 100 nm mHEMT, class A, 4 finger device
- $f_0 = 140\,\mathrm{GHz}$, $P_{\mathrm{in}} = 5.6\,\mathrm{dBm}$
- limit to maximum individual transistor size due to limited available gain at f_0

Using Power Combiners/Splitters (Fig. 135)

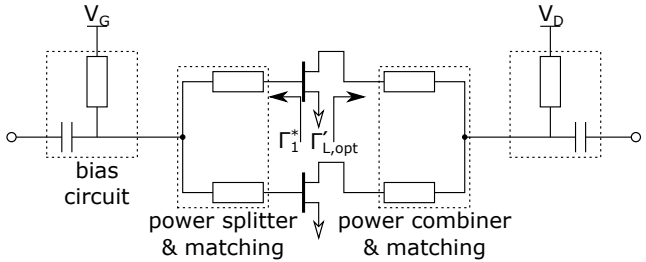


Figure 135: Power combining.

2 parallel FETs

- in-phase combination
- requires symmetrical layout
- requires matched (= identical performance) transistors
- power splitter/combiner can incorporate matching function
- additional loss in combiners/splitters

Using Power Combiners/Splitters (ctd.)

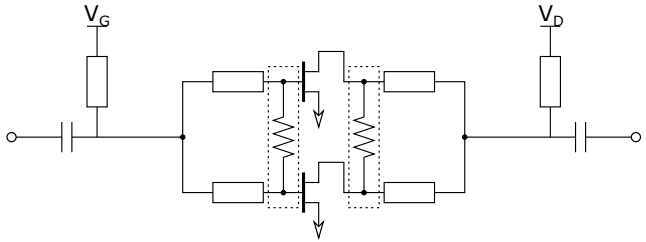


Figure 136: Odd-mode suppression.

odd-mode suppression (Fig. 136)

- possible instability due to odd-mode loop
- suppressed by parallel resistors

resistor dimensioning

- theoretically arbitrary (ideally: pure even-mode)
- stability analysis requires odd-mode excitation or intentional miss-match of transistors

Using Power Combiners/Splitters (ctd.)

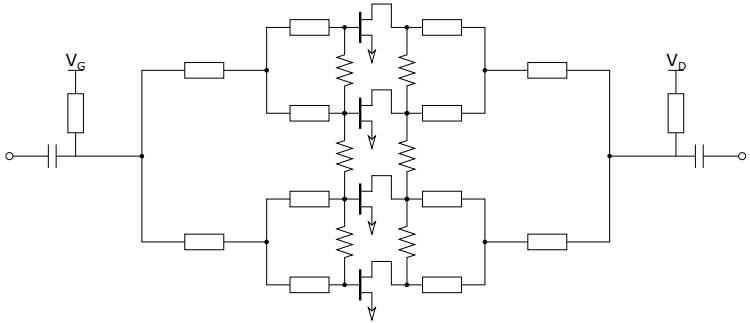


Figure 137: 4 parallel output stages.

4 parallel FETs (Fig. 137)

- in-phase combination
- odd-mode suppression

Example: 110 GHz PA

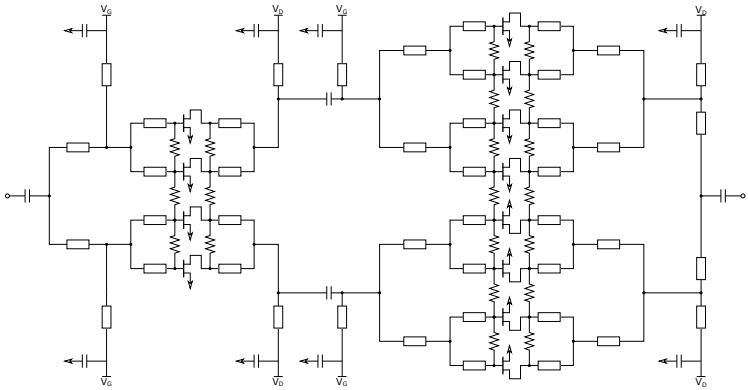


Figure 138: Power amplifier example: schematic.

2-stage topology (Fig. 138)

- individual FET size: $4 \times 40\,\mu m$ in dual-gate conf. = cascode of $2 \times 40\,\mu m$ transistors
- 1.28 mm total gate width in output stage with 8 parallel dual-gate FETs

Example: 110 GHz PA

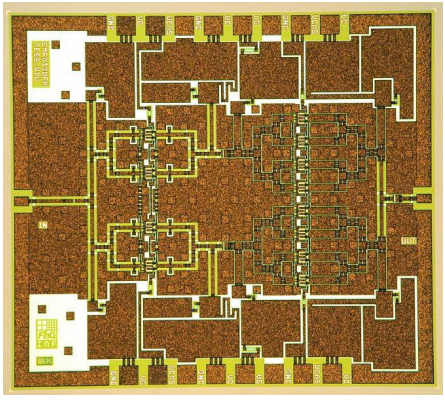


Figure 139: Power amplifier example: layout.

circuit layout (Fig. 139)

- technology: IAF 100 nm mHEMT
- chip size: $2 \times 2.25\,\mathrm{mm}^2$

Example: 110 GHz PA

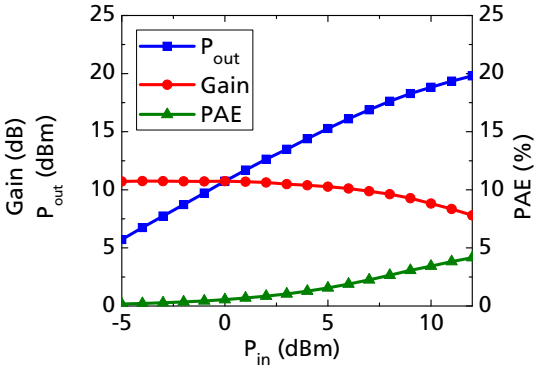


Figure 140: Power amplifier example: measurements.

1-tone power measurement (Fig. 140)

- $P_{\mathrm{sat}} = 20\,\mathrm{dBm},\ PAE_{\max} = 4\%$

Balanced Power Amplifier (Fig. 141)

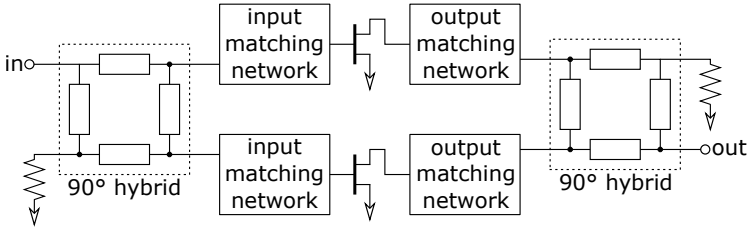


Figure 141: Balanced power amplifier topology.

2 parallel FETs

- FET operate 90° out of phase
- high bandwidth due to inherent 50Ω match

Traveling-Wave Amplifier (Fig. 142)

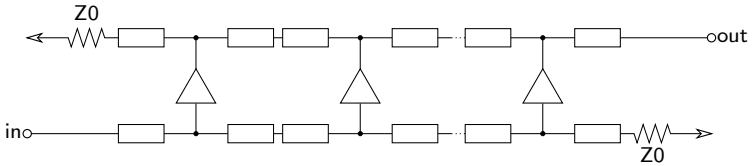


Figure 142: Traveling-wave power amplifier.

n stages

- constructive (in-phase) wave interference
- ultra-high bandwidth possible, hence:
- preferred topology for modulator drivers in high-speed optical communication
- low efficiency (half of P_{out} is lost in reverse-traveling wave, cascode)

PA Design Sequence

single-stage PA design (not switching amplifiers)

- choice of transistor quiesent point (class A to C)
- maximise individual transistor dimensions
  - sufficient gain at center frequency
- required number of parallel transistors
  - required output power
  - choice of topology: parallel, power-combining, TWA ...
- in- and output matching networks
  - output matched for optimum power or PAE (from load-pull analysis)
  - input matched for power under optimum output match conditions
  - minimize loss in matching networks
- check stability
  - in-band
  - out-of-band (e.g. LF terminations)
  - unconditional (arbitrary source and load impedance)

PA Design Sequence

multi-stage PA design

- required gain
  - available input power = output power of preceeding stage
  - required output power
- power partitioning
  - output power of preceeding stage has to drive subsequent stage into desired compression (e.g. maximum PAE)
- check stability
  - internal stability at all active-passive interfaces
  - unconditional stability

Example: Internal Stability Analysis (Fig. 143)

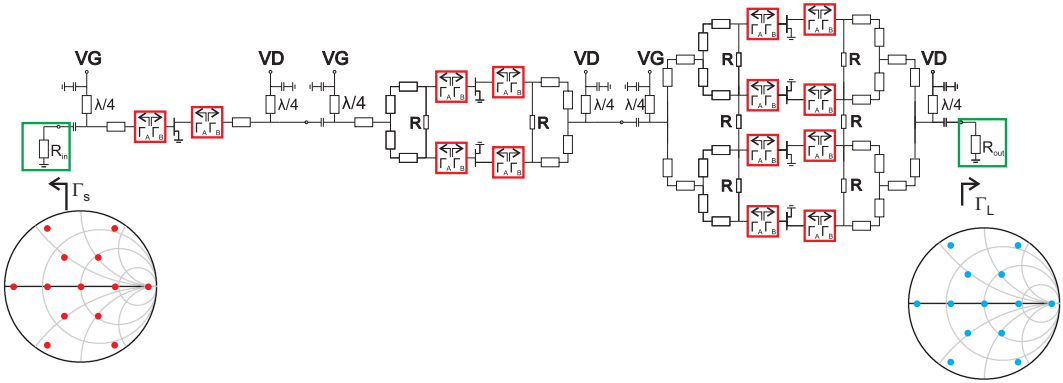


Figure 143: Power amplifier stability analysis [1].

Gamma-Probe test

- check for $|\Gamma_A \cdot \Gamma_B| < 1$ at all passive-active interfaces and for all source and load impedances

Example: Power Partitioning

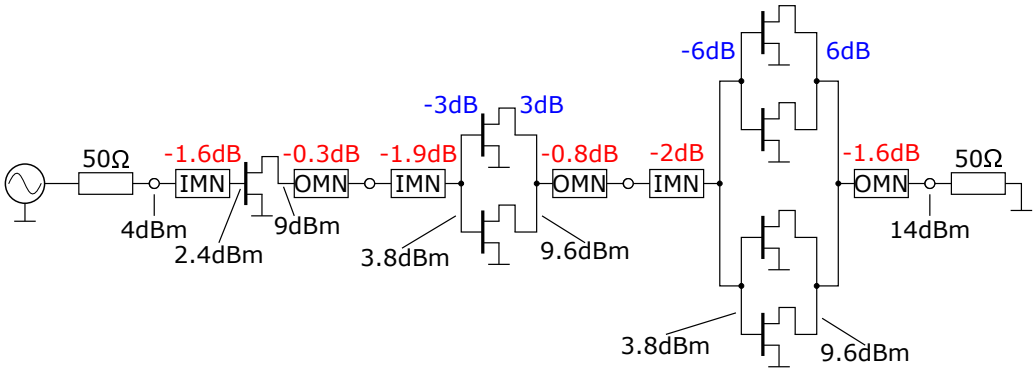


Figure 144: Power amplifier gain and linearity partitioning [1].

gain and power partitioning (Fig. 144)

- all devices operate with approx. identical input power
- include loss in combiner/splitter networks

University of Stuttgart
Germany

Institute of
Robust Power
Semiconductor Systems

Microwave Network Analysis (Appendix)

7

1 Course Outline

2 The Millimeterwave Frequency Spectrum: Applicatio
and Technologies

3 Functional Buildung Blocks of MMICs

4 Low Noise Amplifier

5 Broadband Amplifiers

6 Power Amplifiers

7 Microwave Network Analysis (Appendix)

Lecture Goals

in this part of the lecture, we will...

- introduce the theoretical background and tools for linear microwave circuit design and analysis, in particular S-parameters and the Smith-chart
- introduce the Rollet Stability Factor for two-port stability analysis
- learn how to transform impedances in matching networks to achieve power matching of a two-port to its source and load

7.1 N-Port Parameters
7.2 Scattering Parameters
7.3 The Smith Chart
7.4 Rollet Stability Factor and Power Gains
7.5 Power Matching and Impedance Transformation

N-Port Parameters

7.1

General n-Port Microwave Network

Port-incident and port-leaving voltage and current waves (145)

- total port voltage:

$$V_n = V_n^+ + V_n^- \tag{159}$$

- total port current (incident):

$$I_n = I_n^+ - I_n^- \tag{160}$$

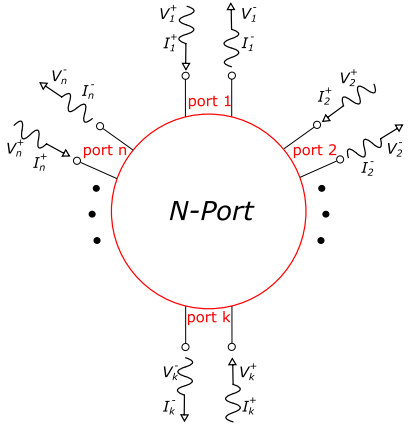


Figure 145: General n-port microwave network representation.

General n-Port Microwave Network

incident wave (or: signal)

- port excitation from an external source

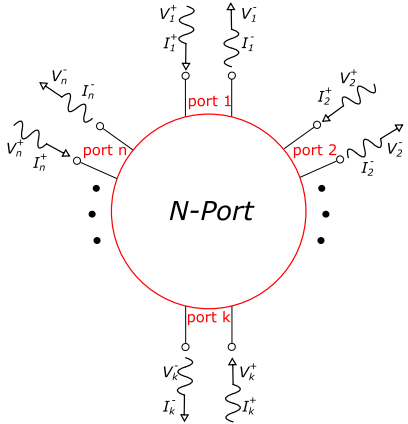


Figure 145: General n-port microwave network representation.

General n-Port Microwave Network

port-leaving wave (or: signal)

- reflection back into the source at the excited port
- transmission into an external load due to excitation at another port
- generation due to an internal source (e.g. oscillator)
- response due to an internal controlled source (e.g. voltage-controlled current source)

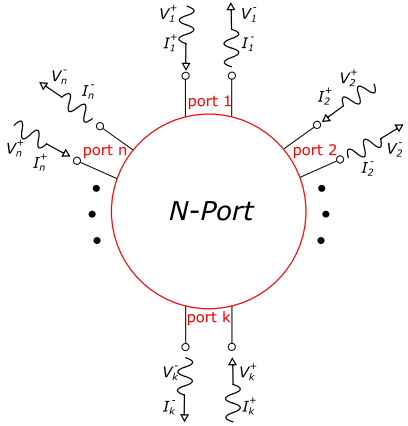


Figure 145: General n-port microwave network representation.

Principles of Microwave Network Analysis

Relate port entities to each other

- port currents and voltages ($\rightarrow Z$, Y, H, ABCD parameters, voltage/current gain etc.)
- port-incident and port-leaving waves ($\rightarrow$ S-parameters, reflection coefficients, power gain etc.)
- reflected and transmitted power ($\rightarrow$ S-parameters, cp. measurement principle of vector network analyzer (VNA)s)

Principles of Microwave Network Analysis

"Think in the frequency domain"

- individual frequency components of a signal experience a transformation of phase and/or magnitude (e.g. along transmission lines)
- networks are typically analyzed by single-tone or multi-tone analysis
- signal reflection occurs with phase and magnitude shift at abrupt impedance discontinuities, cp. time domain reflectormetry
- linear networks "respond" with changes in phase and magnitude of the excitation frequencies, but do not introduce new frequencies
- nonlinear networks introduce signal compression, new frequency components (harmonics and intermodulation), AM-PM and PM-AM conversion

Frequency Domain Description of Microwave Networks

Signals

- consider periodic signals only
- small-signal (linear) single-tone analysis or superposition of multiple tones (multi-tone analysis)
- large-signal (non-linear) periodic waveforms show a discrete frequency spectrum, obtained from Fourier analysis
- time domain waveform obtained from Inverse Fourier Transformation (IFT)

Frequency Domain Description of Microwave Networks

Circuit elements

- ideal resistor: $V = R \cdot I$
- ideal capacitor: $I = C \cdot \frac{dV}{dt} \rightarrow i = j\omega C \cdot v \rightarrow Y_C = j\omega C$
- ideal inductor: $V = L \cdot \frac{dI}{dt} \rightarrow v = j\omega L \cdot i \rightarrow Z_L = j\omega L$
- controlled sources
  - voltage-controlled current source: $i_{\text{out}} = g_{\text{m}} v_{\text{in}} \rightarrow$ "trans-conductance" $Y_t = g_m = \frac{i_{\text{out}}}{v_{\text{in}}}$

Frequency Domain Description of Microwave Networks

Conversion from time to frequency domain

- periodic signals only
- arbitrary waveforms can be represented by Fourier series
- consider one frequency component only:
- $V = \hat{v} \cos 2\pi f t = \hat{v} \cos \omega t = v$
  - note: typically, arbitrary nonlinear voltage and current waveforms are represented by capital letters, small-signal/linear waveforms by minuscule letters
- $\frac{dV}{dt} = \hat{v}\omega \sin \omega t = \hat{v}\omega \cos\left(\omega t - \frac{\pi}{2}\right)$
- ideal capacitor: $I = C \cdot \frac{dV}{dt} = \omega C \cdot \hat{v} \cos\left(\omega t - \frac{\pi}{2}\right)$
  - 90° phase lag of current to voltage in an ideal capacitor

Frequency Domain Description of Microwave Networks

Conversion from time to frequency domain using complex numbers

- Euler's formula: $e^{jx} = \cos x + j \sin x$
- $v = \hat{v} e^{j\omega t}$
- $\frac{dv}{dt} = j\omega \hat{v} e^{j\omega t}$
- ideal capacitor: $I = C \cdot \frac{dV}{dt} = j\omega C \cdot \hat{v} e^{j\omega t} = j\omega C \cdot v$
- admittance: $Y_C = \frac{I}{V} = j\omega C$

Frequency Domain Description of Microwave Networks

Nomenclature

$$Z = R + jX \tag{161}$$

- Z...impedance
- R...resistance
- X...reactance

$$Y = Z^{-1} = G + jB \tag{162}$$

- Y...admittance
- G...conductance
- B...susceptance

Linear Networks

n-port parameters normally describe linear networks

- transfer function: $v_{out} = G(\omega) \cdot v_{in}$
- output signal consists of the same frequency(ies) as input signal
- change only in amplitude and phase: $G(\omega) = |G(\omega)| \cdot e^{j\phi(\omega)}$
- $G(\omega)$ is function of frequency only, not of voltage/current/power
  - excludes AM-PM, PM-AM conversion
  - excludes intermodulation

Linear Networks

can be applied to:

- passive networks
  - within limits: e.g. Ohmic law for resistors, breakdown voltage in dielectrics etc.
- nonlinear devices (transistors etc.) are linearised about their operating point
  - two-port parameters are derived from differential characteristics
  - valid only under small-signal conditions (= small signal amplitudes)

Transistor Linearisation: Transconductance I

Transistor Linearisation: Output Conductance I

Two-Port Representation I

two-port (146)

- two pairs of nodes
- typically referenced to ground → two terminals above ground
- eight independent parameters: currents+voltages or incident+reflected power, each complex (= phase and magnitude)

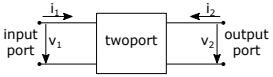


Figure 146: two port with input (p1) and output (p2) ports.

Two-Port Example: Transmission Line / Cable (147)

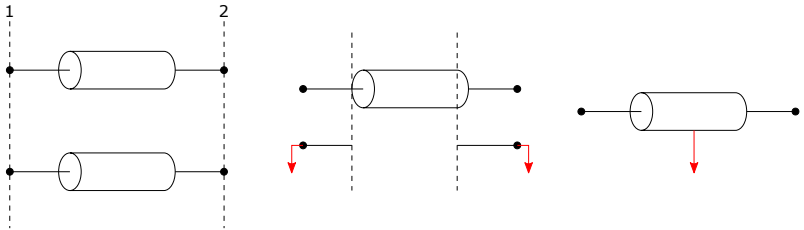


Figure 147: Two-port representation of a transmission line.

Two-Port Example: Lumped Passive R,L,C I

Two-Port Example: Diode I

Transistors as Two-Ports I

The transistor as a three-port device... (148)

- becomes a two-port in the three standard "transistor configurations"
  - common-source/emitter
  - common-gate/base
  - common-drain/collector
- note: in MOSFETs, "bulk" as the fourth port is typically shortened to the source in network analysis

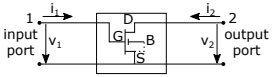


Figure 148: Common-source transistor configuration.

Impedance (Z-)Parameters

Impedance matrix $[z]$

$$\begin{bmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} & \dots & z_{1n} \\ z_{21} & z_{22} & \dots & \dots \\ \dots & \dots & \dots & \dots \\ z_{1n} & \dots & \dots & z_{nn} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ \dots \\ i_n \end{bmatrix} \tag{163}$$

$$[v] = [Z]\,[i] \tag{164}$$

Impedance parameters Z_{ij}

$$z_{ij} = \left.\frac{v_i}{i_j}\right|_{i_k=0, k\neq j} \tag{165}$$

- z_{ij}... drive port j with current i_j, open-circuit all other ports, measure open-circuit voltage at port i
- z_{ii}... input impedance at port i, when all other ports are open-circuited
- z_{ij}... transfer-impedance between ports i and j, when all other ports are open-circuited

Admittance (Y) Parameters

Admittance matrix $[y]$

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \tag{166}$$

$$[i] = [y]\,[v] \tag{167}$$

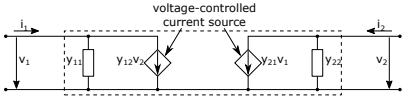


Figure 149: Equivalent circuit representation of Y-parameters.

Admittance (Y) Parameters

Admittance parameters y_{ij}

$$y_{ij} = \left.\frac{i_i}{v_j}\right|_{v_k=0, k\neq j} \tag{168}$$

- $y_{11} = \frac{i_1}{v_1}|_{v_2=0}$ ... short circuit input admittance
- $y_{12} = \frac{i_1}{v_2}|_{v_1=0}$ ... short circuit reverse transadmittance
- $y_{21} = \frac{i_2}{v_1}|_{v_2=0}$ ... short circuit (forward) transadmittance
- $y_{22} = \frac{i_2}{v_2}|_{v_1=0}$ ... short circuit output admittance

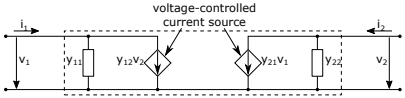


Figure 149: Equivalent circuit representation of Y-parameters.

Example: Resistive Attenuator I

Y-Parameters and FET Models I

FET small-signal model, simple approximation (Fig. 150)

- approximation: neglect Miller capacitance C_{gd} and other elements (series R, inductance etc.)
- $y_{11} = \frac{i_1}{v_1}|_{v_2=0} = j\omega C_{gs} \rightarrow C_{gs} = \text{Im}(y_{11})/\omega$
- $y_{12} = \frac{i_1}{v_2}|_{v_1=0} = 0$
- $y_{21} = \frac{i_2}{v_1}|_{v_2=0} = g_m \rightarrow g_m = \text{Re}(y_{21})$
- $y_{22} = \frac{i_2}{v_2}|_{v_1=0} = g_{ds} + j\omega C_{ds} \rightarrow g_{ds} = \text{Re}(y_{22}),\ C_{ds} = \text{Im}(y_{22})/\omega$

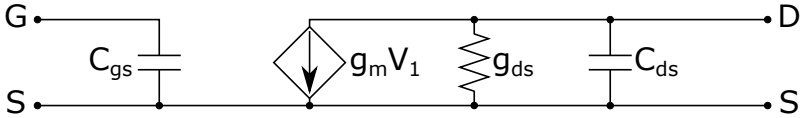


Figure 150: Simplified small-signal equivalent circuit of a common-source transistor.

Hybrid (H) Parameters

Hybrid matrix $[h]$

$$\begin{bmatrix} v_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} i_1 \\ v_2 \end{bmatrix} \tag{169}$$

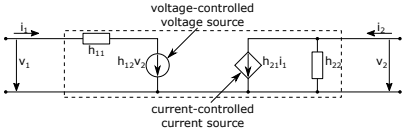


Figure 151: Equivalent circuit representation of H-parameters.

Hybrid (H) Parameters

Hybrid parameters h_{ij}

- $h_{11} = \frac{v_1}{i_1}|_{v_2=0}$ ... short circuit input impedance
- $h_{12} = \frac{v_1}{v_2}|_{i_1=0}$ ... open circuit reverse voltage gain
- $h_{21} = \frac{i_2}{i_1}|_{v_2=0}$ ... short circuit (forward) current gain
- $h_{22} = \frac{i_2}{v_2}|_{i_1=0}$ ... open circuit output conductance

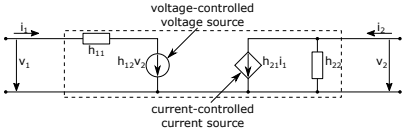


Figure 151: Equivalent circuit representation of H-parameters.

H-Parameters and Transistor Cutoff Frequencies

transit frequency f_T (152)

- $f_\mathrm{T} = f|_{|h_{21}|=1}$ ... "transit frequency" = frequency, where the short-circuit current gain becomes unity (or 0 dB)

$$\omega_\mathrm{T} \approx \frac{g_\mathrm{m}}{C_\mathrm{gs}} \tag{170}$$

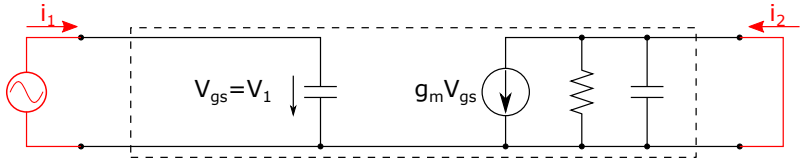


Figure 152: Short-circuit current gain of a common-source transistor.

Chain (ABCD) Parameters

ABCD matrix

$$\begin{bmatrix} v_1 \\ i_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} v_2 \\ -i_2 \end{bmatrix} \tag{171}$$

ABCD parameters

$$[\mathrm{ABCD}]_{\mathrm{cascade}} = [\mathrm{ABCD}]_1 \cdot [\mathrm{ABCD}]_2 \tag{172}$$

- multiply chain matrices for cascaded two-ports to obtain chain matrix of the resulting two port

7.1 N-Port Parameters
7.2 Scattering Parameters
7.3 The Smith Chart
7.4 Rollet Stability Factor and Power Gains
7.5 Power Matching and Impedance Transformation

Scattering Parameters

7.2

Concept of Normalized Voltage/Current Waves

normalised voltage and current waves

- $a = \frac{V^+}{\sqrt{Z_0}} = \frac{Z_0 I^+}{\sqrt{Z_0}} = \sqrt{Z_0} I^+$ ... incident normalized wave
- $b = \frac{V^-}{\sqrt{Z_0}} = \frac{Z_0 I^-}{\sqrt{Z_0}} = \sqrt{Z_0} I^-$ ... reflected normalized wave
- Z_0 ...normalizing impedance

Concept of Normalized Voltage/Current Waves

Power delivered to the load

$$\begin{aligned} P &= \mathrm{Re}\left(VI^*\right) = \mathrm{Re}\left(\sqrt{Z_0}\,(a+b)\,\tfrac{1}{\sqrt{Z_0}}\,(a-b)^*\right) \\ &= \mathrm{Re}\left((a+b)\,(a^*-b^*)\right) \\ &= \mathrm{Re}\left(aa^*-bb^*+ba^*-ab^*\right) = \mathrm{Re}\left(aa^*-bb^*\right) \\ &= |a|^2-|b|^2 \end{aligned}$$

- using

$$\begin{aligned} ba^*-ab^* &= (b_r+jb_i)\,(a_r-ja_i)-(a_r+ja_i)\,(b_r-jb_i) \\ &= b_ra_r-b_ra_r+b_ia_i-b_ia_i+j\left(-b_ra_i+a_rb_i+a_rb_i-b_ra_i\right) \\ &= 2j\,(a_rb_i-b_ra_i) \end{aligned}$$

- $|a|^2$... incident power, $|b|^2$... reflected power

Normalizing Impedance Z_0

$50\,\Omega$ in microwaves

- originated shortly after WW2
- best compromise/geometrical mean value between low loss ($\approx 70\,\Omega$) and power handling ($\approx 30\,\Omega$) in air coaxial transmission lines

Normalizing Impedance Z_0

general case: Z_0 ...

- ...can be an arbitrary complex normalizing impedance
- ...can be different for individual ports
- ...is often chosen as the source impedance of the (external) measurement terminal
- ...is often chosen as the characteristic impedance[a] of access or connecting transmission lines

[a] dt. Wellenwiderstand

Scattering (S) Parameters

Scattering matrix (153)

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} s_{11} & s_{12} \\ s_{21} & s_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \tag{173}$$

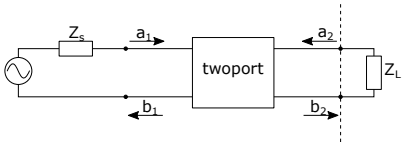


Figure 153: Two port with normalized voltage waves, source and load.

Scattering (S) Parameters

Scattering parameter s_{11}

$$s_{11} = \frac{b_1}{a_1}|_{a_2=0} \tag{174}$$

- ... input reflection coefficient, output terminated into non-reflecting load

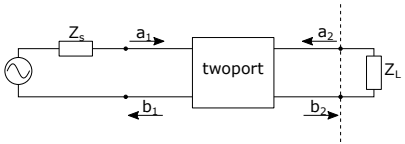


Figure 153: Two port with normalized voltage waves, source and load.

Scattering (S) Parameters

Scattering parameter s_{12}

$$s_{12} = \frac{b_1}{a_2}\,|_{a_1=0} \tag{175}$$

- ... reverse transmission (isolation $= s_{12}^{-1}$), input terminated into non-reflecting load

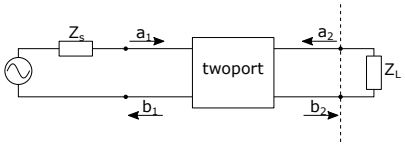


Figure 153: Two port with normalized voltage waves, source and load.

Scattering (S) Parameters

Scattering parameter s_{21}

$$s_{21} = \frac{b_2}{a_1}\,|_{a_2=0} \tag{176}$$

- ... forward transmission, output terminated into non-reflecting load

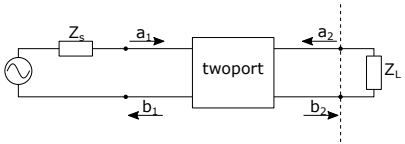


Figure 153: Two port with normalized voltage waves, source and load.

Scattering (S) Parameters

Scattering parameter s_{22}

$$s_{22} = \frac{b_2}{a_2}|_{a_1=0} \tag{177}$$

- ... output reflection coefficient, input terminated into non-reflecting load

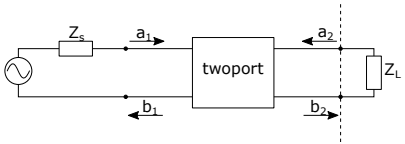


Figure 153: Two port with normalized voltage waves, source and load.

Special Two-port Networks

reciprocal

- $s_{12} = s_{21}$
- n-port: $(S)^T = (S)$

Special Two-port Networks

unilateral

- $s_{12} = 0$
- no feedback from output to input
  - this is often assumed in analog circuit design, cp. operational amplifiers
- at microwaves, this assumption breaks down
  - the (terminating) load impedance has an impact on the input impedance of the device

Special Two-port Networks

lossless

- $|s_{21}|^2 + |s_{11}|^2 = 1$
- $|s_{12}|^2 + |s_{22}|^2 = 1$
- n-port: $\sum_{i=1}^{n} |s_{ni}|^2 = 1$ for all $i = 1..n$
- interpretation: in a lossless network, incident power has to be identical to the sum of reflected and transmitted power

Reflection Coefficient

Reflection coefficient Γ

$$\Gamma = \frac{b}{a} \tag{178}$$

$$\begin{aligned} \Gamma = \quad \frac{b}{a} &= \quad \frac{V^-}{\sqrt{Z_0}} \left(\frac{V^+}{\sqrt{Z_0}} \right)^{-1} = \frac{V^-}{V^+} \\ &= \quad \frac{\sqrt{Z_0} I^-}{\sqrt{Z_0} I^+} = \frac{I^-}{I^+} \end{aligned} \tag{179}$$

- ...ratio of reflected to incident voltage and current waves
  - ... complex entity with magnitude and phase
  - ...function of space and frequency
- $|\Gamma|^2 = \frac{|b^2|}{|a^2|}$ ... (scalar) ratio of reflected power to incident power

Reflection Coefficient

Impedance and reflection coefficient

- $Z = \frac{V}{I} = \frac{\sqrt{Z_0}(a+b)}{\frac{1}{\sqrt{Z_0}}(a-b)} = Z_0 \frac{1+\Gamma}{1-\Gamma}$

$$z = \frac{1+\Gamma}{1-\Gamma} \tag{180}$$

...impedance in terms of Γ

- $\Gamma = \frac{Z/Z_0-1}{Z/Z_0+1} = \frac{Z-Z_0}{Z+Z_0}$

$$\Gamma = \frac{z-1}{z+1} \tag{181}$$

...reflection coefficient in terms of impedance

Reflection Coefficient

Admittance and reflection coefficient

- $Y = \frac{I}{V} = \frac{\frac{1}{\sqrt{Z_0}}(a-b)}{\sqrt{Z_0}(a+b)} = Y_0 \frac{1-\Gamma}{1+\Gamma}$

$$y = \frac{1-\Gamma}{1+\Gamma} \tag{182}$$

...admittance in terms of Γ, $Y_0 = \frac{1}{Z_0}$

- $\Gamma = \frac{1-Y/Y_0}{1+Y/Y_0} = \frac{Y_0-Y}{Y_0+Y}$

$$\Gamma = \frac{1-y}{1+y} \tag{183}$$

...reflection coefficient in terms of admittance

Transfer (T) Parameters

Transfer matrix

$$\begin{bmatrix} b_1 \\ a_1 \end{bmatrix} = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} \begin{bmatrix} a_2 \\ b_2 \end{bmatrix} \tag{184}$$

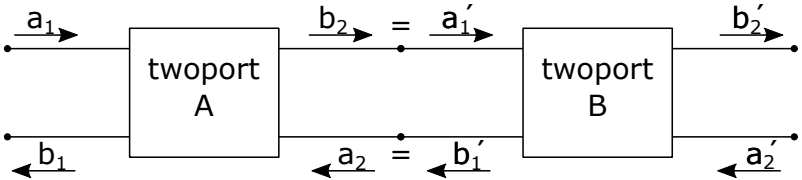


Figure 154: Cascaded two-ports and transfer parameters.

Transfer (T) Parameters

Cascaded two-ports (154)

$$\begin{bmatrix} b_1 \\ a_1 \end{bmatrix} = \begin{bmatrix} t_{11}^A & t_{12}^A \\ t_{21}^A & t_{22}^A \end{bmatrix} \begin{bmatrix} t_{11}^B & t_{12}^B \\ t_{21}^B & t_{22}^B \end{bmatrix} \begin{bmatrix} a_2' \\ b_2' \end{bmatrix} \tag{185}$$

- Multiply T-matrices for cascaded two-ports

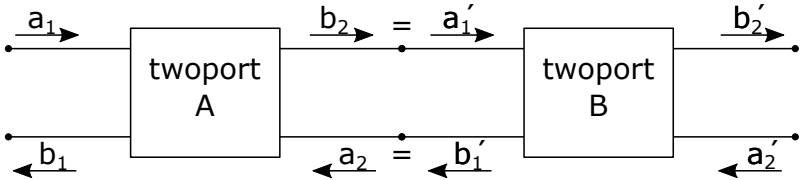


Figure 154: Cascaded two-ports and transfer parameters.

Transfer (T) Parameters

Conversion from S to T-parameters

$$
\begin{aligned}
t_{11} &= 1/s_{21} \\
t_{12} &= -s_{22}/s_{21} \\
t_{21} &= s_{11}/s_{21} \\
t_{22} &= s_{12} - s_{11}s_{22}/s_{21}
\end{aligned}
$$

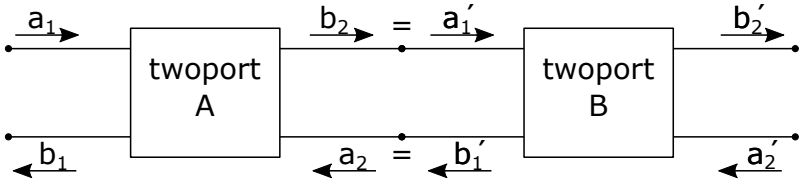


Figure 154: Cascaded two-ports and transfer parameters.

Network Parameter Conversion

Network parameter conversion (155)

- all (linear) network parameters can be converted into each other by simple linear relationships

TABLE 4.2 Conversions Between Two-Port Network Parameters

	S	Z	Y	$ABCD$
S_{11}	S_{11}	$\frac{(Z_{11}-Z_0)(Z_{22}+Z_0)-Z_{12}Z_{21}}{\Delta Z}$	$\frac{(Y_0-Y_{11})(Y_0+Y_{22})+Y_{12}Y_{21}}{\Delta Y}$	$\frac{A+B/Z_0-CZ_0-D}{A+B/Z_0+CZ_0+D}$
S_{12}	S_{12}	$\frac{2Z_{12}Z_0}{\Delta Z}$	$\frac{-2Y_{12}Y_0}{\Delta Y}$	$\frac{2(AD-BC)}{A+B/Z_0+CZ_0+D}$
S_{21}	S_{21}	$\frac{2Z_{21}Z_0}{\Delta Z}$	$\frac{-2Y_{21}Y_0}{\Delta Y}$	$\frac{2}{A+B/Z_0+CZ_0+D}$
S_{22}	S_{22}	$\frac{(Z_{11}+Z_0)(Z_{22}-Z_0)-Z_{12}Z_{21}}{\Delta Z}$	$\frac{(Y_0+Y_{11})(Y_0-Y_{22})+Y_{12}Y_{21}}{\Delta Y}$	$\frac{-A+B/Z_0-CZ_0+D}{A+B/Z_0+CZ_0+D}$
Z_{11}	$Z_0\frac{(1+S_{11})(1-S_{22})+S_{12}S_{21}}{(1-S_{11})(1-S_{22})-S_{12}S_{21}}$	Z_{11}	$\frac{Y_{22}}{\lvert Y\rvert}$	$\frac{A}{C}$
Z_{12}	$Z_0\frac{2S_{12}}{(1-S_{11})(1-S_{22})-S_{12}S_{21}}$	Z_{12}	$\frac{-Y_{12}}{\lvert Y\rvert}$	$\frac{AD-BC}{C}$
Z_{21}	$Z_0\frac{2S_{21}}{(1-S_{11})(1-S_{22})-S_{12}S_{21}}$	Z_{21}	$\frac{-Y_{21}}{\lvert Y\rvert}$	$\frac{1}{C}$
Z_{22}	$Z_0\frac{(1-S_{11})(1+S_{22})+S_{12}S_{21}}{(1-S_{11})(1-S_{22})-S_{12}S_{21}}$	Z_{22}	$\frac{Y_{11}}{\lvert Y\rvert}$	$\frac{D}{C}$
Y_{11}	$Y_0\frac{(1-S_{11})(1+S_{22})+S_{12}S_{21}}{(1+S_{11})(1+S_{22})-S_{12}S_{21}}$	$\frac{Z_{22}}{\lvert Z\rvert}$	Y_{11}	$\frac{D}{B}$
Y_{12}	$Y_0\frac{-2S_{12}}{(1+S_{11})(1+S_{22})-S_{12}S_{21}}$	$\frac{-Z_{12}}{\lvert Z\rvert}$	Y_{12}	$\frac{BC-AD}{B}$
Y_{21}	$Y_0\frac{-2S_{21}}{(1+S_{11})(1+S_{22})-S_{12}S_{21}}$	$\frac{-Z_{21}}{\lvert Z\rvert}$	Y_{21}	$\frac{-1}{B}$
Y_{22}	$Y_0\frac{(1+S_{11})(1-S_{22})+S_{12}S_{21}}{(1+S_{11})(1+S_{22})-S_{12}S_{21}}$	$\frac{Z_{11}}{\lvert Z\rvert}$	Y_{22}	$\frac{A}{B}$
A	$\frac{(1+S_{11})(1-S_{22})+S_{12}S_{21}}{2S_{21}}$	$\frac{Z_{11}}{Z_{21}}$	$\frac{-Y_{22}}{Y_{21}}$	A
B	$Z_0\frac{(1+S_{11})(1+S_{22})-S_{12}S_{21}}{2S_{21}}$	$\frac{\lvert Z\rvert}{Z_{21}}$	$\frac{-1}{Y_{21}}$	B
C	$\frac{1}{Z_0}\frac{(1-S_{11})(1-S_{22})-S_{12}S_{21}}{2S_{21}}$	$\frac{1}{Z_{21}}$	$\frac{-\lvert Y\rvert}{Y_{21}}$	C
D	$\frac{(1-S_{11})(1+S_{22})+S_{12}S_{21}}{2S_{21}}$	$\frac{Z_{22}}{Z_{21}}$	$\frac{-Y_{11}}{Y_{21}}$	D

$|Z| = Z_{11}Z_{22} - Z_{12}Z_{21}$; $|Y| = Y_{11}Y_{22} - Y_{12}Y_{21}$; $\Delta Y = (Y_{11}+Y_0)(Y_{22}+Y_0) - Y_{12}Y_{21}$; $\Delta Z = (Z_{11}+Z_0)(Z_{22}+Z_0) - Z_{12}Z_{21}$; $Y_0 = 1/Z_0$.

Figure 155: Conversion table for two-port network parameters [2].

Measuring Scattering (S) Parameters

S-parameter measurement (156)

- Using vectorial network analysers (VNA)
- S-parameters can today be measured up to $\sim$1 THz. Commercial systems for on-wafer measurements operate from 0-110 GHz and in waveguide bands (using frequency extention T/R modules), e.g. G-band 140-220 GHz (photo), H-band 220-340 GHz.

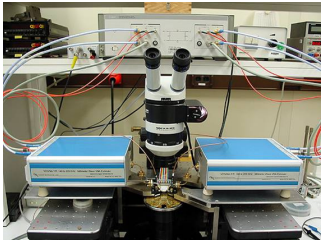


Figure 156: On-wafer S-parameter measurement setup with waveguide frequency extentions.

S-Parameter On-Wafer Measurements

Probe configurations (157)

- coplanar (ground-signal-ground) RF probes
- DC probes with integrated RF-shunting capacitors, e.g. GPPGPPG

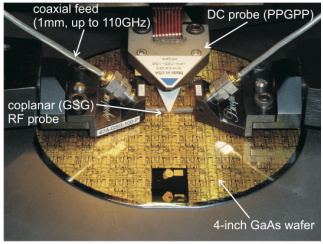


Figure 157: On-wafer probing for S-parameter measurements.

Working With Network Parameters

Model parameter extraction

- For a given equivalent circuit model network, choose appropriate (best suited) two-port parameters
  - Y for mainly parallel networks (e.g. FET)
  - Z for mainly serial networks
  - ABCD for cascaded networks
- Convert measured S-parameters to chosen network parameters
- Calculate (or fit) equivalent circuit parameters to measured network parameters

7.1 N-Port Parameters
7.2 Scattering Parameters
7.3 The Smith Chart
7.4 Rollet Stability Factor and Power Gains
7.5 Power Matching and Impedance Transformation

The Smith Chart

7.3

The Smith Chart

graphical representation of relation $Z, Y = f(\Gamma)$

- $\Gamma = \frac{Z-Z_0}{Z+Z_0} = \frac{\frac{Z}{Z_0}-1}{\frac{Z}{Z_0}+1} = \frac{z-1}{z+1}$
  - impedance chart: map normalized Z- to Γ-plane
- $\Gamma = \frac{Y_0-Y}{Y_0+Y} = \frac{1-\frac{Y}{Y_0}}{1+\frac{Y}{Y_0}} = \frac{1-y}{1+y}$
  - admittance chart: map normalized Y- to Γ-plane

The Smith Chart

Z to Γ transformation

- $\Gamma = \frac{z-1}{z+1} = |\Gamma| e^{j\Theta} = \Gamma_r + j\Gamma_i$
  - $z = \frac{1+|\Gamma|e^{j\Theta}}{1-|\Gamma|e^{j\Theta}} = r + jx = \frac{1+\Gamma_r+j\Gamma_i}{1-\Gamma_r-j\Gamma_i}$
  - $r = \frac{1-\Gamma_r^2-\Gamma_i^2}{(1-\Gamma_r^2)+\Gamma_i^2}$
  - $x = \frac{2\Gamma_i}{(1-\Gamma_r^2)+\Gamma_i^2}$

The Smith Chart

circles in the Γ-plane

$$\left(\Gamma_r - \frac{r}{1+r}\right)^2 + \Gamma_i^2 = \left(\frac{1}{1+r}\right)^2 \tag{186}$$

- resistance circles (center on the horizontal $\Gamma_i = 0$ line)

$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x}\right)^2 = \left(\frac{1}{x}\right)^2 \tag{187}$$

- reactance circles (center on the vertical $\Gamma_r = 1$ line)

Graphical Z to Γ Transformation (158)

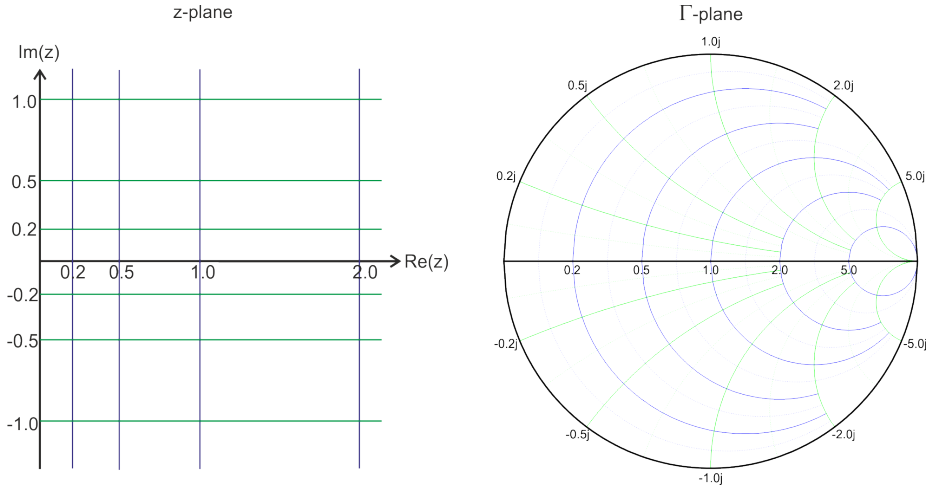


Figure 158: Graphical Z to Γ transformation.

The Impedance Smith Chart

Smith chart in "impedance form" (159)

- polar coordinates: reflection coefficient $\Gamma = |\Gamma| e^{j\Theta}$
- blue circles = impedance of constant real part (normalized to Z_0, typically $50\,\Omega$)
- green circles = impedance of constant imaginary part (normalized to Z_0)

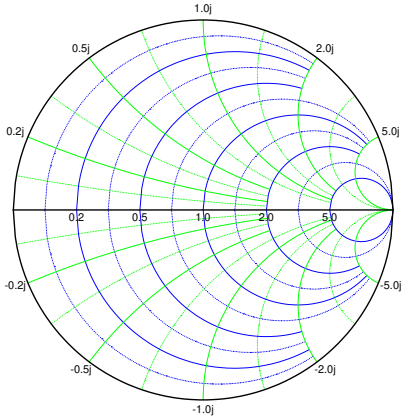


Figure 159: The impedance Smith chart.

The Admittance Smith Chart

Smith chart in "admittance form" (160)

- polar coordinates: reflection coefficient
- blue circles = admittance constant real part (normalized to $Y_0 = Z_0^{-1}$, typically $20\,\mathrm{mS}$)
- green circles = admittance constant imaginary part (normalized to Y_0)

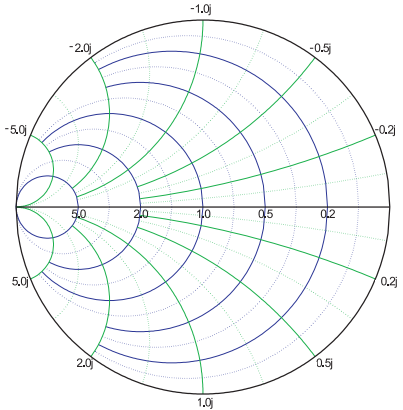


Figure 160: The admittance Smith chart.

Combined Impedance Admittance Smith Chart

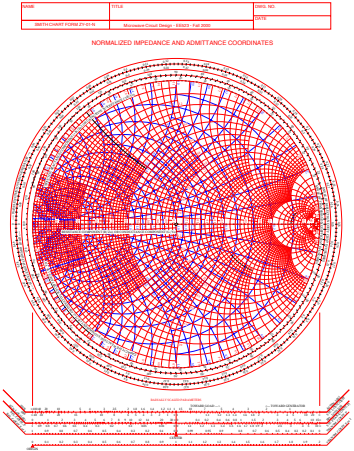


Figure 161: Combined impedance and admittance Smith charts.

7.1 N-Port Parameters
7.2 Scattering Parameters
7.3 The Smith Chart
7.4 Rollet Stability Factor and Power Gains
7.5 Power Matching and Impedance Transformation

Rollet Stability Factor and Power Gains

7.4

Motivation: Network/Circuit Stability Analysis

"design for manufacturing" (Fig. 162)

- stability analysis is an integral part and takes a significant fraction of the total time and resource effort in a full MMIC design cycle

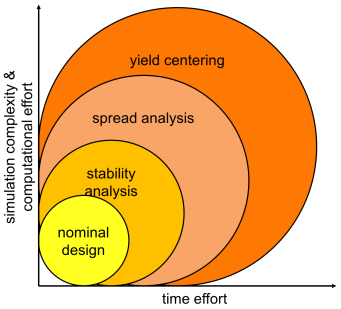


Figure 162: Design for manufacturing - MMIC design flow.

Rollet Stability Factor k

Unconditional stability (Fig. 163)

- $|\Gamma_1| < 1$ for any passive load impedance Z_L or load reflection factor Γ_L

and ($\wedge$)

- $|\Gamma_2| < 1$ for any passive source impedance Z_S or source reflection factor Γ_S

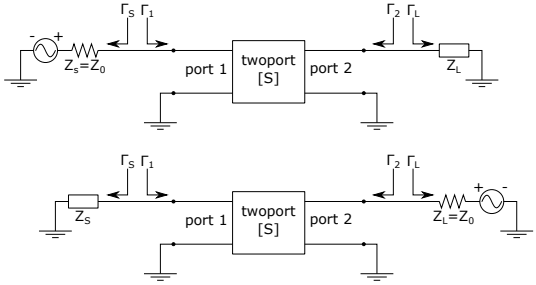


Figure 163: Approach of the Rollet stability factor.

Rollet Stability Factor k

"Passive" impedance

$$\mathrm{Re}\,(Z) \geq 0 \tag{188}$$

$$|\Gamma| \leq 1 \tag{189}$$

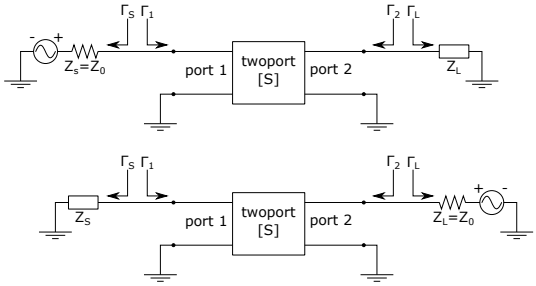


Figure 163: Approach of the Rollet stability factor.

Rollet Stability Factor k

- @input (@output): find load (source) impedances which lead to $|\Gamma_1| > 1$ ($|\Gamma_2| > 1$)
- requires a transformation of the load into the source plane and vice versa

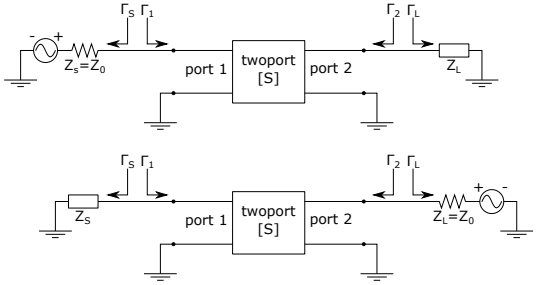


Figure 163: Approach of the Rollet stability factor.

Transformation of L- into G-plane and vice versa

derivation of $\Gamma_{1,2} = f\left([S], \Gamma_{L,S}\right)$

$$b_1 = S_{11}a_1 + S_{12}a_2 = S_{11}a_1 + S_{12}\Gamma_L b_2 \tag{190}$$

$$b_2 = S_{21}a_1 + S_{22}a_2 = S_{21}a_1 + S_{22}\Gamma_L b_2 \tag{191}$$

$$\Gamma_L = \frac{a_2}{b_2} \tag{192}$$

- eliminate b_2:

$$b_2\left(1 - S_{22}\Gamma_L\right) = S_{21}a_1 \tag{193}$$

$$b_2 = \frac{S_{21}a_1}{1 - S_{22}\Gamma_L} \tag{194}$$

$$b_1 = S_{11}a_1 + S_{12}\Gamma_L\frac{S_{21}a_1}{1 - S_{22}\Gamma_L} = a_1\left(S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L}\right) \tag{195}$$

$$\frac{b_1}{a_1} = \Gamma_1 = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L} \tag{196}$$

for input stability, consider locus of $\Gamma_1 = 1$ in load (Γ_L) plane (164)

- $\Gamma_1 = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1-S_{22}\Gamma_L} = 1$ produces circle with
  - center $\Gamma_{L,C} = \frac{\Delta S^* S_{11} - S_{22}^*}{|\Delta S|^2 - |S_{22}|^2}$
  - radius $r_L = \frac{|S_{12}S_{21}|}{|\Delta S^2 - S_{22}^2|}$,
    $\Delta S = S_{11}S_{22} - S_{12}S_{21}$

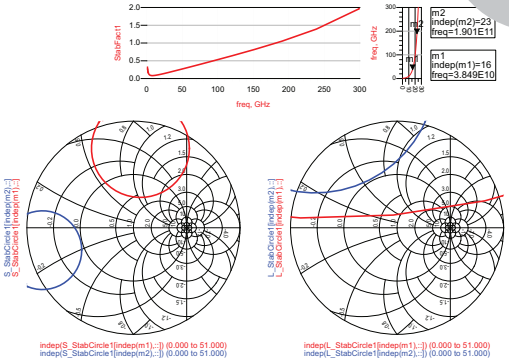


Figure 164: Input and output stability circles.

stable region

- consider e.g. load plane stability circle
- consider $Z_\mathrm{L} = Z_0 \rightarrow \Gamma_\mathrm{L} = 0$
- then $|\Gamma_1| = \left|S_{11} + \frac{S_{12}S_{21}\Gamma_\mathrm{L}}{1-S_{22}\Gamma_\mathrm{L}}\right| = |S_{11}|$
- if $|S_{11}| < 1$, then $|\Gamma_1| < 1$, therefore $\Gamma_\mathrm{L} = 0$ lies in the stable region

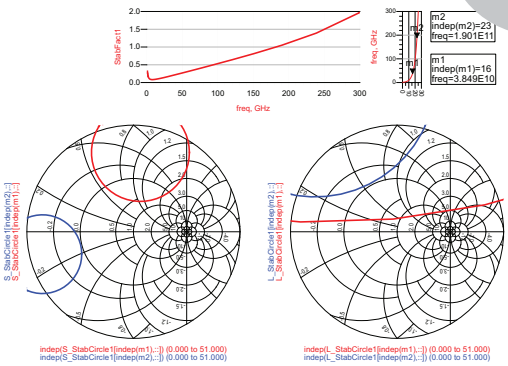


Figure 164: Input and output stability circles.

unconditional stability = circles outside $\Gamma_{G,L} = 1$ regions (164)

- load plane: $\big||\Gamma_{L,C}| - r_\mathrm{L}\big| > 1$ for $|S_{11}| < 1$
- source plane: $\big||\Gamma_{G,C}| - r_\mathrm{G}\big| > 1$ for $|S_{22}| < 1$

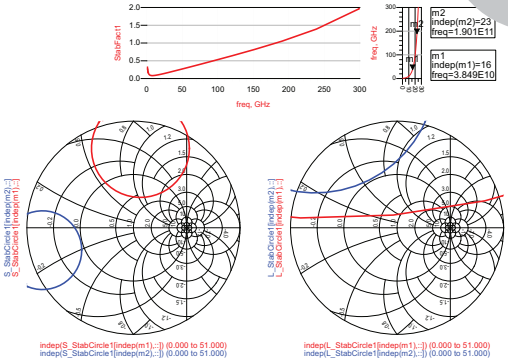


Figure 164: Input and output stability circles.

Rollet stability factor k

- fulfilled, if

$$k = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |D|^2}{2\,|S_{12}|\,|S_{21}|} > 1 \qquad (197)$$

and

$$|S_{12}|\,|S_{21}| < 1 - |S_{11}|^2 \qquad (198)$$

and

$$|S_{12}|\,|S_{21}| < 1 - |S_{22}|^2 \qquad (199)$$

- $D = S_{11}S_{22} - S_{12}S_{21}$, normally $|D| < 1$, then $k > 1$ is sufficient

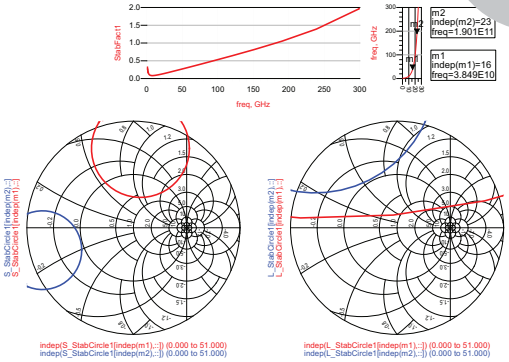


Figure 164: Input and output stability circles.

interpretation

- when unconditionally stable, the two-port remains stable under all passive load and source terminations.
- when conditionally stable, combinations of source and load impedance exist which lead to oscillation.

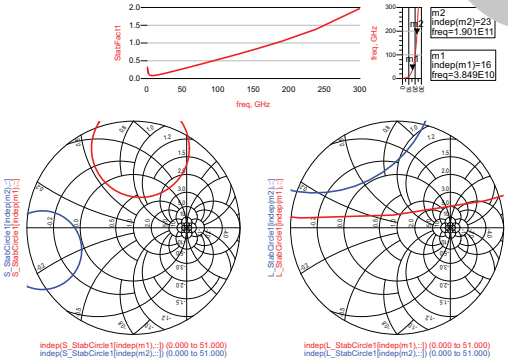


Figure 164: Input and output stability circles.

in practice

- ...extreme source and load impedances $|\Gamma| \to 1$ may occur when an amplifier is DC ramped up before being connected at its RF input or output (typically $50\,\Omega$).
- ...the increasing loss of access or connecting lines at increasing frequency helps in avoiding extreme impedances

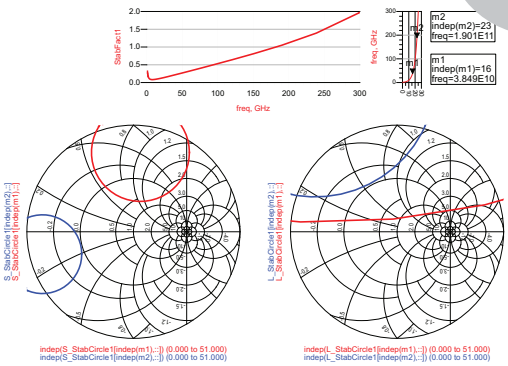


Figure 164: Input and output stability circles.

Power Gain Definitions (165)

available power gain

$$G_A = \frac{P_{avo}}{P_G} = f\left(\Gamma_G, S\right) \tag{200}$$

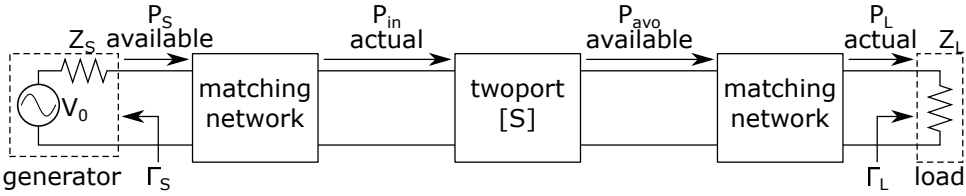


Figure 165: Power gain definitions.

Power Gain Definitions (165)

power gain

$$G = \frac{P_L}{P_{in}} = f\left(\Gamma_L, S\right) \tag{201}$$

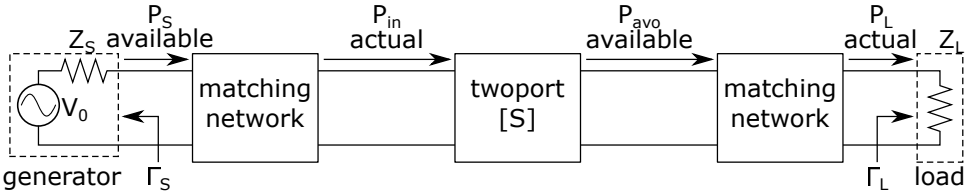


Figure 165: Power gain definitions.

Power Gain Definitions (165)

transducer gain

$$G_T = \frac{P_L}{P_G} = f\left(\Gamma_G, \Gamma_L, S\right) \tag{202}$$

- for arbitrary $\Gamma_G(Z_G)$ and $\Gamma_L(Z_L)$:

$$G_T = \frac{\left(1 - |\Gamma_G|^2\right)|S_{21}|^2\left(1 - |\Gamma_L|^2\right)}{|(1 - S_{11}\Gamma_G)(1 - S_{22}\Gamma_L) - S_{12}S_{21}\Gamma_L\Gamma_G|^2} \tag{203}$$

- in 50Ω system: $G_T = |S_{21}|^2$...which is why $|S_{21}|_{\mathrm{dB}} = 20\log|S_{21}|$ is typically called "gain"

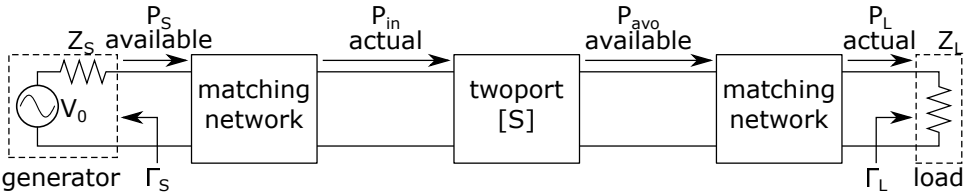


Figure 165: Power gain definitions.

Power Gain Definitions (165)

maximum achievable gain

- maximum available gain (for $k \geq 1$): $\mathrm{MAG} = \left|\frac{S_{21}}{S_{12}}\right| \left(k - \sqrt{k^2 - 1}\right)$
- maximum stable gain MSG (for $k < 1$): $\mathrm{MSG} = \left|\frac{S_{21}}{S_{12}}\right|$

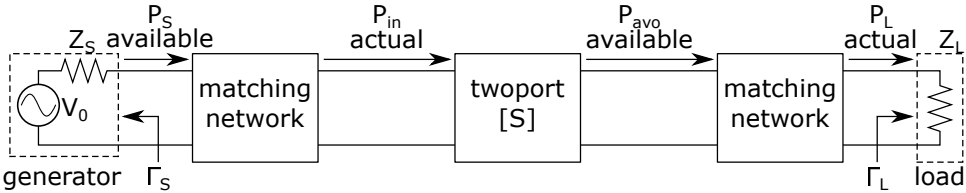


Figure 165: Power gain definitions.

Unilateral Gain

unilateral gain (166)

- unilateralisation ($S_{12} = 0$) by feedback network
- conjugate power match at in- and output
- unilateral power gain: $U = \frac{1}{2} \frac{\left|\frac{S_{21}}{S_{12}} - 1\right|^2}{k\left|\frac{S_{21}}{S_{12}}\right| - \mathrm{Re}\left(\frac{S_{21}}{S_{12}}\right)}$

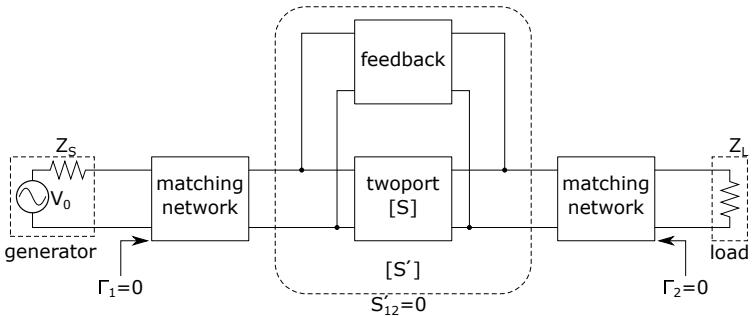


Figure 166: Unilateralisation of a two-port.

7.1 N-Port Parameters
7.2 Scattering Parameters
7.3 The Smith Chart
7.4 Rollet Stability Factor and Power Gains
7.5 Power Matching and Impedance Transformation

Power Matching and Impedance Transformation

7.5

Power Matching

Power transfer from source to load ((167))

- $V_L = V_0 \frac{Z_L}{Z_L + Z_0}$
- $P_L = \frac{1}{2}\mathrm{Re}\left(\mathrm{V_L} \cdot \mathrm{I_L^*}\right) = \frac{1}{2} V_0^2 \frac{R_L}{(R_0+R_L)^2+(X_0+X_L)^2}$

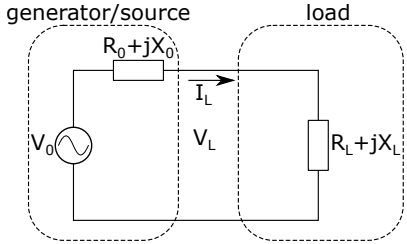


Figure 167: Source and load impedance for maximum power transfer.

Power Matching

Maximum power transfer = power matching

- $\frac{\partial P_L}{\partial R_L} = 0,\ \frac{\partial P_L}{\partial X_L} = 0$
- $X_L = -X_0$
- $R_L = R_0$

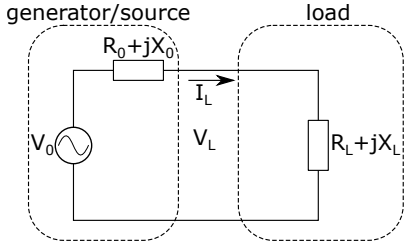


Figure 167: Source and load impedance for maximum power transfer.

Power Matching

Conjugate complex matching for maximum power transfer

$$Z_L = Z_0^* \tag{204}$$

$$Y_L = Y_0^* \tag{205}$$

$$\Gamma_L = \Gamma_0^* \tag{206}$$

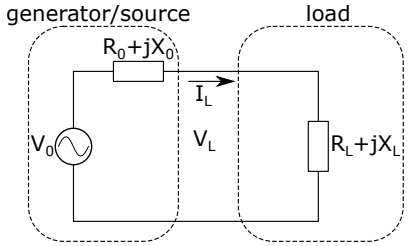


Figure 167: Source and load impedance for maximum power transfer.

Input and Output Matching Networks (168)

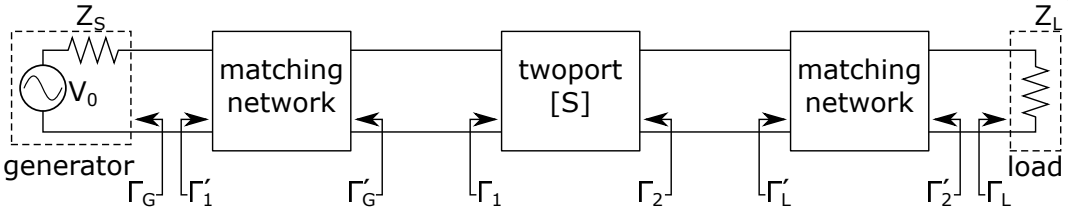


Figure 168: Input and output matching networks.

Two-port

Gain cell

- FET/BJT configuration: CS/CE, CG/CB, CD/CC, cascode, ...
- Bias condition: class A/B/C
- Input reflection coeff. Γ_1 to be matched to generator impedance Γ_G
- Output reflection coeff. Γ_2 to be matched to load impedance Γ_L

Two-port

Two-port S-parameter matrix $[S]$

- $\Gamma_1 = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1-S_{22}\Gamma_L}$
  - input reflection coeff. (and therefore also input impedance Z_1) depends on load if $S_{12} \neq 0$ (finite isolation) and $\Gamma_L \neq 0$ (load mismatch/reflection)
- $\Gamma_2 = S_{22} + \frac{S_{12}S_{21}\Gamma_G}{1-S_{11}\Gamma_G}$
  - output reflection coeff. (and therefore also output impedance Z_2) depends on load if $S_{12} \neq 0$ (finite isolation) and $\Gamma_G \neq 0$ (generator mismatch/reflection)

Input matching network (IMN)

Input matching network (IMN)

- transforms Γ_1 into Γ_1' (looking into IMN towards twoport)
- transforms Γ_G into Γ_G' (looking into IMN towards generator)
- passive
  - typically reactive (L, C, TRL) for low loss
  - R only for stability
- active:
  - e.g. common-gate/common-base stage (transform low to high impedance)

Input matching network (IMN)

Matching case: maximum power transfer

- conjugate complex input matching
- $\Gamma'_G = \Gamma_1^*$
- $\Gamma'_1 = \Gamma_G^*$
- $Z_G = 50\,\Omega$ (typical)
  - $Z_0 = 50\,\Omega \rightarrow \Gamma_G = \Gamma^*{}_G = \Gamma'_1 = 0$

Input matching network (IMN)

Matching case: low noise amplification

- input matching for optimum (low noise) source impedance
- $\Gamma'_G = \Gamma_{\mathrm{G,opt}}$ ...cf. noise parameters in chapter LNA design

Output matching network (OMN)

Output matching network (OMN)

- transforms Γ_2 into Γ_2' (looking into OMN towards twoport)
- transforms Γ_L into Γ_L' (looking into OMN towards load)
- passive
  - typically reactive (L, C, TRL) for low loss
  - R only for stability
- active:
  - e.g. source/emitter follower (= common-drain/common-collector stage), transform high to low impedance

Output matching network (OMN)

Matching case: maximum power transfer

- conjugate complex output matching
- $\Gamma'_L = \Gamma_2^*$
- $\Gamma'_2 = \Gamma_L^*$
- $Z_L = 50\,\Omega$ (typical)
  - $Z_0 = 50\,\Omega \rightarrow \Gamma_L = \Gamma^*{}_L = \Gamma'_2 = 0$

Output matching network (OMN)

Matching case: maximum output power

- output matching for maximum output power
- $\Gamma'_L = \Gamma_{\mathrm{L,opt}}$ ...cf. load pulling in section PA design

Impedance Transformations in the Smith Chart (169)

ideal lumped reactive elements: series L

- $Z' = Z_0 + j\omega L$
- $\mathrm{Im}\,(Z) \mathrel{+}= \omega L$
- the real part of the impedance (resistance) stays constant, the imaginary part (reactance) becomes more positive (inductive)

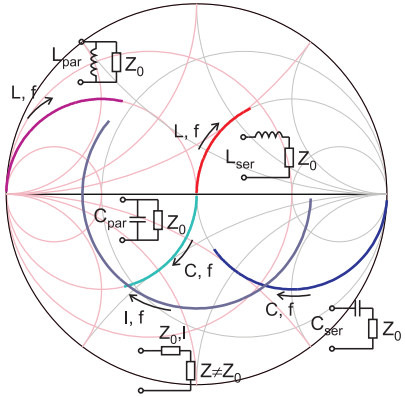


Figure 169: Impedance transformations in the Smith chart.

Impedance Transformations in the Smith Chart (169)

ideal lumped reactive elements: series C

- $Z' = Z_0 + \frac{1}{j\omega C}$
- $\mathrm{Im}\,(Z) - = 1/\omega C$
- the real part of the impedance (resistance) stays constant, the imaginary part (reactance) becomes more negative (capacitive)
- parallel L: $\mathrm{Im}\,(Y) - = 1/\omega L$, parallel C: $\mathrm{Im}\,(Y) + = \omega C$

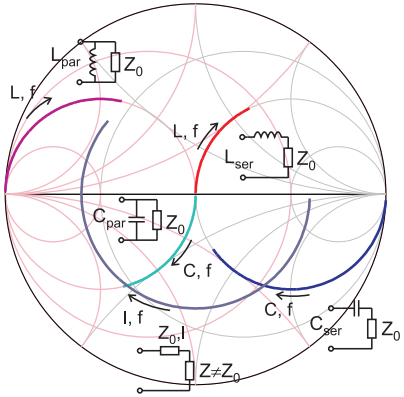


Figure 169: Impedance transformations in the Smith chart.

Impedance Transformations in the Smith Chart (169)

ideal lumped reactive elements: parallel L

- $Y' = Y_0 + \frac{1}{j\omega L}$
- $\mathrm{Im}\,(Y) - = 1/\omega L$
- the real part of the admittance (conductance) stays constant, the imaginary part (susceptance) becomes more negative (inductive)
- parallel L: $\mathrm{Im}\,(Y) - = 1/\omega L$, parallel C: $\mathrm{Im}\,(Y) + = \omega C$

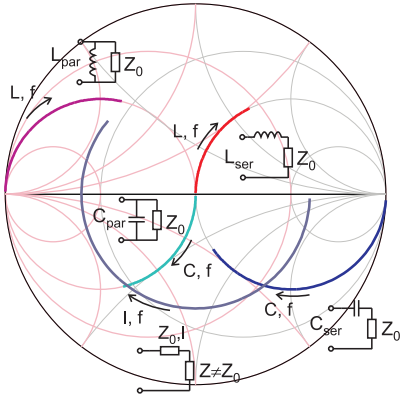


Figure 169: Impedance transformations in the Smith chart.

Impedance Transformations in the Smith Chart (169)

ideal lumped reactive elements: parallel C

- $Y' = Y_0 + j\omega C$
- $\mathrm{Im}\,(Y) \mathrel{+}= \omega C$
- the real part of the admittance (conductance) stays constant, the imaginary part (susceptance) becomes more positive (capacitive)

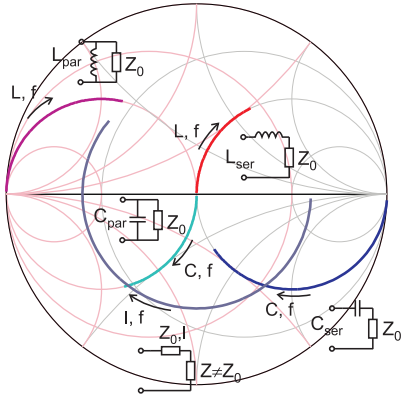


Figure 169: Impedance transformations in the Smith chart.

Impedance Transformations in the Smith Chart (169)

ideal lumped resistive elements: series resistor

- $Z' = Z_0 + R$
- $\mathrm{Re}\,(Z) \mathrel{+}= R$
- the imaginary part of the impedance (reactance) stays constant, the real part (resistance) becomes more positive (resistive)

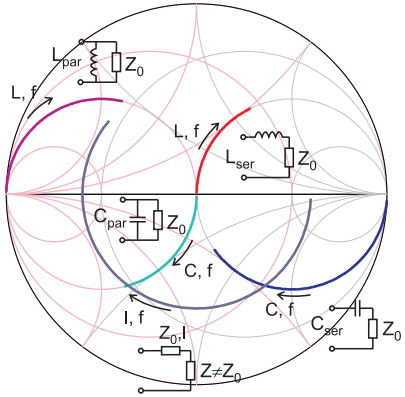


Figure 169: Impedance transformations in the Smith chart.

Impedance Transformations in the Smith Chart (169)

ideal lumped resistive elements: parallel resistor

- $Y' = Y_0 + \frac{1}{R}$
- $\mathrm{Re}\,(Y) + = \frac{1}{R}$
- the imaginary part of the admittance (susceptance) stays constant, the real part (conductance) becomes more positive (conductive)

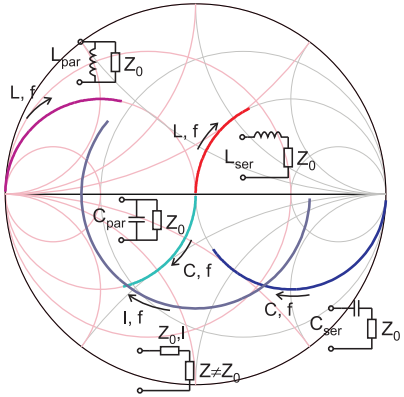


Figure 169: Impedance transformations in the Smith chart.

Impedance Transformations in the Smith Chart (169)

ideal lossless transmission lines

- signals undergo a phase shift (delay)
- transformation along a circle with center Z_{line}, if $Z_{\text{line}} = Z_0$
- cp. chapter on TRL theory

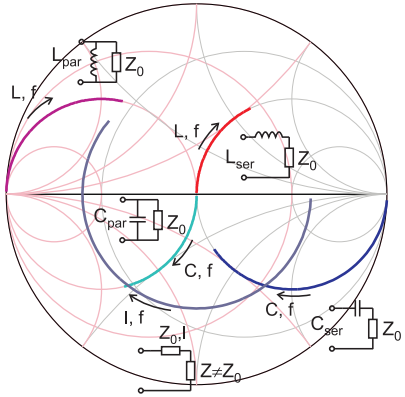


Figure 169: Impedance transformations in the Smith chart.

Impedance Transformations in the Smith Chart (169)

real elements with effective series resistance

- along the transformation path, the resistance increases

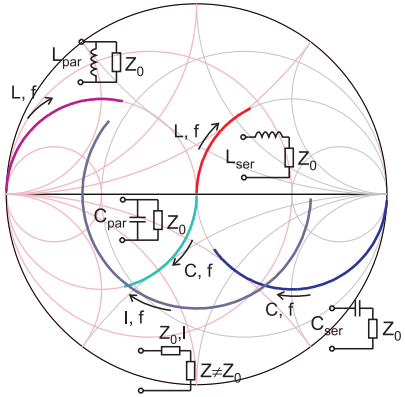


Figure 169: Impedance transformations in the Smith chart.

Impedance Transformations in the Smith Chart (169)

lossy transmission line

- how does the transformation path look like?

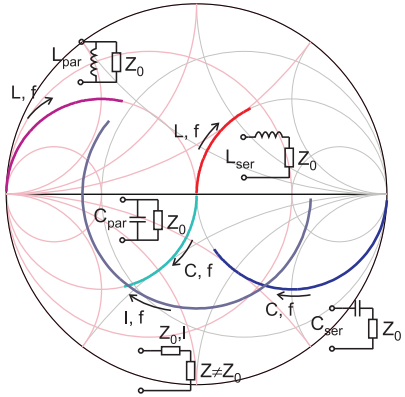


Figure 169: Impedance transformations in the Smith chart.

Planar Transmission Lines

8

1 Course Outline

2 The Millimeterwave Frequency Spectrum: Applicatio
and Technologies

3 Functional Buildung Blocks of MMICs

4 Low Noise Amplifier

5 Broadband Amplifiers

6 Power Amplifiers

7 Microwave Network Analysis (Appendix)

8.1 Introduction
8.2 Coplanar Waveguides
8.3 Transmission Line Theory
8.4 Applications of Transmission Lines in Circuit Design

Introduction

8.1

Transmission Lines

transmission lines = medium for guiding electro-magnetic waves

- cables: coaxial, twisted pair
- waveguides: metallic, dielectric
- planar: microstrip, coplanar, triplate, slot line etc.

Transmission Lines

Characteristics

- phase shift (frequency domain), time delay (time domain)
- attenuation
- (characteristic) impedance = ratio between voltage and current amplitudes along the line

Standard mmW waveguides

coax

- “1 mm” coax up to 110 GHz, air dielectric

Standard mmW waveguides

(Rectangular) waveguide

band	waveguide	freq.	dimension
V	WR-15	50 to 75	0.148 x 0.074
W	WR-10	75 to 110	0.100 x 0.050
D	WR-6	110 to 170	0.0650 x 0.0325
G	WR-5	140 to 220	0.0510 x 0.0255
(H)	WR-3	220 to 325	0.0340 x 0.0170
(Y)	WR-2	325 to 500	0.0200 x 0.0100

Table 7: Rectangular waveguide dimensions.

TRL Loss Mechanisms

conductor loss

- ohmic due to finite conductivity of conductor (e.g. Au) and adhesive (e.g. Cr, Ta) layers
- skin effect
- surface roughness

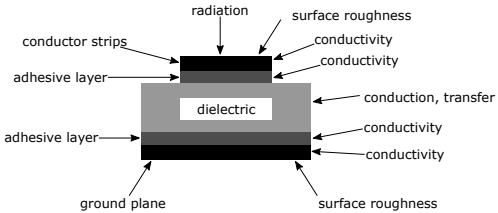


Figure 170: TRL loss mechanism.

TRL Loss Mechanisms

Dielectric Loss

- Molecules with a dipole moment get excited by the electromagnetic field and absorb energy.
- This loss mechanism can be represented by a complex permittivity:

$$\epsilon_r = \epsilon_r^{'} - j\epsilon_r^{''} \tag{207}$$

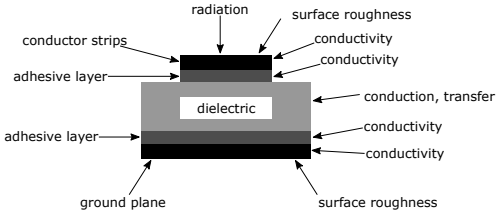


Figure 171: TRL loss mechanism.

TRL Loss Mechanisms

Dielectric Loss Tangent

$$\tan\delta = \frac{\epsilon_r''}{\epsilon_r'} \tag{208}$$

- $\tan\delta$ is typically included in data sheets of dielectric materials.
- $\tan\delta = \tan\delta(f)$ ... dispersive (= function of frequency), typically increases with frequency.

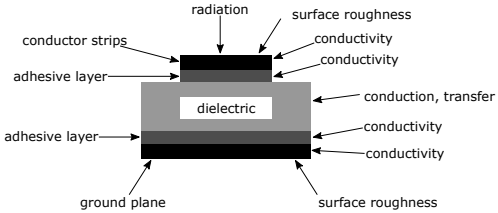


Figure 172: TRL loss mechanism.

TRL Loss Mechanisms

Substrate Loss

- ohmic due to non-zero substrate resistivity

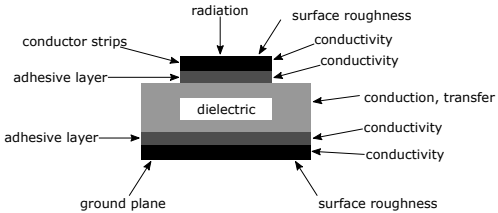


Figure 173: TRL loss mechanism.

TRL Loss Mechanisms

Radiation

- into free space (e.g. at discontinuities)
- into substrate (substrate modes)

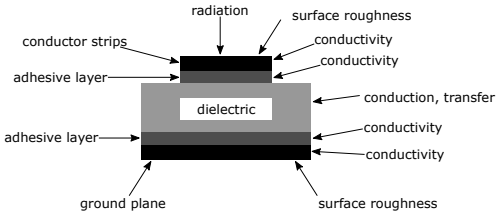


Figure 174: TRL loss mechanism.

Semiconductor Substrates

Semiconductor Specific Resistivity

$$\rho = \frac{1}{\sum_{n,p} \mu_{\text{carrier}}\, q\, n_{\text{carrier}}} \tag{209}$$

- $[\rho] = \Omega\text{cm}$
- μ ... carrier mobility, holes (p) and electrons (n)
- n ... carrier concentration

Semiconductor Substrates

Intentionally Doped Semiconductor

- e.g. p-doped Si:

$$\rho = \frac{1}{\mu_\mathrm{p}\, q\, N_\mathrm{A}} \tag{210}$$

- N_A ... acceptor doping concentration

Semiconductor Substrates

Substrate Categories

- $\rho > 1000\,\Omega\mathrm{cm}$ high resistivity (semi-insulating)
- $\rho = 100...1000\,\Omega\mathrm{cm}$ medium resistivity
- $\rho < 100\,\Omega\mathrm{cm}$ low resistivity

Semiconductor Substrates

GaAs versus Si

- GaAs is semi-insulating
- Si normally low-resistivity: cheap + large (up to 10") wafers
- high-resistivity Si: expensive wafers, early onset of inversion (surface channels)
- hence: Silicon on isolator (SOI) processes for high-speed Si CMOS

Skin Effect

Homogenous Conductor with Circular Cross-Section

- skin depth:

$$\delta = \sqrt{\frac{2}{\omega \mu_0 \sigma}} \tag{211}$$

- μ_0 ... permeability in free-space, σ ... specific conductivity

- distance from conductor surface where normalised current density has decreased to $\frac{1}{e}$

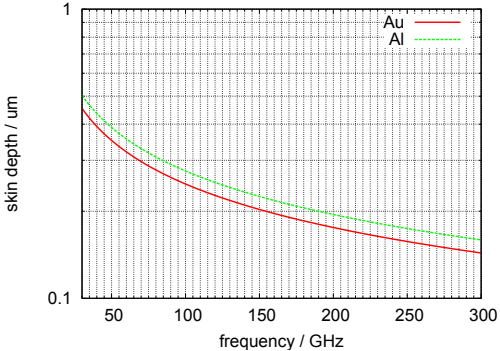


Figure 175: Skin depth.

Skin Effect

Implications

- At elevated frequencies, a thicker metallisation does not lead to reduced loss
- The effect of surface roughness becomes more dominant at elevated frequencies

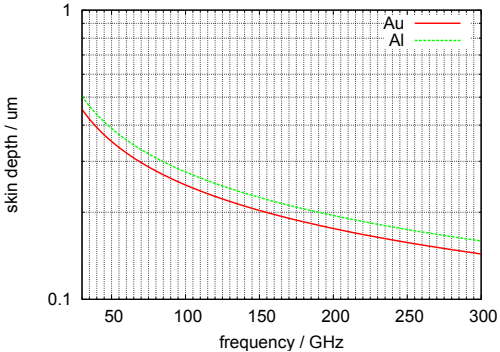


Figure 176: Skin depth.

TRL Loss

Example: TRL loss of 50-Ω microstrip lines

- substrate thickness $100\,\mu\mathrm{m}$, metal thickness $1\,\mu\mathrm{m}$
- neglects surface roughness

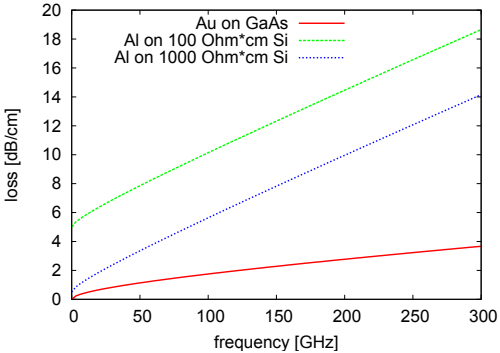


Figure 177: TRL loss.

Microstrip Transmission Lines (MSL)

Advantages

- most popular technique with largest body of CAD models
- radiation may be used for on-chip antennas

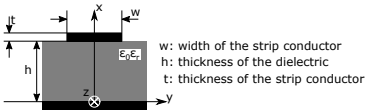


Figure 178: Cross-section of a microstrip transmission line.

MSL

Disadvantages

- requires via-holes and backside metallisation
- hot vias (e.g. in transistor source) introduce parasitic inductance
- radiation introduces loss and strong cross-coupling effects

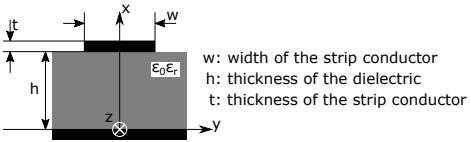


Figure 179: Cross-section of a microstrip transmission line.

Thin-Film Microstrip

MSL on low-loss dielectrics (silicon nitride (SiN), silicon oxide (SiO_2), benzocyclobutene (BCB))

- Used mainly in Si-based technologies with several metal levels
- But: dielectric thickness becomes comparable to metal thickness
- Therefore, zero-thickness approximation is no longer valid and standard empirical models are not applicable

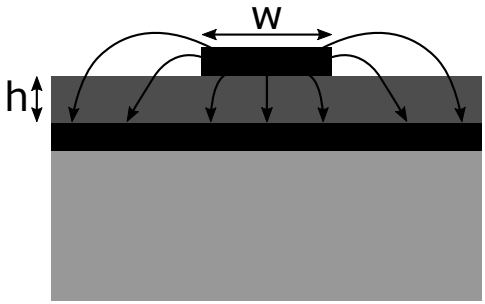


Figure 180: Thin film microstrip line.

Field Distribution in Microstrip Lines

Fundamental wave mode: HE_0

- Lower cutoff $f_{\mathrm{g}0} = 0$
- Onset of first higher order wave HE_1 for $\frac{w}{h} > 1$:

$$f_{\mathrm{g}1} \approx \frac{c_0}{2w\sqrt{\epsilon_r}} \tag{212}$$

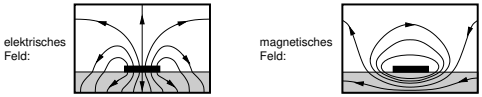


Figure 181: Field distribution in MSL.

Field Distribution in Microstrip Lines

Higher order modes: $\mathrm{HE_n}$

- Onset of n-th higher order mode:

$$f_{\mathrm{gn}} \approx \frac{nc_0}{2w\sqrt{\epsilon_r}} \tag{213}$$

- Onset of radiation loss

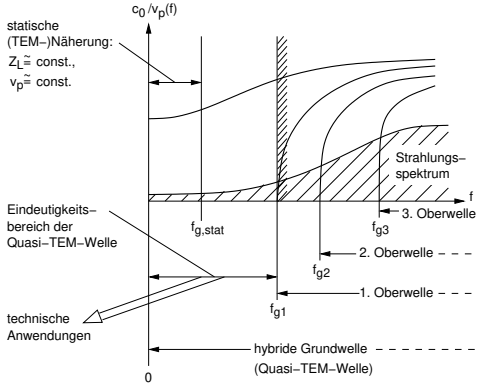


Figure 182: Higher order modes in MSL.

TRL: CAD Models

Geometrical Parameters

- MSL: substrate thickness, signal line width, signal line thickness, ...
- coplanar waveguide (CPW): spacing signal line to ground plane, signal line width, ground-to-ground spacing, ...

TRL Parameters

- Characteristic impedance, propagation constant
- Attenuation constant, loss tangent, dispersion...

Modelling Approaches

1. Physical: TRL parameters as function of geometry and material parameters (see next slide)
2. Empirical: Extract TRL parameters from measurement (S-parameters)

Thin-Film Microstrip: Conformal Mapping

Conformal mapping of MSL

- Transformation to a parallel-plate capacitor with w_{eff} and $\epsilon_{\mathrm{r,eff}}$
- Effective width for same capacitance per unit area:

$$C' = \epsilon_0 \epsilon_{\mathrm{r,eff}} \frac{w_{\mathrm{eff}}}{h} \tag{214}$$

Assumptions

- Ideal (lossless) dielectric and conductors
- Zero conductor thickness
- No dispersion

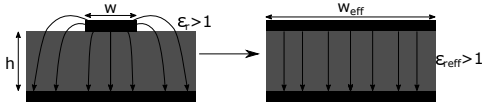


Figure 183: Conformal mapping of a microstrip line to an MIM capacitor.

Thin-Film Microstrip: Design Equations

Wheeler (1977):

$$Z_{\mathrm{L}} = \frac{42.4}{\sqrt{\epsilon_r + 1}} \ln\left(1 + \frac{4h}{w}\left(\left(\frac{14 + \frac{8}{\epsilon_r}}{11}\right)\frac{4h}{w} + \sqrt{\left(\frac{14 + \frac{8}{\epsilon_r}}{11}\right)^2\left(\frac{4h}{w}\right)^2 + \frac{\pi^2}{2}\left(1 + \frac{1}{\epsilon_r}\right)}\right)\right) \tag{215}$$

Hammerstad/Jensen (1980):

$$Z_{\mathrm{L}0} = Z_{\mathrm{L},\epsilon_r=1} = \frac{Z_{\mathrm{F}0}}{2\pi} \ln\left(\frac{F_1 h}{w} + \sqrt{1 + \left(\frac{2w}{h}\right)^2}\right) \tag{216}$$

$Z_{\mathrm{F}0} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\pi\Omega$ ... free space characteristic impedance

$F_1 = 6 + (2\pi - 6)\exp\left(-\left(30.666\frac{h}{w}\right)^{0.7528}\right)$

$\epsilon_{\mathrm{r,eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2}\left(1 + \frac{10h}{w}\right)^{-ab}$

$a = 1 + \frac{1}{49}\ln\left(\frac{(\frac{w}{h})^4 + (\frac{w}{52h})^2}{(\frac{w}{h})^4 + 0.432}\right) + \frac{1}{18.7}\ln\left(1 + (\frac{w}{18.1h})^3\right)$

$b = 0.564 \cdot \left(\frac{\epsilon_r - 0.9}{\epsilon_r + 3}\right)^{0.053}$

$Z_0 = Z_{\mathrm{L}0}\frac{1}{\sqrt{\epsilon_{\mathrm{r,eff}}}}$

Thin-Film Microstrip: Design Equations

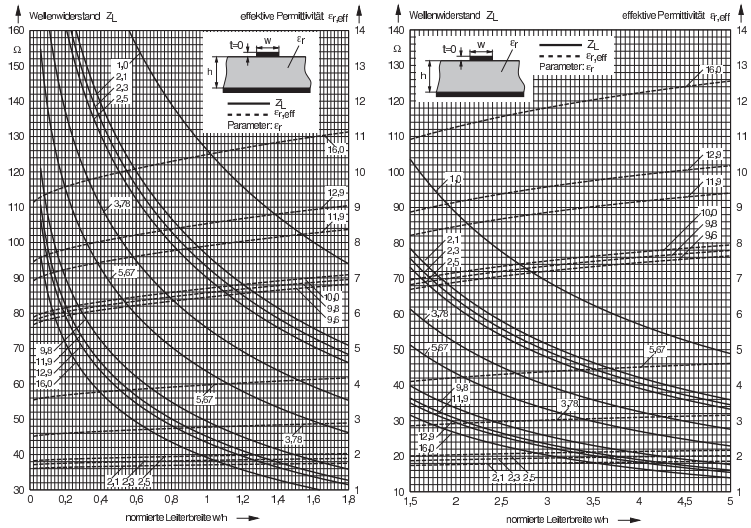


Figure 184: Numerically calculated characteristic impedance Z_L and effective permittivity $\epsilon_{r,eff}$.

8.1 Introduction
8.2 Coplanar Waveguides
8.3 Transmission Line Theory
8.4 Applications of Transmission Lines in Circuit Design

Coplanar Waveguides

8.2

Coplanar Waveguide (CPW)

Advantages

- no backplane or via-holes needed
- suitable for on-wafer characterization during fabrication process (coplanar probe tips)
- center conductor may be varied without changing the characteristic impedance:

$$Z_{0,\text{CPW}} = f(s, w) \tag{217}$$

- lower cross-coupling than MSL

Disadvantages

- higher loss than MSL due to skin effect and field concentration at conductor edges
- higher area consumption due to ground planes
- odd-mode propagation requires airbridges

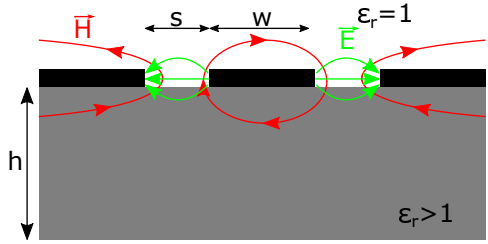


Figure 185: Cross-section of a coplanar waveguide.

Parasitic Mode Suppression in Coplanar Waveguides

Odd-mode in CPW lines

- odd-mode: ground planes on different potential

Odd-mode suppression in CPW lines

- avoid by repeated insertion of short circuits across TRL
- requires:
  - second metal + airbridge process
  - or bond wires

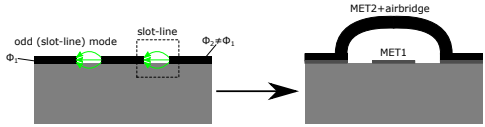


Figure 186: Odd-mode suppression in CPW lines.

Parasitic Mode Suppression in Coplanar Waveguides

Substrate modes in CPW lines

- power loss through propagation in substrate (waveguide modes) occurs beyond approx. 100 GHz
- can be avoided by wafer thinning and insertion of via-holes
- often referred to as “Through-Substrate Vias” (Through-Substrate-Via (TSV))
- requires backside process → grounded CPW

Grounded Coplanar: grounded coplanar waveguide (GCPW)

- CPW-to-MSL transitions and on-chip combinations possible
- compatibility to microstrip mounting technology

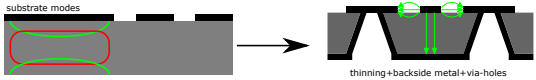


Figure 187: Substrate mode suppression in CPW lines.

8.1 Introduction
8.2 Coplanar Waveguides
8.3 Transmission Line Theory
8.4 Applications of Transmission Lines in Circuit Design

Transmission Line Theory

8.3

Transmission Line Equations

TRL equivalent circuit

- C' ... capacitance per unit length
- L' ... inductance per unit length
- R' ... resistance per unit length
- G' ... conductance per unit length
- $Z' = R' + j\omega L'$
- $Y' = G' + j\omega C'$

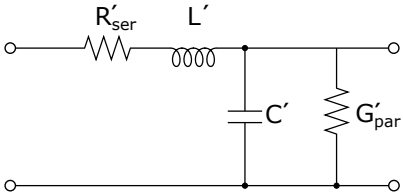


Figure 188: TRL equivalent circuit.

Ansatz:

$$\frac{dV}{dx} = IZ' \tag{218}$$

$$\frac{dI}{dx} = VY' \tag{219}$$

- set of coupled differential equations

Transmission Line Equations

Solution of transmission line differential equations

$$V = V^+ + V^- = V_0^+ e^{-\gamma x} + V_0^- e^{\gamma x} \tag{220}$$

$$I = I^+ - I^- = \frac{V_0^+}{Z_0} e^{-\gamma x} - \frac{V_0^-}{Z_0} e^{\gamma x} \tag{221}$$

- V^+, I^+ ... voltage/current wave travelling towards load
- V^-, I^- ... voltage/current wave travelling towards source
- Z_0 ... characteristic (line) impedance

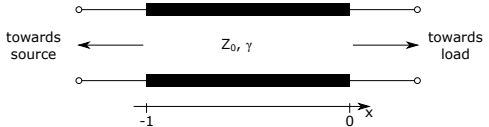


Figure 189: Signals along a transmission line.

Transmission Line Equations

$$\gamma = \alpha + j\beta = \sqrt{(R' + j\omega L')(G' + j\omega C')} \qquad (222)$$

- γ ... propagation constant
- α ... attenuation constant
- β ... phase constant

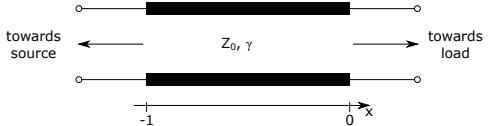


Figure 190: Signals along a transmission line.

Transmission Line Equations

$$Z_0 = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}} \tag{223}$$

- Z_0 ... characteristic (line) impedance

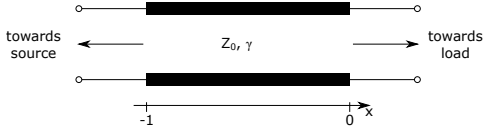


Figure 191: Signals along a transmission line.

Lossless / Weakly Lossy Transmission Line

$$V = V^+ + V^- = V_0^+ e^{-j\beta x} + V_0^- e^{j\beta x} \tag{224}$$

$$I = I^+ - I^- = \frac{V_0^+}{Z_0} e^{-j\beta x} - \frac{V_0^-}{Z_0} e^{j\beta x} \tag{225}$$

$$\beta = \omega\sqrt{L'C'} = \frac{2\pi}{\lambda'} \tag{226}$$

- λ' ... wavelength on the line
- $Z_0 = \sqrt{L'/C'}$ ... characteristic impedance

Lossless / Weakly Lossy Transmission Line

- phase velocity:

$$v_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{L'C'}} \tag{227}$$

- group velocity:

$$v_g = \frac{d\omega}{d\beta} \tag{228}$$

- group delay:

$$\tau = \frac{d(\beta l)}{df} = \frac{2\pi}{v_g} l \tag{229}$$

Lossless / Weakly Lossy Transmission Line

- excl. dispersion:

$$v_g = v_p = \frac{c_0}{\sqrt{\epsilon_{\text{eff}}}} \tag{230}$$

- wavelength on the line:

$$\lambda' = \frac{c_0}{\sqrt{\epsilon_{\text{eff}}} f} = \frac{\lambda_0}{\sqrt{\epsilon_{\text{eff}}}} \tag{231}$$

- ϵ_{eff} ... effective dielectric constant
- dependent on propagation medium, e.g. permittivity of dielectric in a coaxial transmission line

Dielectric constant of selected materials in mmW Technology

Material	ϵ_r
bulk Halbleiter	
Si	11.7
GaAs	12.9
Dielektrikum / Isolator	
SiO_2	4.5
SiN	ca. 6
BCB	2.7
MIC (MIC)-Platinen	
Quartz	3.9
flame retardant (FR)-4	ca. 4.6
Rogers 4003	3.4

Table 8: Dielectric constant of selected materials in mmW Technology.

Wavelength on Substrate

Free-space propagation:

$$\lambda = \frac{c_0}{\nu}, \quad c_0 = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \tag{232}$$

Propagation on substrate:

$$c' = \frac{c_0}{\sqrt{\epsilon_{\mathrm{r,eff}}}}, \quad \lambda' = \frac{\lambda}{\sqrt{\epsilon_{\mathrm{r,eff}}}} \tag{233}$$

velocity factor: $\frac{1}{\sqrt{\epsilon_{\mathrm{r,eff}}}}$

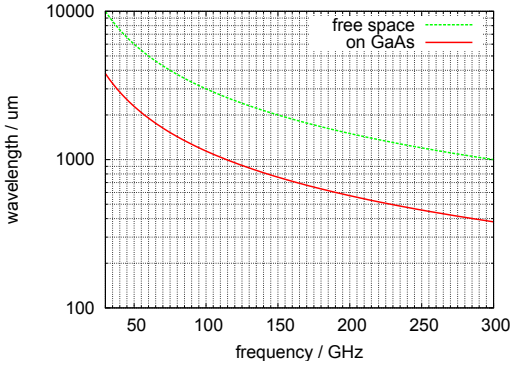


Figure 192: Wavelength on typical semiconductor substrates in mmW.

Applications of Transmission Lines in Circuit Design

8.4

8.1 Introduction
8.2 Coplanar Waveguides
8.3 Transmission Line Theory
8.4 Applications of Transmission Lines in Circuit Design

Terminated Lossless Transmission Line

lossless TRL

$$V(x) = V^+(x) + V^-(x) = V_0^+ e^{-j\beta x} + V_0^- e^{j\beta x} \quad (234)$$

$$I(x) = I^+(x) - I^-(x) = \frac{V_0^+}{Z_0} e^{-j\beta x} - \frac{V_0^-}{Z_0} e^{j\beta x} \quad (235)$$

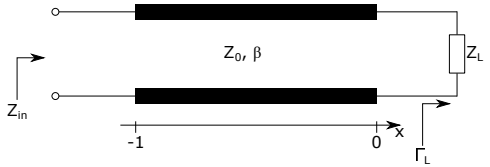


Figure 193: Impedance transformation along a transmission line

Terminated Lossless Transmission Line

at load $x = 0$

$$Z_L = \frac{V(0)}{I(0)} = \frac{V_0^+ + V_0^-}{V_0^+ - V_0^-} Z_0 \tag{236}$$

$$V_0^- = \frac{Z_L - Z_0}{Z_L + Z_0} V_0^+ \tag{237}$$

$$\Gamma_L = \frac{V^-}{V^+} = \frac{Z_L - Z_0}{Z_L + Z_0} \tag{238}$$

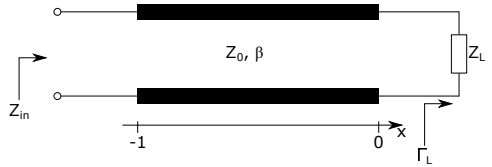


Figure 194: Impedance transformation along a transmission line

Terminated Lossless Transmission Line

Voltage and Current Waves

$$V(x) = V_0^+ e^{-j\beta x} + V_0^- e^{j\beta x} = V_0^+ \left(e^{-j\beta x} + \Gamma_L e^{j\beta x} \right) \tag{239}$$

$$I(x) = \frac{V_0^+}{Z_0} \left(e^{-j\beta x} - \Gamma_L e^{j\beta x} \right) \tag{240}$$

- Superposition of incident and reflected wave = standing wave

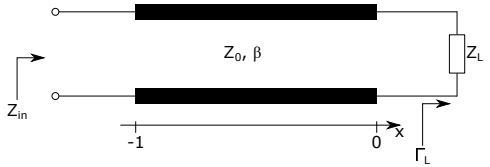


Figure 195: Impedance transformation along a transmission line

Terminated Lossless Transmission Line

Reflection Cases

- $Z_L = Z_0$ (matched load): $\Gamma = 0$ (no reflection)
- $Z_L = 0$ (electrical short): $\Gamma = -1$ (full reflection with phase reversal)
- $Z_L = \infty$ (electrical open): $\Gamma = 1$ (full reflection without phase reversal)

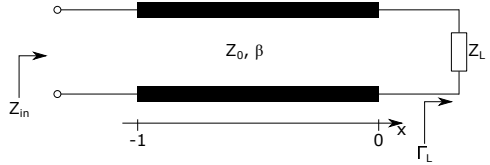


Figure 196: Impedance transformation along a transmission line

Terminated Lossless Transmission Line

Voltage magnitude along the line

- $|V(x)| = \left|V_0^+\right| \left|e^{-j\beta x} + \Gamma e^{j\beta x}\right| = \left|V_0^+\right| \left|1 + \Gamma e^{j2\beta x}\right| = \left|V_0^+\right| \left|1 + \Gamma e^{-j2\beta l}\right|$
- $\Gamma = |\Gamma| e^{j\Theta}$

$$|V(x)| = \left|V_0^+\right| \left|1 + |\Gamma| e^{j(\Theta - 2\beta l)}\right| \qquad (241)$$

- ... "standing wave"

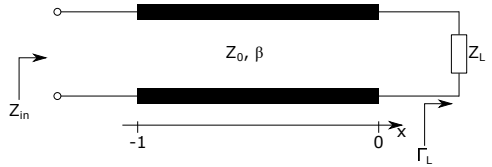


Figure 197: Impedance transformation along a transmission line

Terminated Lossless Transmission Line

voltage standing wave ratio (VSWR)

- Maximum voltage: $V_{\max} = |V_0^+|(1 + |\Gamma|)$
- Minimum voltage: $V_{\min} = |V_0^+|(1 - |\Gamma|)$
- Voltage standing wave ratio: $\mathrm{VSWR} = \frac{V_{\max}}{V_{\min}} = \frac{1+|\Gamma|}{1-|\Gamma|}$
- $1 \leq \mathrm{VSWR} \leq \infty$

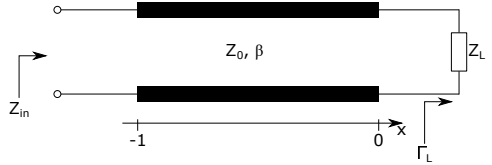


Figure 198: Impedance transformation along a transmission line

Terminated Lossless Transmission Line

Standing waves along the line

- Full reflection: $V_{\max} = 2|V_0^+|$, $V_{\min} = 0$
- No reflection: $V_{\max} = V_{\min} = |V_0^+|$
- Amplitude extrema occur at: $\Theta - 2\beta l = n\frac{\pi}{2}$

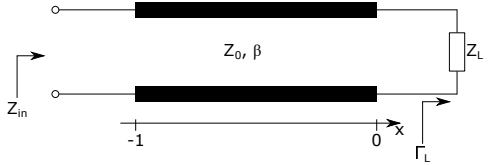


Figure 199: Impedance transformation along a transmission line

Terminated Lossless Transmission Line

Input impedance

- Impedance along the line:
  $Z(x) = \frac{V(x)}{I(x)} = Z_0 \frac{e^{-j\beta x} + \Gamma_L e^{j\beta x}}{e^{-j\beta x} - \Gamma_L e^{j\beta x}}$
- At $x = -l$: $Z_{\text{in}} = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l}$

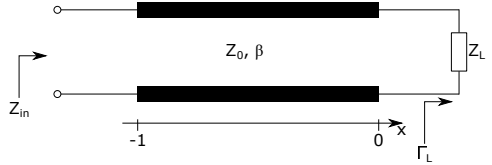


Figure 200: Impedance transformation along a transmission line

Terminated Lossless Transmission Line

Reflection coefficient along the line

- $V(x) = V_0^+ \left(e^{-j\beta x} + \Gamma e^{j\beta x}\right)$
- Reflection coefficient:
  $\Gamma(x) = \frac{V(x)^-}{V(x)^+} = \frac{V_0^-}{V_0^+} e^{j2\beta x} = \Gamma_L e^{j2\beta x}$
- After a line of length l: $\Gamma(-l) = \Gamma_L e^{-j2\beta l}$
- $|\Gamma| = \text{const.}$, phase change $2\beta l$ (clockwise in Smith chart)
- Periodicity: line of length $\lambda/2 \rightarrow$ phase change $2\beta l = 2\pi$ (full rotation in Smith chart)

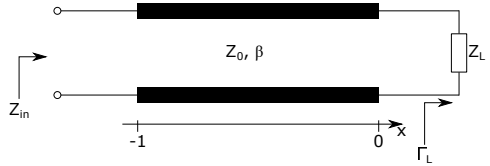


Figure 201: Impedance transformation along a transmission line

Impedance Transformation by Lossless Transmission Line

- $Z_0 = 50\,\Omega$
- $Z_L = 50\,\Omega + j100\,\Omega \rightarrow z_L = 1 + j2$
- line with $Z_0 = 50\,\Omega$ of length 0.2λ
  - $z_{in} = 0.3 - j0.8$

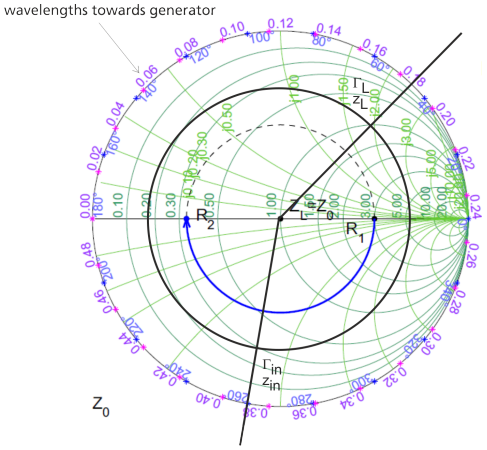


Figure 202: Impedance transformation along a transmission line.

Quarter-Wave Transformer

$\lambda/4$ line

- $Z_{\text{in}} = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l} = Z_0 \frac{Z_L + jZ_0 \tan \frac{2\pi}{\lambda'} l}{Z_0 + jZ_L \tan \frac{2\pi}{\lambda'} l}, \quad l = \frac{\lambda'}{4}$

$$Z_{\text{in}} = Z_0 \frac{Z_L + jZ_0 \tan \frac{\pi}{2}}{Z_0 + jZ_L \tan \frac{\pi}{2}} = \frac{Z_0^2}{Z_L} \tag{242}$$

- using $\lim_{x \to \frac{\pi}{2}} \tan x = \infty$

$\lambda/4$ transformer

- a short circuit is transformed into an open circuit and vice versa
- to transform R_{A} into R_{B}, use a $\lambda/4$ transformer with characteristic impedance

$$Z_0 = \sqrt{R_{\text{A}} R_{\text{B}}} \tag{243}$$

$\lambda/4$ Wavelength on Substrate

electrical length

- Consider velocity factor $\frac{1}{\sqrt{\epsilon_{\mathrm{r,eff}}}}$ of the transmission line

interconnect lines of electronic circuits

- even at moderate frequencies, interconnections between circuit nodes may have a non-negligible effect on impedance transformation

end effect

- electrical length $\neq$ geometrical length for a termination into open or short (strong longitudinal field component)

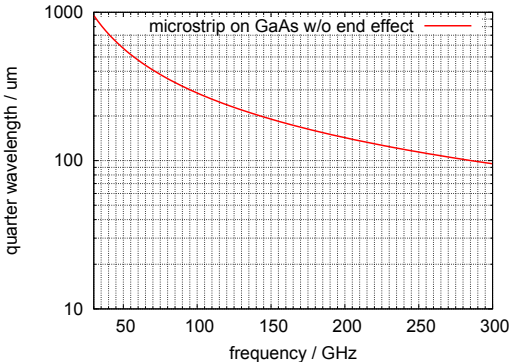


Figure 203: $\lambda/4$ wavelength on substrate

$\lambda/4$ transformer in biasing circuits

$\lambda/4$ transformers in biasing circuits

- de-couple influence of off-chip elements (e.g., bond-wire, probes, cables, etc.) from the circuit

attention: out-of-band characteristics

- short-to-open transformation valid only at rf center frequency
- at low frequencies, the transmission line represents a short circuit
- always consider out-of-band impedances, e.g., for stability analysis

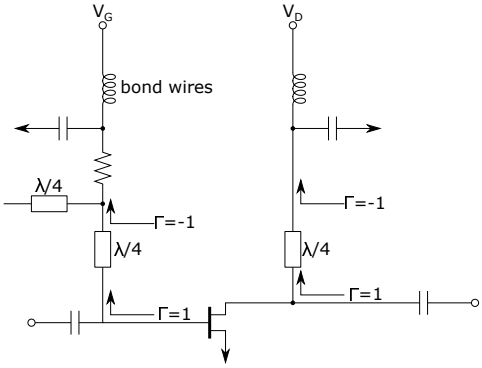


Figure 204: $\lambda/4$ transformer in biasing circuits

$\lambda/4$ Transformer as Coplanar-to-Microstrip Transition

- on-wafer probing of microstrip circuits
- requires CPW (probe) to MSL (circuit) transition
- use $\lambda/4$ stubs to provide coplanar (RF) ground
- broadband version: radial stubs

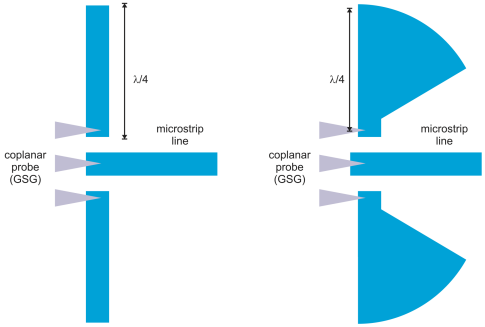


Figure 205: Radial stubs for coplanar-to-microstrip transitions.

Open Stubs

- open stub input impedance
- $Z_{\text{in}} = Z_0 \frac{Z_{\text{L}} + jZ_0 \tan \beta l}{Z_0 + jZ_{\text{L}} \tan \beta l} = Z_0 \frac{Z_{\text{L}} + jZ_0 \tan \frac{2\pi}{\lambda'} l}{Z_0 + jZ_{\text{L}} \tan \frac{2\pi}{\lambda'} l}$, $Z_{\text{L}} = \infty$

$$Z_{\text{in}} = Z_0 \frac{1}{j \tan \frac{2\pi}{\lambda'} l} = -jZ_0 \cot \frac{2\pi}{\lambda'} l \qquad (244)$$

- if $\beta l < \frac{\pi}{2}$, i.e. $l < \frac{\lambda'}{4}$, input impedance of open stub is capacitive: $C_{\text{eq}} = \frac{\tan \frac{2\pi}{\lambda'} l}{\omega Z_0}$
- for short stubs ($\tan x \approx x$): $C_{\text{eq}} = \frac{\sqrt{\epsilon_{\text{eff}}} l}{c_0 Z_0}$
- frequency-independent

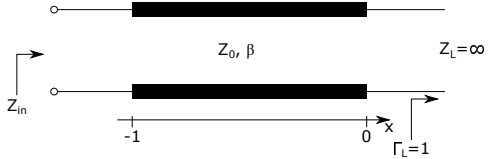


Figure 206: Impedance transformation by an open stub.

Shorted Stubs

- shorted stub input impedance
- $Z_{\text{in}} = Z_0 \frac{Z_{\text{L}} + jZ_0 \tan \beta l}{Z_0 + jZ_{\text{L}} \tan \beta l} = Z_0 \frac{Z_{\text{L}} + jZ_0 \tan \frac{2\pi}{\lambda'} l}{Z_0 + jZ_{\text{L}} \tan \frac{2\pi}{\lambda'} l}$, $Z_{\text{L}} = 0$

$$Z_{\text{in}} = Z_0 \frac{jZ_0 \tan \frac{2\pi}{\lambda'} l}{Z_0} = jZ_0 \tan \frac{2\pi}{\lambda'} l \qquad (245)$$

- if $\beta l < \frac{\pi}{2}$, i.e. $l < \frac{\lambda'}{4}$, input impedance of shorted stub is inductive: $L_{\text{eq}} = \frac{Z_0 \tan \frac{2\pi}{\lambda'} l}{\omega}$
- for short stubs ($\tan x \approx x$): $L_{\text{eq}} = Z_0 \frac{\sqrt{\epsilon_{\text{eff}}} l}{c_0}$
- frequency-independent

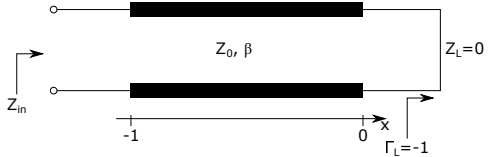


Figure 207: Impedance transformation by a shorted stub.

Stubs in MMICs

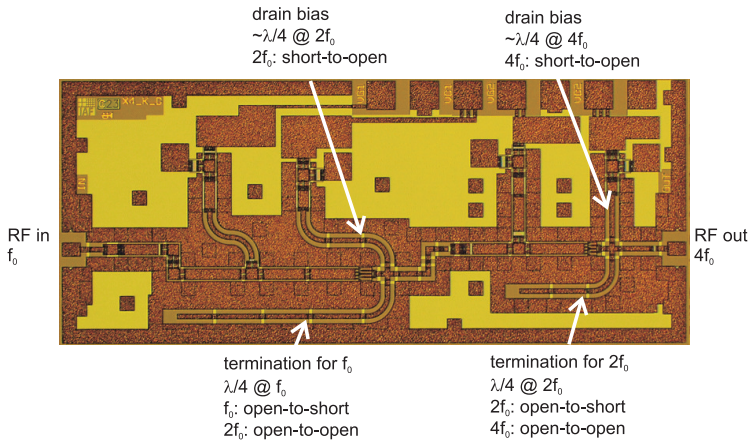


Figure 208: Stubs in MMICs: FET frequency multiplier from 35 to 140 GHz

Open Stub End Effect

- field reaches beyond end of TRL
- virtual lengthening Δl_{L} of TRL

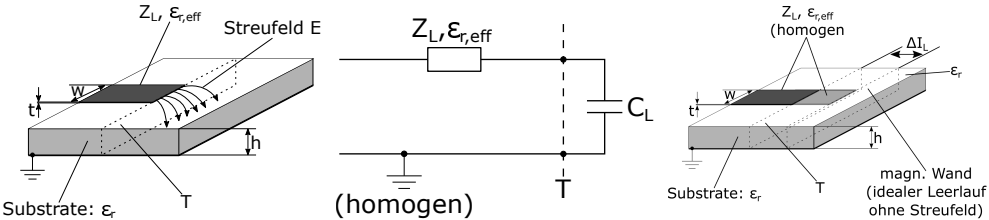


Figure 209: Open stub end effect and its equivalent circuit

Open Stub End Effect

Equivalent circuits:

- shunt capacitor C_{L}
- virtual TRL of length Δl_{L}

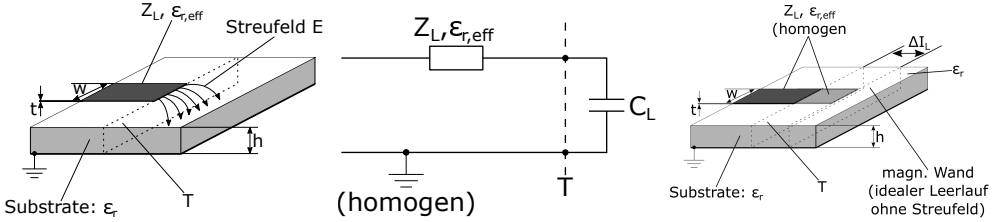


Figure 210: Open stub equivalent circuits

Shorted Stub End Effect

shorted stub end effect in MSL

- via to backside ground metal → parasitic inductance

shorted stub end effect in CPW

- less impact, small field perturbation at end of line

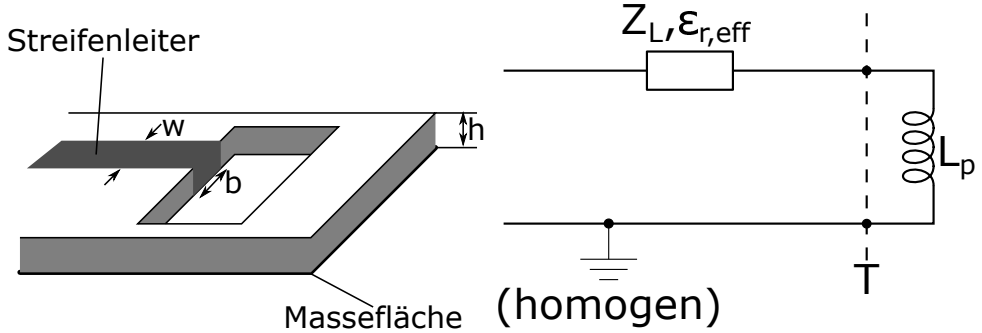


Figure 211: Shorted stub end effect

Serial Stubs

abrupt discontinuities of the line impedance ((??))

- modeled by series inductance and parallel capacitor (TRL)

popular applications

- TRL filters

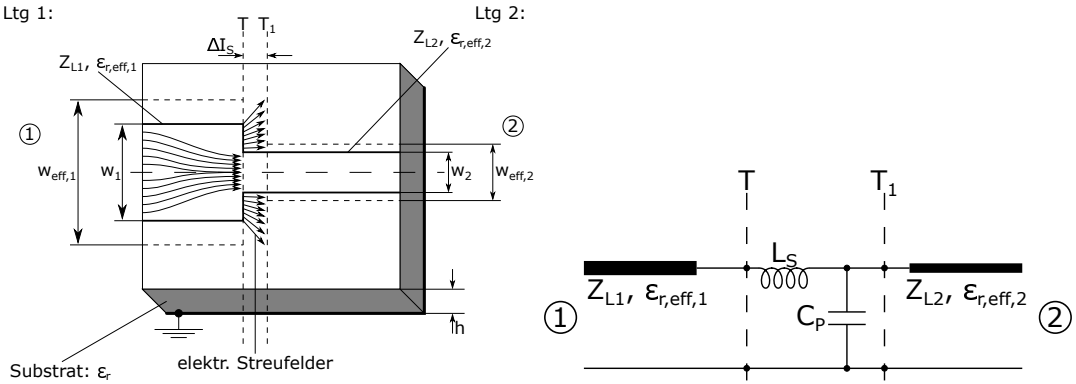


Figure 212: End effect in serial stubs and their equivalent circuit model

Tapered Lines

Tapers: continuous change of the line impedance

- $Z_{\mathrm{L}} = f(z)$ ... linear or exponential taper shape

popular applications

- Bond pad to circuit transitions

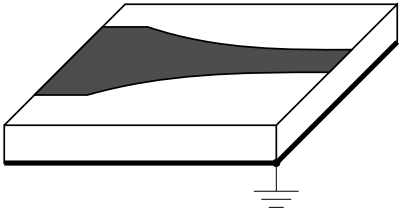


Figure 213: Tapered microstrip line

Couplers

Analysis Principle

- Analytical solution of coupled lines and couplers via even- and odd-mode analysis (not included here)
- cf. e.g. [**pozar2004microwave**]

Couplers

Intuitive derivation of coupler S-matrix

- Reflection: matched ports: $S_{ii} = 0$ (no reflection at port i, when all other ports are terminated with matched loads)
- Transmission
  - isolated ports: $S_{ij} = 0$
  - coupled port magnitude
    - 3-dB case: $|S_{ij}| = \frac{1}{\sqrt{2}}$ (half of the power is transmitted from port j to i)
  - coupled port phase
    - 90° phase shift: $\angle S_{ij} = -j = -90°$ (negative phase, because transmitted signal at port i arrives delayed w.r.t. incident signal at port j)
    - 180° phase shift: $\angle S_{ij} = -1 = -180°$
- reciprocity

Rat-race coupler (180° hybrid)

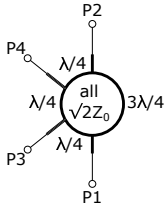
S-parameters of the ideal rat-race coupler:

$$S = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & j & -j & 0 \\ j & 0 & 0 & -j \\ -j & 0 & 0 & -j \\ 0 & -j & -j & 0 \end{bmatrix} \tag{246}$$

- isolation of ports P1-P4 and P2-P3 (symmetry)
- sum (P3) and difference (P2) ports, if input at P1 and P4
- dimensioning of $Z_{\mathrm{L}} = \sqrt{2} Z_0$ for port matching

popular applications:

- monostatic transceivers



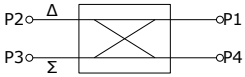


Figure 214: Rat-race coupler

(90°) Hybrid Coupler, Branchline Coupler

s-parameters of the ideal 3-dB branchline coupler:

$$S = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -j & -1 & 0 \\ -j & 0 & 0 & -1 \\ -1 & 0 & 0 & -j \\ 0 & -1 & -j & 0 \end{bmatrix} \tag{247}$$

- isolated port (P4)
- 90° phase shift between ports P2-P3, if input at P1
- dimensioning of Z_{L}: port match, degree of coupling typically -3 dB

popular applications:

- balanced power amplifiers
- IQ mixers
- single-sideband (ssb) mixers

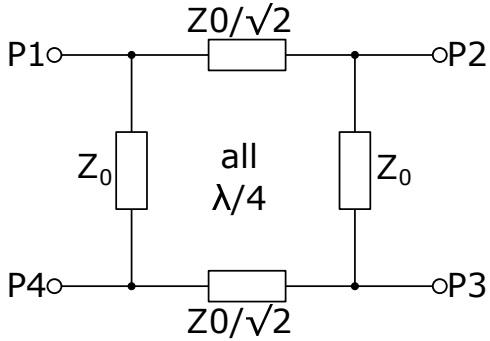


Figure 215: Branchline coupler

Directional coupler (90°)

S-parameters of the ideal 3-dB directional coupler:

$$S = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -j & -1 & 0 \\ -j & 0 & 0 & -1 \\ -1 & 0 & 0 & -j \\ 0 & -1 & -j & 0 \end{bmatrix} \tag{248}$$

- isolated port (P4)
- 90° phase shift between ports P2-P3, if input at P1
- dimensioning of Z_{L} and line distance: degree of coupling, port match

popular applications:

- power detectors (typ. as 10-dB coupler)
- network analysers: distinguish incident (a) and reflected (b) waves

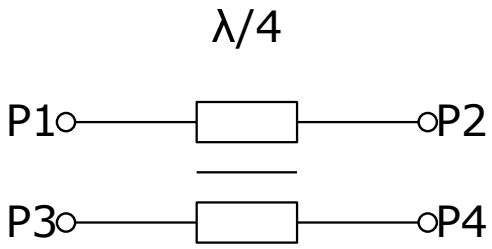


Figure 216: Directional coupler

Marchand-BALUN (180°)

S-parameters of the ideal 3-dB Marchand Balun:

$$S = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -j & j \\ -j & 0 & 0 \\ j & 0 & 0 \end{bmatrix} \tag{249}$$

- unbalanced (single-ended, P1) to balanced (differential, P2-P3) transformer (UNBAL, BAL)

popular applications:

- balanced frequency multipliers and mixers

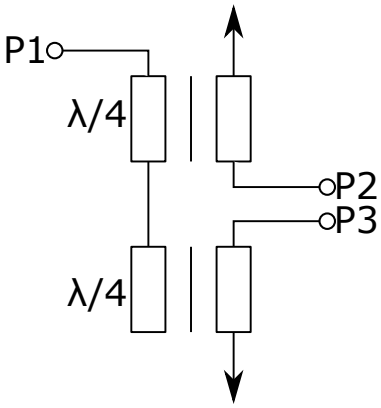


Figure 217: Marchand Balun

Wilkinson Divider (0°)

S-parameters of the ideal 3-dB wilkinson power divider/combiner:

$$S = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -j & -j \\ -j & 0 & 0 \\ -j & 0 & 0 \end{bmatrix} \qquad (250)$$

- power splitter/combiner
- dimensioning of $Z_{\mathrm{L}} = \sqrt{2}Z_0$ for port match
- isolation resistor $2Z_0$

popular applications:

- in-phase power splitting and combining in power amplifiers
- diplexing of signals in the same frequency range

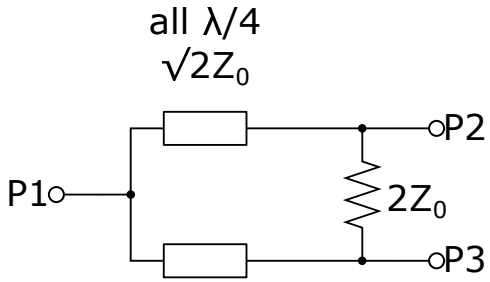


Figure 218: Wilkinson power divider/combiner

Universität Stuttgart

© Ingmar Kallfass
ILH

eMail	ingmar.kallfass@ilh.uni-stuttgart.de
Telephone	+49 711 685 68747
Facsimile	+49 711 685 58747

Bibliography I

[1] P. Pahl. “Power Amplifier MMICs for Frequencies Beyond 100 GHz”. Diplomarbeit. Germany: Universität Karlsruhe, 2008.

[2] David M. Pozar. *Microwave Engineering*. 3rd ed. Hoboken, NJ, USA: Wiley, 2004.